

We shall now define another product of two vectors. This product is a vector. Two important quantities in the study of rotational motion, namely, moment of a force and angular momentum, are defined as vector products.

Definition of Vector Product

A vector product of two vectors **a** and **b** is a vector **c** such that

- magnitude of $\mathbf{c} = c = ab \sin \theta$ where a and b are magnitudes of **a** and **b** and θ is the angle between the two vectors.
- c** is perpendicular to the plane containing **a** and **b**.
- if we take a right handed screw with its head lying in the plane of **a** and **b** and the screw perpendicular to this plane, and if we turn the head in the direction from **a** to **b**, then the tip of the screw advances in the direction of **c**. This right handed screw rule is illustrated in Fig. 7.15a.

Alternately, if one curls up the fingers of right hand around a line perpendicular to the plane of the vectors **a** and **b** and if the fingers are curled up in the direction from **a** to **b**, then the stretched thumb points in the direction of **c**, as shown in Fig. 7.15b.

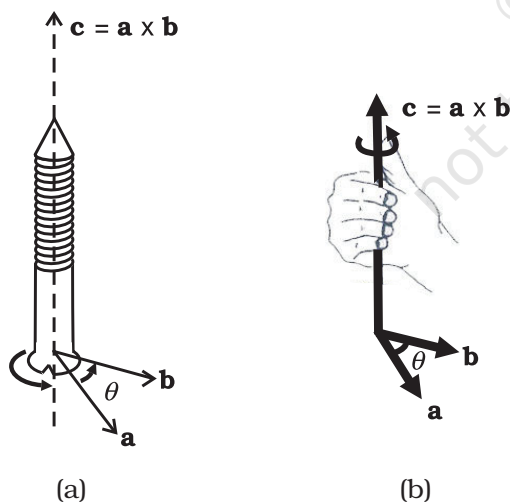


Fig. 7.15 (a) Rule of the right handed screw for defining the direction of the vector product of two vectors.

(b) Rule of the right hand for defining the direction of the vector product.

A simpler version of the right hand rule is the following : Open up your right hand palm and curl the fingers pointing from **a** to **b**. Your stretched thumb points in the direction of **c**.

It should be remembered that there are two angles between any two vectors **a** and **b**. In Fig. 7.15 (a) or (b) they correspond to θ (as shown) and $(360^\circ - \theta)$. While applying either of the above rules, the rotation should be taken through the smaller angle ($<180^\circ$) between **a** and **b**. It is θ here.

Because of the cross ($\times$) used to denote the vector product, it is also referred to as cross product.

- Note that scalar product of two vectors is commutative as said earlier, $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$

The vector product, however, is not commutative, i.e. $\mathbf{a} \times \mathbf{b} \neq \mathbf{b} \times \mathbf{a}$

The magnitude of both $\mathbf{a} \times \mathbf{b}$ and $\mathbf{b} \times \mathbf{a}$ is the same ($ab \sin \theta$); also, both of them are perpendicular to the plane of **a** and **b**. But the rotation of the right-handed screw in case of $\mathbf{a} \times \mathbf{b}$ is from **a** to **b**, whereas in case of $\mathbf{b} \times \mathbf{a}$ it is from **b** to **a**. This means the two vectors are in opposite directions. We have

$$\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$$

- Another interesting property of a vector product is its behaviour under reflection. Under reflection (i.e. on taking the plane mirror image) we have $x \rightarrow -x, y \rightarrow -y$ and $z \rightarrow -z$. As a result all the components of a vector change sign and thus $\mathbf{a} \rightarrow -\mathbf{a}, \mathbf{b} \rightarrow -\mathbf{b}$. What happens to $\mathbf{a} \times \mathbf{b}$ under reflection?

$$\mathbf{a} \times \mathbf{b} \rightarrow (-\mathbf{a}) \times (-\mathbf{b}) = \mathbf{a} \times \mathbf{b}$$

Thus, $\mathbf{a} \times \mathbf{b}$ does not change sign under reflection.

- Both scalar and vector products are distributive with respect to vector addition. Thus,

$$\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$$

$$\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$$

- We may write $\mathbf{c} = \mathbf{a} \times \mathbf{b}$ in the component form. For this we first need to obtain some elementary cross products:

(i) $\mathbf{a} \times \mathbf{a} = \mathbf{0}$ (**0** is a null vector, i.e. a vector with zero magnitude)

This follows since magnitude of $\mathbf{a} \times \mathbf{a}$ is $a^2 \sin 0^\circ = 0$.

From this follow the results

$$(i) \hat{i} \times \hat{i} = \mathbf{0}, \hat{j} \times \hat{j} = \mathbf{0}, \hat{k} \times \hat{k} = \mathbf{0}$$

$$(ii) \hat{i} \times \hat{j} = \hat{k}$$

Note that the magnitude of $\hat{i} \times \hat{j}$ is $\sin 90^\circ$ or 1, since $\hat{i}$ and $\hat{j}$ both have unit magnitude and the angle between them is 90° . Thus, $\hat{i} \times \hat{j}$ is a unit vector. A unit vector perpendicular to the plane of $\hat{i}$ and $\hat{j}$ and related to them by the right hand screw rule is $\hat{k}$. Hence, the above result. You may verify similarly,

$$\hat{j} \times \hat{k} = \hat{i} \text{ and } \hat{k} \times \hat{i} = \hat{j}$$

From the rule for commutation of the cross product, it follows:

$$\hat{j} \times \hat{i} = -\hat{k}, \hat{k} \times \hat{j} = -\hat{i}, \hat{i} \times \hat{k} = -\hat{j}$$

Note if $\hat{i}, \hat{j}, \hat{k}$ occur cyclically in the above vector product relation, the vector product is positive. If $\hat{i}, \hat{j}, \hat{k}$ do not occur in cyclic order, the vector product is negative.

Now,

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= (a_x \hat{i} + a_y \hat{j} + a_z \hat{k}) \times (b_x \hat{i} + b_y \hat{j} + b_z \hat{k}) \\ &= a_x b_y \hat{k} - a_x b_z \hat{j} - a_y b_x \hat{k} + a_y b_z \hat{i} + a_z b_x \hat{j} - a_z b_y \hat{i} \\ &= (a_y b_z - a_z b_y) \hat{i} + (a_z b_x - a_x b_z) \hat{j} + (a_x b_y - a_y b_x) \hat{k} \end{aligned}$$

We have used the elementary cross products in obtaining the above relation. The expression for $\mathbf{a} \times \mathbf{b}$ can be put in a determinant form which is easy to remember.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

► **Example 7.4** Find the scalar and vector products of two vectors. $\mathbf{a} = (3\hat{i} - 4\hat{j} + 5\hat{k})$ and $\mathbf{b} = (-2\hat{i} + \hat{j} - 3\hat{k})$

Answer

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= (3\hat{i} - 4\hat{j} + 5\hat{k}) \cdot (-2\hat{i} + \hat{j} - 3\hat{k}) \\ &= -6 - 4 - 15 \\ &= -25 \end{aligned}$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -4 & 5 \\ -2 & 1 & -3 \end{vmatrix} = 7\hat{i} - \hat{j} - 5\hat{k}$$

$$\text{Note } \mathbf{b} \times \mathbf{a} = -7\hat{i} + \hat{j} + 5\hat{k}$$

7.6 ANGULAR VELOCITY AND ITS RELATION WITH LINEAR VELOCITY

In this section we shall study what is angular velocity and its role in rotational motion. We have seen that every particle of a rotating body moves in a circle. The linear velocity of the particle is related to the angular velocity. The relation between these two quantities involves a vector product which we learnt about in the last section.

Let us go back to Fig. 7.4. As said above, in rotational motion of a rigid body about a fixed axis, every particle of the body moves in a circle,

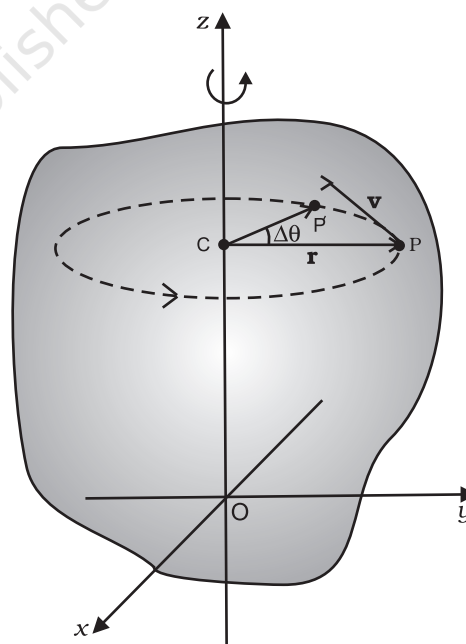


Fig. 7.16 Rotation about a fixed axis. (A particle (P) of the rigid body rotating about the fixed (z-) axis moves in a circle with centre (C) on the axis.)

which lies in a plane perpendicular to the axis and has its centre on the axis. In Fig. 7.16 we redraw Fig. 7.4, showing a typical particle (at a point P) of the rigid body rotating about a fixed axis (taken as the z-axis). The particle describes

a circle with a centre C on the axis. The radius of the circle is r , the perpendicular distance of the point P from the axis. We also show the linear velocity vector $\mathbf{v}$ of the particle at P . It is along the tangent at P to the circle.

Let P' be the position of the particle after an interval of time Δt (Fig. 7.16). The angle PCP' describes the angular displacement $\Delta\theta$ of the particle in time Δt . The average angular velocity of the particle over the interval Δt is $\Delta\theta/\Delta t$. As Δt tends to zero (i.e. takes smaller and smaller values), the ratio $\Delta\theta/\Delta t$ approaches a limit which is the instantaneous angular velocity $d\theta/dt$ of the particle at the position P . We denote the **instantaneous angular velocity** by ω (the Greek letter omega). We know from our study of circular motion that the magnitude of linear velocity v of a particle moving in a circle is related to the angular velocity of the particle ω by the simple relation $v = \omega r$, where r is the radius of the circle.

We observe that at any given instant the relation $v = \omega r$ applies to all particles of the rigid body. Thus for a particle at a perpendicular distance r_i from the fixed axis, the linear velocity at a given instant v_i is given by

$$v_i = \omega r_i \quad (7.19)$$

The index i runs from 1 to n , where n is the total number of particles of the body.

For particles on the axis, $r = 0$, and hence $v = \omega r = 0$. Thus, particles on the axis are stationary. This verifies that the axis is *fixed*.

Note that we use the same angular velocity ω for all the particles. **We therefore, refer to ω as the angular velocity of the whole body.**

We have characterised pure translation of a body by all parts of the body having the same velocity at any instant of time. Similarly, we may characterise pure rotation by all parts of the body having the same angular velocity at any instant of time. Note that this characterisation of the rotation of a rigid body about a fixed axis is **just another way** of saying as in Sec. 7.1 that each particle of the body moves in a circle, which lies in a plane perpendicular to the axis and has the centre on the axis.

In our discussion so far the angular velocity appears to be a scalar. In fact, it is a vector. We shall not justify this fact, but we shall accept it. For rotation about a fixed axis, the angular velocity vector lies along the axis of rotation,

and points out in the direction in which a right handed screw would advance, if the head of the screw is rotated with the body. (See Fig. 7.17a).

The magnitude of this vector is $\omega = d\theta/dt$ referred as above.

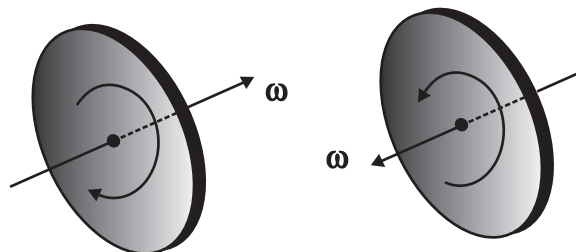


Fig. 7.17 (a) If the head of a right handed screw rotates with the body, the screw advances in the direction of the angular velocity ω . If the sense (clockwise or anticlockwise) of rotation of the body changes, so does the direction of ω .

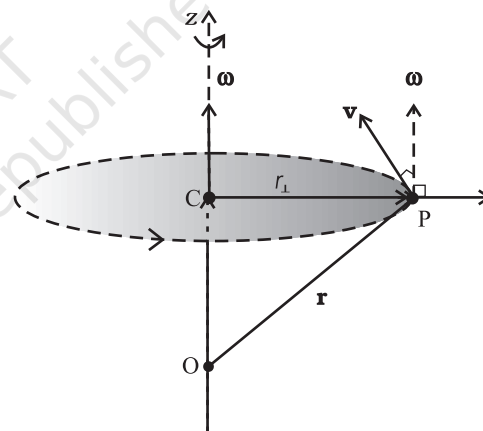


Fig. 7.17 (b) The angular velocity vector ω is directed along the fixed axis as shown. The linear velocity of the particle at P is $\mathbf{v} = \omega \times \mathbf{r}$. It is perpendicular to both ω and $\mathbf{r}$ and is directed along the tangent to the circle described by the particle.

We shall now look at what the vector product $\omega \times \mathbf{r}$ corresponds to. Refer to Fig. 7.17(b) which is a part of Fig. 7.16 reproduced to show the path of the particle P . The figure shows the vector ω directed along the fixed (z -) axis and also the position vector $\mathbf{r} = \mathbf{OP}$ of the particle at P of the rigid body with respect to the origin O . Note that the origin is chosen to be on the axis of rotation.

Now $\boldsymbol{\omega} \times \mathbf{r} = \boldsymbol{\omega} \times \mathbf{OP} = \boldsymbol{\omega} \times (\mathbf{OC} + \mathbf{CP})$
 But $\boldsymbol{\omega} \times \mathbf{OC} = \mathbf{0}$ as $\boldsymbol{\omega}$ is along $\mathbf{OC}$
 Hence $\boldsymbol{\omega} \times \mathbf{r} = \boldsymbol{\omega} \times \mathbf{CP}$

The vector $\boldsymbol{\omega} \times \mathbf{CP}$ is perpendicular to $\boldsymbol{\omega}$, i.e. to the z-axis and also to $\mathbf{CP}$, the radius of the circle described by the particle at P. It is therefore, along the tangent to the circle at P. Also, the magnitude of $\boldsymbol{\omega} \times \mathbf{CP}$ is ω (CP) since $\boldsymbol{\omega}$ and $\mathbf{CP}$ are perpendicular to each other. We shall denote $\mathbf{CP}$ by $\mathbf{r}_\perp$ and not by $\mathbf{r}$, as we did earlier.

Thus, $\boldsymbol{\omega} \times \mathbf{r}$ is a vector of magnitude $\omega r_\perp$ and is along the tangent to the circle described by the particle at P. The linear velocity vector $\mathbf{v}$ at P has the same magnitude and direction. Thus,

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} \quad (7.20)$$

In fact, the relation, Eq. (7.20), holds good even for rotation of a rigid body with one point fixed, such as the rotation of the top [Fig. 7.6(a)]. In this case $\mathbf{r}$ represents the position vector of the particle with respect to the fixed point taken as the origin.

We note that **for rotation about a fixed axis, the direction of the vector $\boldsymbol{\omega}$ does not change with time. Its magnitude may, however, change from instant to instant. For the more general rotation, both the magnitude and the direction of $\boldsymbol{\omega}$ may change from instant to instant.**

7.6.1 Angular acceleration

You may have noticed that we are developing the study of rotational motion along the lines of the study of translational motion with which we are already familiar. Analogous to the kinetic variables of linear displacement ($\mathbf{s}$) and velocity ($\mathbf{v}$) in translational motion, we have angular displacement ($\boldsymbol{\theta}$) and angular velocity ($\boldsymbol{\omega}$) in rotational motion. It is then natural to define in rotational motion the concept of angular acceleration in analogy with linear acceleration defined as the time rate of change of velocity in translational motion. We define angular acceleration $\boldsymbol{\alpha}$ as the time rate of change of angular velocity; Thus,

$$\boldsymbol{\alpha} = \frac{d\boldsymbol{\omega}}{dt} \quad (7.21)$$

If the axis of rotation is fixed, the direction of $\boldsymbol{\omega}$ and hence, that of $\boldsymbol{\alpha}$ is fixed. In this case the vector equation reduces to a scalar equation

$$\alpha = \frac{d\omega}{dt} \quad (7.22)$$

7.7 TORQUE AND ANGULAR MOMENTUM

In this section, we shall acquaint ourselves with two physical quantities (torque and angular momentum) which are defined as vector products of two vectors. These as we shall see, are especially important in the discussion of motion of systems of particles, particularly rigid bodies.

7.7.1 Moment of force (Torque)

We have learnt that the motion of a rigid body, in general, is a combination of rotation and translation. If the body is fixed at a point or along a line, it has only rotational motion. We know that force is needed to change the translational state of a body, i.e. to produce linear acceleration. We may then ask, what is the analogue of force in the case of rotational motion? To look into the question in a concrete situation let us take the example of opening or closing of a door. A door is a rigid body which can rotate about a fixed vertical axis passing through the hinges. What makes the door rotate? It is clear that unless a force is applied the door does not rotate. But any force does not do the job. A force applied to the hinge line cannot produce any rotation at all, whereas a force of given magnitude applied at right angles to the door at its outer edge is most effective in producing rotation. It is not the force alone, but how and where the force is applied is important in rotational motion.

The rotational analogue of force in linear motion is **moment of force**. It is also referred to as **torque** or **couple**. (We shall use the words moment of force and torque interchangeably.) We shall first define the moment of force for the special case of a single particle. Later on we shall extend the concept to systems of particles including rigid bodies. We shall also relate it to a change in the state of rotational motion, i.e. is angular acceleration of a rigid body.

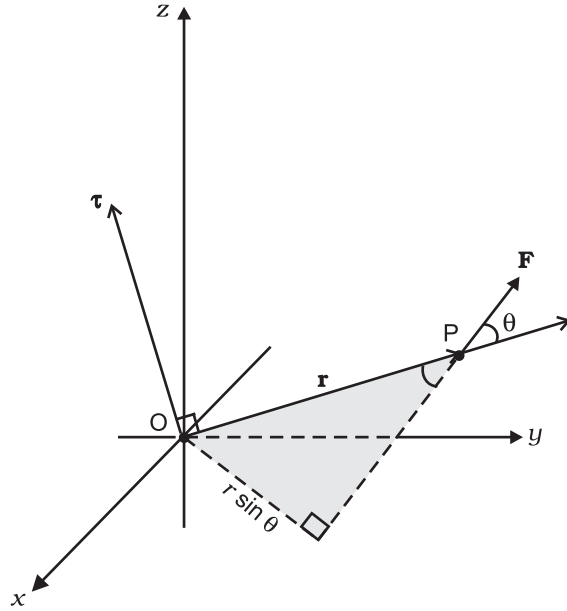


Fig. 7.18 $\tau = \mathbf{r} \times \mathbf{F}$, τ is perpendicular to the plane containing $\mathbf{r}$ and $\mathbf{F}$, and its direction is given by the right handed screw rule.

If a force acts on a single particle at a point P whose position with respect to the origin O is given by the position vector $\mathbf{r}$ (Fig. 7.18), the moment of the force acting on the particle with respect to the origin O is defined as the vector product

$$\tau = \mathbf{r} \times \mathbf{F} \quad (7.23)$$

The moment of force (or torque) is a vector quantity. The symbol τ stands for the Greek letter *tau*. The magnitude of τ is

$$\tau = r F \sin \theta \quad (7.24a)$$

where r is the magnitude of the position vector $\mathbf{r}$, i.e. the length OP, F is the magnitude of force $\mathbf{F}$ and θ is the angle between $\mathbf{r}$ and $\mathbf{F}$ as shown.

Moment of force has dimensions $M L^2 T^{-2}$. Its dimensions are the same as those of work or energy. It is, however, a very different physical quantity than work. Moment of a force is a vector, while work is a scalar. The SI unit of moment of force is newton metre (N m). The magnitude of the moment of force may be written

$$\tau = (r \sin \theta) F = r_{\perp} F \quad (7.24b)$$

$$\text{or } \tau = r F \sin \theta = r F_{\perp} \quad (7.24c)$$

where $r_{\perp} = r \sin \theta$ is the perpendicular distance

of the line of action of $\mathbf{F}$ from the origin and $F_{\perp} (= F \sin \theta)$ is the component of $\mathbf{F}$ in the direction perpendicular to $\mathbf{r}$. Note that $\tau = 0$ if $r = 0$, $F = 0$ or $\theta = 0^\circ$ or 180° . Thus, the moment of a force vanishes if either the magnitude of the force is zero, or if the line of action of the force passes through the origin.

One may note that since $\mathbf{r} \times \mathbf{F}$ is a vector product, properties of a vector product of two vectors apply to it. If the direction of $\mathbf{F}$ is reversed, the direction of the moment of force is reversed. If directions of both $\mathbf{r}$ and $\mathbf{F}$ are reversed, the direction of the moment of force remains the same.

7.7.2 Angular momentum of a particle

Just as the moment of a force is the rotational analogue of force in linear motion, the quantity angular momentum is the rotational analogue of linear momentum. We shall first define angular momentum for the special case of a single particle and look at its usefulness in the context of single particle motion. We shall then extend the definition of angular momentum to systems of particles including rigid bodies.

Like moment of a force, angular momentum is also a vector product. It could also be referred to as moment of (linear) momentum. From this term one could guess how angular momentum is defined.

Consider a particle of mass m and linear momentum $\mathbf{p}$ at a position $\mathbf{r}$ relative to the origin O. The angular momentum $\mathbf{l}$ of the particle with respect to the origin O is defined to be

$$\mathbf{l} = \mathbf{r} \times \mathbf{p} \quad (7.25a)$$

The magnitude of the angular momentum vector is

$$l = r p \sin \theta \quad (7.26a)$$

where p is the magnitude of $\mathbf{p}$ and θ is the angle between $\mathbf{r}$ and $\mathbf{p}$. We may write

$$l = r p_{\perp} \text{ or } r_{\perp} p \quad (7.26b)$$

where $r_{\perp} (= r \sin \theta)$ is the perpendicular distance of the directional line of $\mathbf{p}$ from the origin and $p_{\perp} (= p \sin \theta)$ is the component of $\mathbf{p}$ in a direction perpendicular to $\mathbf{r}$. We expect the angular momentum to be zero ($l = 0$), if the linear momentum vanishes ($p = 0$), if the particle is at the origin ($r = 0$), or if the directional line of $\mathbf{p}$ passes through the origin $\theta = 0^\circ$ or 180° .

The physical quantities, moment of a force and angular momentum, have an important relation between them. It is the rotational analogue of the relation between force and linear momentum. For deriving the relation in the context of a single particle, we differentiate $\mathbf{l} = \mathbf{r} \times \mathbf{p}$ with respect to time,

$$\frac{d\mathbf{l}}{dt} = \frac{d}{dt}(\mathbf{r} \times \mathbf{p})$$

Applying the product rule for differentiation to the right hand side,

$$\frac{d}{dt}(\mathbf{r} \times \mathbf{p}) = \frac{d\mathbf{r}}{dt} \times \mathbf{p} + \mathbf{r} \times \frac{d\mathbf{p}}{dt}$$

Now, the velocity of the particle is $\mathbf{v} = d\mathbf{r}/dt$ and $\mathbf{p} = m\mathbf{v}$

$$\text{Because of this } \frac{d\mathbf{r}}{dt} \times \mathbf{p} = \mathbf{v} \times m\mathbf{v} = 0,$$

as the vector product of two parallel vectors vanishes. Further, since $d\mathbf{p}/dt = \mathbf{F}$,

$$\mathbf{r} \times \frac{d\mathbf{p}}{dt} = \mathbf{r} \times \mathbf{F} = \boldsymbol{\tau}$$

$$\text{Hence } \frac{d}{dt}(\mathbf{r} \times \mathbf{p}) = \boldsymbol{\tau}$$

$$\text{or } \frac{d\mathbf{l}}{dt} = \boldsymbol{\tau} \quad (7.27)$$

Thus, the time rate of change of the angular momentum of a particle is equal to the torque acting on it. This is the rotational analogue of the equation $\mathbf{F} = d\mathbf{p}/dt$, which expresses Newton's second law for the translational motion of a single particle.

Torque and angular momentum for a system of particles

To get the total angular momentum of a system of particles about a given point we need to add vectorially the angular momenta of individual particles. Thus, for a system of n particles,

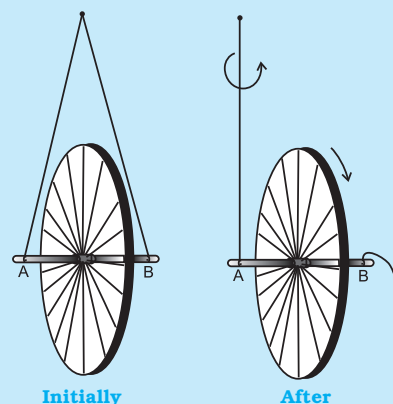
$$\mathbf{L} = \mathbf{l}_1 + \mathbf{l}_2 + \dots + \mathbf{l}_n = \sum_{i=1}^n \mathbf{l}_i$$

The angular momentum of the i^{th} particle is given by

$$\mathbf{l}_i = \mathbf{r}_i \times \mathbf{p}_i$$

where $\mathbf{r}_i$ is the position vector of the i^{th} particle with respect to a given origin and $\mathbf{p} = (m_i \mathbf{v}_i)$ is the linear momentum of the particle. (The

An experiment with the bicycle rim



Take a bicycle rim and extend its axle on both sides. Tie two strings at both ends A and B, as shown in the adjoining figure. Hold both the strings together in

one hand such that the rim is vertical. If you leave one string, the rim will tilt. Now keeping the rim in vertical position with both the strings in one hand, put the wheel in fast rotation around the axle with the other hand. Then leave one string, say B, from your hand, and observe what happens.

The rim keeps rotating in a vertical plane and the plane of rotation turns around the string A which you are holding. We say that the axis of rotation of the rim or equivalently its angular momentum precesses about the string A.

The rotating rim gives rise to an angular momentum. Determine the direction of this angular momentum. When you are holding the rotating rim with string A, a torque is generated. (We leave it to you to find out how the torque is generated and what its direction is.) The effect of the torque on the angular momentum is to make it precess around an axis perpendicular to both the angular momentum and the torque. Verify all these statements.

particle has mass m_i and velocity $\mathbf{v}_i$.) We may write the total angular momentum of a system of particles as

$$\mathbf{L} = \sum \mathbf{l}_i = \sum \mathbf{r}_i \times \mathbf{p}_i \quad (7.25b)$$

This is a generalisation of the definition of angular momentum (Eq. 7.25a) for a single particle to a system of particles.

Using Eqs. (7.23) and (7.25b), we get

$$\frac{d\mathbf{L}}{dt} = \frac{d}{dt}(\sum \mathbf{l}_i) = \sum \frac{d\mathbf{l}_i}{dt} = \sum \boldsymbol{\tau}_i \quad (7.28a)$$

where $\boldsymbol{\tau}_i$ is the torque acting on the i^{th} particle;

$$\boldsymbol{\tau}_i = \mathbf{r}_i \times \mathbf{F}_i$$

The force $\mathbf{F}_i$ on the i^{th} particle is the vector sum of external forces $\mathbf{F}_i^{\text{ext}}$ acting on the particle and the internal forces $\mathbf{F}_i^{\text{int}}$ exerted on it by the other particles of the system. We may therefore separate the contribution of the external and the internal forces to the total torque

$$\boldsymbol{\tau} = \sum_i \boldsymbol{\tau}_i = \sum_i \mathbf{r}_i \times \mathbf{F}_i \quad \text{as}$$

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{\text{ext}} + \boldsymbol{\tau}_{\text{int}},$$

$$\text{where } \boldsymbol{\tau}_{\text{ext}} = \sum_i \mathbf{r}_i \times \mathbf{F}_i^{\text{ext}}$$

$$\text{and } \boldsymbol{\tau}_{\text{int}} = \sum_i \mathbf{r}_i \times \mathbf{F}_i^{\text{int}}$$

We shall assume not only Newton's third law of motion, i.e. the forces between any two particles of the system are equal and opposite, but also that these forces are directed along the line joining the two particles. In this case the contribution of the internal forces to the total torque on the system is zero, since the torque resulting from each action-reaction pair of forces is zero. We thus have, $\boldsymbol{\tau}_{\text{int}} = \mathbf{0}$ and therefore $\boldsymbol{\tau} = \boldsymbol{\tau}_{\text{ext}}$.

Since $\boldsymbol{\tau} = \sum_i \boldsymbol{\tau}_i$, it follows from Eq. (7.28a) that

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}_{\text{ext}} \quad (7.28 \text{ b})$$

Thus, **the time rate of the total angular momentum of a system of particles about a point** (taken as the origin of our frame of reference) **is equal to the sum of the external torques** (i.e. the torques due to external forces) **acting on the system taken about the same point.** Eq. (7.28 b) is the generalisation of the single particle case of Eq. (7.23) to a system of particles. Note that when we have only one particle, there are no internal forces or torques. Eq.(7.28 b) is the rotational analogue of

$$\frac{d\mathbf{P}}{dt} = \mathbf{F}_{\text{ext}} \quad (7.17)$$

Note that like Eq.(7.17), Eq.(7.28b) holds good for any system of particles, whether it is a rigid body or its individual particles have all kinds of internal motion.

Conservation of angular momentum

If $\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$, Eq. (7.28b) reduces to

$$\frac{d\mathbf{L}}{dt} = 0$$

$$\text{or } \mathbf{L} = \text{constant.} \quad (7.29 \text{ a})$$

Thus, if the total external torque on a system of particles is zero, then the total angular momentum of the system is conserved, i.e. remains constant. Eq. (7.29a) is equivalent to three scalar equations,

$$L_x = K_1, L_y = K_2 \text{ and } L_z = K_3 \quad (7.29 \text{ b})$$

Here K_1 , K_2 and K_3 are constants; L_x , L_y and L_z are the components of the total angular momentum vector $\mathbf{L}$ along the x, y and z axes respectively. The statement that the total angular momentum is conserved means that each of these three components is conserved.

Eq. (7.29a) is the rotational analogue of Eq. (7.18a), i.e. the conservation law of the total linear momentum for a system of particles. Like Eq. (7.18a), it has applications in many practical situations. We shall look at a few of the interesting applications later on in this chapter.

► **Example 7.5** Find the torque of a force $7\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 5\hat{\mathbf{k}}$ about the origin. The force acts on a particle whose position vector is $\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$.

Answer Here $\mathbf{r} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$

$$\text{and } \mathbf{F} = 7\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 5\hat{\mathbf{k}}.$$

We shall use the determinant rule to find the torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$

$$\boldsymbol{\tau} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & -1 & 1 \\ 7 & 3 & -5 \end{vmatrix} = (5 - 3)\hat{\mathbf{i}} - (-5 - 7)\hat{\mathbf{j}} + (3 - (-7))\hat{\mathbf{k}}$$

$$\text{or } \boldsymbol{\tau} = 2\hat{\mathbf{i}} + 12\hat{\mathbf{j}} + 10\hat{\mathbf{k}} \quad \blacktriangleleft$$

► **Example 7.6** Show that the angular momentum about any point of a single particle moving with constant velocity remains constant throughout the motion.

Answer Let the particle with velocity $\mathbf{v}$ be at point P at some instant t . We want to calculate the angular momentum of the particle about an arbitrary point O.

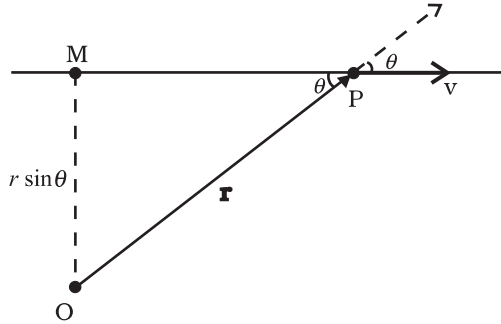


Fig 7.19

The angular momentum is $\mathbf{l} = \mathbf{r} \times m\mathbf{v}$. Its magnitude is $mvr \sin \theta$, where θ is the angle between $\mathbf{r}$ and $\mathbf{v}$ as shown in Fig. 7.19. Although the particle changes position with time, the line of direction of $\mathbf{v}$ remains the same and hence $OM = r \sin \theta$, is a constant.

Further, the direction of $\mathbf{l}$ is perpendicular to the plane of $\mathbf{r}$ and $\mathbf{v}$. It is into the page of the figure. This direction does not change with time.

Thus, $\mathbf{l}$ remains the same in magnitude and direction and is therefore conserved. Is there any external torque on the particle?

7.8 EQUILIBRIUM OF A RIGID BODY

We are now going to concentrate on the motion of rigid bodies rather than on the motion of general systems of particles.

We shall recapitulate what effect the external forces have on a rigid body. (Henceforth we shall omit the adjective 'external' because unless stated otherwise, we shall deal with only external forces and torques.) The forces change the translational state of the motion of the rigid body, i.e. they change its total linear momentum in accordance with Eq. (7.17). But this is not the only effect the forces have. The total torque on the body may not vanish. Such a torque changes the rotational state of motion of the rigid body, i.e. it changes the total angular momentum of the body in accordance with Eq. (7.28 b).

A rigid body is said to be in mechanical equilibrium, if both its linear momentum and angular momentum are not changing with time, or equivalently, the body has neither linear

acceleration nor angular acceleration. This means

- (1) the total force, i.e. the vector sum of the forces, on the rigid body is zero;

$$\mathbf{F}_1 + \mathbf{F}_2 + \dots + \mathbf{F}_n = \sum_{i=1}^n \mathbf{F}_i = \mathbf{0} \quad (7.30a)$$

If the total force on the body is zero, then the total linear momentum of the body does not change with time. Eq. (7.30a) gives the condition for the translational equilibrium of the body.

- (2) The total torque, i.e. the vector sum of the torques on the rigid body is zero,

$$\boldsymbol{\tau}_1 + \boldsymbol{\tau}_2 + \dots + \boldsymbol{\tau}_n = \sum_{i=1}^n \boldsymbol{\tau}_i = \mathbf{0} \quad (7.30b)$$

If the total torque on the rigid body is zero, the total angular momentum of the body does not change with time. Eq. (7.30 b) gives the condition for the rotational equilibrium of the body.

One may raise a question, whether the rotational equilibrium condition [Eq. 7.30(b)] remains valid, if the origin with respect to which the torques are taken is shifted. One can show that if the translational equilibrium condition [Eq. 7.30(a)] holds for a rigid body, then such a shift of origin does not matter, i.e. the rotational equilibrium condition is independent of the location of the origin about which the torques are taken. Example 7.7 gives a proof of this result in a special case of a couple, i.e. two forces acting on a rigid body in translational equilibrium. The generalisation of this result to n forces is left as an exercise.

Eq. (7.30a) and Eq. (7.30b), both, are vector equations. They are equivalent to three scalar equations each. Eq. (7.30a) corresponds to

$$\sum_{i=1}^n F_{ix} = 0, \quad \sum_{i=1}^n F_{iy} = 0 \quad \text{and} \quad \sum_{i=1}^n F_{iz} = 0 \quad (7.31a)$$

where F_{ix} , F_{iy} and F_{iz} are respectively the x , y and z components of the forces $\mathbf{F}_i$. Similarly, Eq. (7.30b) is equivalent to three scalar equations

$$\sum_{i=1}^n \tau_{ix} = 0, \quad \sum_{i=1}^n \tau_{iy} = 0 \quad \text{and} \quad \sum_{i=1}^n \tau_{iz} = 0 \quad (7.31b)$$

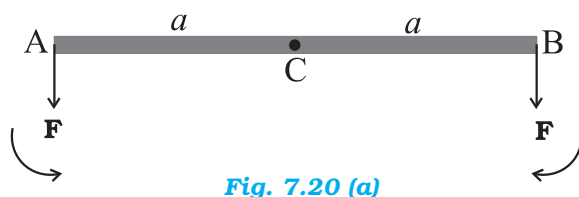
where τ_{ix} , τ_{iy} and τ_{iz} are respectively the x , y and z components of the torque $\boldsymbol{\tau}_i$.

Eq. (7.31a) and (7.31b) give six independent conditions to be satisfied for mechanical equilibrium of a rigid body. In a number of problems all the forces acting on the body are coplanar. Then we need only three conditions to be satisfied for mechanical equilibrium. Two of these conditions correspond to translational equilibrium; the sum of the components of the forces along any two perpendicular axes in the plane must be zero. The third condition corresponds to rotational equilibrium. The sum of the components of the torques along any axis perpendicular to the plane of the forces must be zero.

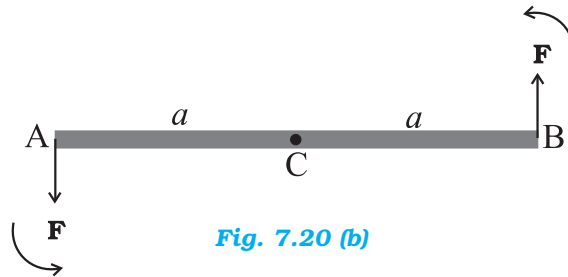
The conditions of equilibrium of a rigid body may be compared with those for a particle, which we considered in earlier chapters. Since consideration of rotational motion does not apply to a particle, only the conditions for translational equilibrium (Eq. 7.30 a) apply to a particle. Thus, for equilibrium of a particle the vector sum of all the forces on it must be zero. Since all these forces act on the single particle, they must be concurrent. Equilibrium under concurrent forces was discussed in the earlier chapters.

A body may be in partial equilibrium, i.e., it may be in translational equilibrium and not in rotational equilibrium, or it may be in rotational equilibrium and not in translational equilibrium.

Consider a light (i.e. of negligible mass) rod (AB) as shown in Fig. 7.20(a). At the two ends (A and B) of which two parallel forces, both equal in magnitude and acting along same direction are applied perpendicular to the rod.



Let C be the midpoint of AB, $CA = CB = a$. the moment of the forces at A and B will both be equal in magnitude (aF), but opposite in sense as shown. The net moment on the rod will be zero. The system will be in rotational equilibrium, but it will not be in translational equilibrium; $\sum \mathbf{F} \neq \mathbf{0}$



The force at B in Fig. 7.20(a) is reversed in Fig. 7.20(b). Thus, we have the same rod with two forces of equal magnitude but acting in opposite directions applied perpendicular to the rod, one at end A and the other at end B. Here the moments of both the forces are equal, but they are not opposite; they act in the same sense and cause anticlockwise rotation of the rod. The total force on the body is zero; so the body is in translational equilibrium; but it is not in rotational equilibrium. Although the rod is not fixed in any way, it undergoes pure rotation (i.e. rotation without translation).

A pair of forces of equal magnitude but acting in opposite directions with different lines of action is known as a **couple** or **torque**. A couple produces rotation without translation.

When we open the lid of a bottle by turning it, our fingers are applying a couple to the lid [Fig. 7.21(a)]. Another known example is a compass needle in the earth's magnetic field as shown in the Fig. 7.21(b). The earth's magnetic field exerts equal forces on the north and south poles. The force on the North Pole is towards the north, and the force on the South Pole is toward the south. Except when the needle points in the north-south direction; the two forces do not have the same line of action. Thus there is a **couple** acting on the needle due to the earth's magnetic field.



Fig. 7.21(a) Our fingers apply a couple to turn the lid.

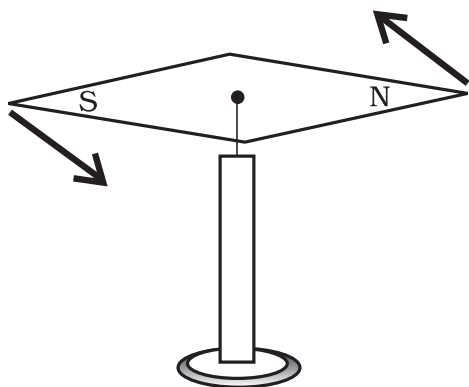


Fig. 7.21(b) The Earth's magnetic field exerts equal and opposite forces on the poles of a compass needle. These two forces form a couple.

► **Example 7.7** Show that moment of a couple does not depend on the point about which you take the moments.

Answer

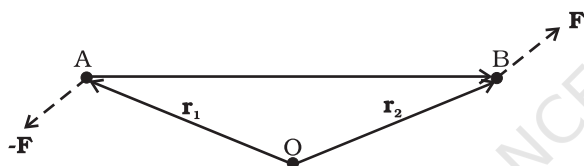


Fig. 7.22

Consider a couple as shown in Fig. 7.22 acting on a rigid body. The forces $\mathbf{F}$ and $-\mathbf{F}$ act respectively at points B and A. These points have position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ with respect to origin O. Let us take the moments of the forces about the origin.

The moment of the couple = sum of the moments of the two forces making the couple

$$\begin{aligned} &= \mathbf{r}_1 \times (-\mathbf{F}) + \mathbf{r}_2 \times \mathbf{F} \\ &= \mathbf{r}_2 \times \mathbf{F} - \mathbf{r}_1 \times \mathbf{F} \\ &= (\mathbf{r}_2 - \mathbf{r}_1) \times \mathbf{F} \end{aligned}$$

But $\mathbf{r}_1 + \mathbf{AB} = \mathbf{r}_2$, and hence $\mathbf{AB} = \mathbf{r}_2 - \mathbf{r}_1$.

The moment of the couple, therefore, is $\mathbf{AB} \times \mathbf{F}$.

Clearly this is independent of the origin, the point about which we took the moments of the forces. ◀

7.8.1 Principle of moments

An ideal lever is essentially a light (i.e. of negligible mass) rod pivoted at a point along its

length. This point is called the fulcrum. A see-saw on the children's playground is a typical example of a lever. Two forces F_1 and F_2 , parallel to each other and usually perpendicular to the lever, as shown here, act on the lever at distances d_1 and d_2 respectively from the fulcrum as shown in Fig. 7.23.

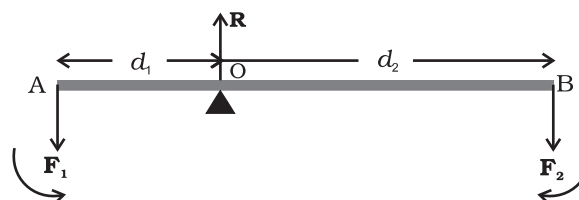


Fig. 7.23

The lever is a system in mechanical equilibrium. Let $\mathbf{R}$ be the reaction of the support at the fulcrum; $\mathbf{R}$ is directed opposite to the forces F_1 and F_2 . For translational equilibrium,

$$R - F_1 - F_2 = 0 \quad (\text{i})$$

For considering rotational equilibrium we take the moments about the fulcrum; the sum of moments must be zero,

$$d_1 F_1 - d_2 F_2 = 0 \quad (\text{ii})$$

Normally the anticlockwise (clockwise) moments are taken to be positive (negative). Note $\mathbf{R}$ acts at the fulcrum itself and has zero moment about the fulcrum.

In the case of the lever force F_1 is usually some weight to be lifted. It is called the *load* and its distance from the fulcrum d_1 is called the *load arm*. Force F_2 is the *effort* applied to lift the load; distance d_2 of the effort from the fulcrum is the *effort arm*.

Eq. (ii) can be written as

$$d_1 F_1 = d_2 F_2 \quad (7.32a)$$

or load arm $\times$ load = effort arm $\times$ effort

The above equation expresses the principle of moments for a lever. Incidentally the ratio F_1/F_2 is called the Mechanical Advantage (M.A.);

$$\text{M.A.} = \frac{F_1}{F_2} = \frac{d_2}{d_1} \quad (7.32b)$$

If the effort arm d_2 is larger than the load arm, the mechanical advantage is greater than one. Mechanical advantage greater than one means that a small effort can be used to lift a large load. There are several examples of a lever around you besides the see-saw. The beam of a balance is a lever. Try to find more such

examples and identify the fulcrum, the effort and effort arm, and the load and the load arm of the lever in each case.

You may easily show that the principle of moment holds even when the parallel forces F_1 and F_2 are not perpendicular, but act at some angle, to the lever.

7.8.2 Centre of gravity

Many of you may have the experience of balancing your notebook on the tip of a finger. Figure 7.24 illustrates a similar experiment that you can easily perform. Take an irregular-shaped cardboard having mass M and a narrow tipped object like a pencil. You can locate by trial and error a point G on the cardboard where it can be balanced on the tip of the pencil. (The cardboard remains horizontal in this position.) This point of balance is the centre of gravity (CG) of the cardboard. The tip of the pencil provides a vertically upward force due to which the cardboard is in mechanical equilibrium. As shown in the Fig. 7.24, the reaction of the tip is equal and opposite to $M\mathbf{g}$ and hence the cardboard is in translational equilibrium. It is also in rotational equilibrium; if it were not so, due to the unbalanced torque it would tilt and fall. There are torques on the card board due to the forces of gravity like $m_1\mathbf{g}$, $m_2\mathbf{g}$ etc, acting on the individual particles that make up the cardboard.

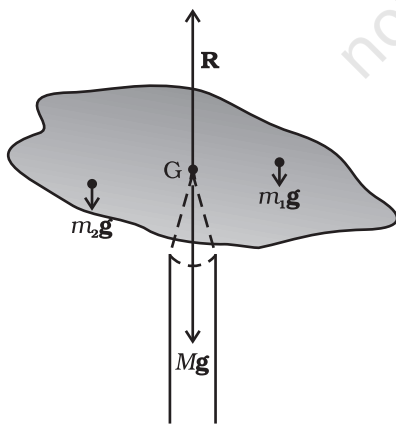


Fig. 7.24 Balancing a cardboard on the tip of a pencil. The point of support, G , is the centre of gravity.

The CG of the cardboard is so located that the total torque on it due to the forces $m_1\mathbf{g}$, $m_2\mathbf{g}$ etc. is zero.

If $\mathbf{r}_i$ is the position vector of the i th particle of an extended body with respect to its CG, then the torque about the CG, due to the force of gravity on the particle is $\boldsymbol{\tau}_i = \mathbf{r}_i \times m_i\mathbf{g}$. The total gravitational torque about the CG is zero, i.e.

$$\boldsymbol{\tau}_g = \sum \boldsymbol{\tau}_i = \sum \mathbf{r}_i \times m_i\mathbf{g} = \mathbf{0} \quad (7.33)$$

We may therefore, define the CG of a body as that point where the total gravitational torque on the body is zero.

We notice that in Eq. (7.33), $\mathbf{g}$ is the same for all particles, and hence it comes out of the summation. This gives, since $\mathbf{g}$ is non-zero,

$\sum m_i\mathbf{r}_i = \mathbf{0}$. Remember that the position vectors ($\mathbf{r}_i$) are taken with respect to the CG. Now, in accordance with the reasoning given below Eq. (7.4a) in Sec. 7.2, if the sum is zero, the origin must be the centre of mass of the body. Thus, the centre of gravity of the body coincides with the centre of mass in uniform gravity or gravity-free space. We note that this is true because the body being small, $\mathbf{g}$ does not

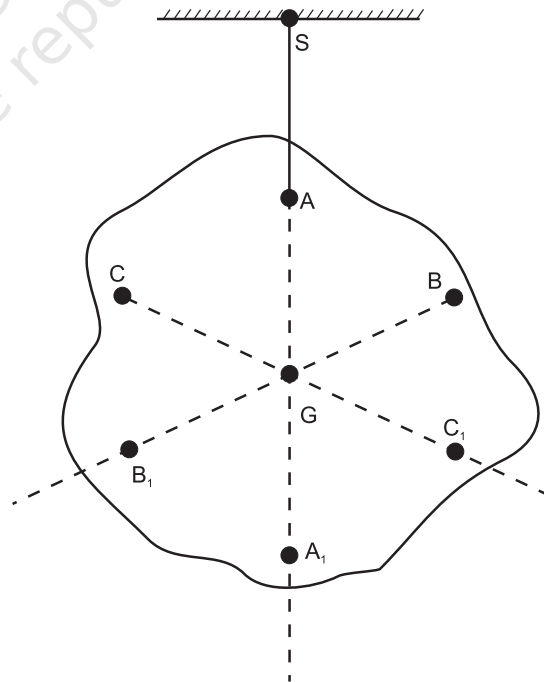


Fig. 7.25 Determining the centre of gravity of a body of irregular shape. The centre of gravity G lies on the vertical AA_1 through the point of suspension of the body A .

vary from one point of the body to the other. If the body is so extended that g varies from part to part of the body, then the centre of gravity and centre of mass will not coincide. Basically, the two are different concepts. The centre of mass has nothing to do with gravity. It depends only on the distribution of mass of the body.

In Sec. 7.2 we found out the position of the centre of mass of several regular, homogeneous objects. Obviously the method used there gives us also the centre of gravity of these bodies, if they are small enough.

Figure 7.25 illustrates another way of determining the CG of an irregular shaped body like a cardboard. If you suspend the body from some point like A, the vertical line through A passes through the CG. We mark the vertical AA_1 . We then suspend the body through other points like B and C. The intersection of the verticals gives the CG. Explain why the method works. Since the body is small enough, the method allows us to determine also its centre of mass.

► **Example 7.8** A metal bar 70 cm long and 4.00 kg in mass supported on two knife-edges placed 10 cm from each end. A 6.00 kg load is suspended at 30 cm from one end. Find the reactions at the knife-edges. (Assume the bar to be of uniform cross section and homogeneous.)

Answer

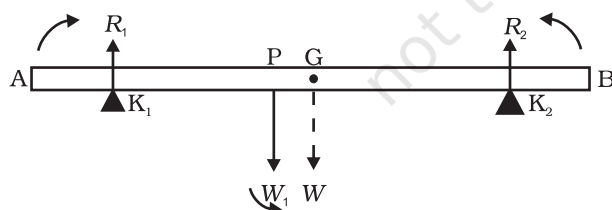


Fig. 7.26

Figure 7.26 shows the rod AB, the positions of the knife edges K_1 and K_2 , the centre of gravity of the rod at G and the suspended load at P.

Note the weight of the rod W acts at its centre of gravity G. The rod is uniform in cross section and homogeneous; hence G is at the centre of the rod; $AB = 70$ cm. $AG = 35$ cm, $AP = 30$ cm, $PG = 5$ cm, $AK_1 = BK_2 = 10$ cm and $K_1G = K_2G = 25$ cm. Also, $W =$ weight of the rod =

4.00 kg and $W_1 =$ suspended load = 6.00 kg; R_1 and R_2 are the normal reactions of the support at the knife edges.

For translational equilibrium of the rod,

$$R_1 + R_2 - W_1 - W = 0 \quad (i)$$

Note W_1 and W act vertically down and R_1 and R_2 act vertically up.

For considering rotational equilibrium, we take moments of the forces. A convenient point to take moments about is G. The moments of R_2 and W_1 are anticlockwise (+ve), whereas the moment of R_1 is clockwise (-ve).

For rotational equilibrium,

$$-R_1(K_1G) + W_1(PG) + R_2(K_2G) = 0 \quad (ii)$$

It is given that $W = 4.00g$ N and $W_1 = 6.00g$ N, where $g =$ acceleration due to gravity. We take $g = 9.8$ m/s².

With numerical values inserted, from (i)

$$\begin{aligned} R_1 + R_2 - 4.00g - 6.00g &= 0 \\ \text{or } R_1 + R_2 &= 10.00g \text{ N} \quad (iii) \\ &= 98.00 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{From (ii), } -0.25 R_1 + 0.05 W_1 + 0.25 R_2 &= 0 \\ \text{or } R_1 - R_2 &= 1.2g \text{ N} = 11.76 \text{ N} \quad (iv) \end{aligned}$$

$$\begin{aligned} \text{From (iii) and (iv), } R_1 &= 54.88 \text{ N,} \\ R_2 &= 43.12 \text{ N} \end{aligned}$$

Thus the reactions of the support are about 55 N at K_1 and 43 N at K_2 .

► **Example 7.9** A 3m long ladder weighing 20 kg leans on a frictionless wall. Its feet rest on the floor 1 m from the wall as shown in Fig. 7.27. Find the reaction forces of the wall and the floor.

Answer

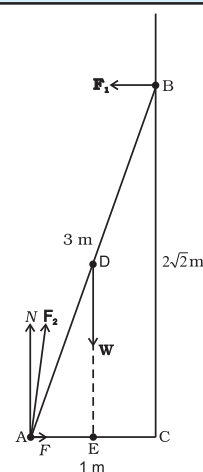


Fig. 7.27

The ladder AB is 3 m long, its foot A is at distance AC = 1 m from the wall. From

Pythagoras theorem, $BC = 2\sqrt{2}$ m. The forces on the ladder are its weight W acting at its centre of gravity D , reaction forces F_1 and F_2 of the wall and the floor respectively. Force F_1 is perpendicular to the wall, since the wall is frictionless. Force F_2 is resolved into two components, the normal reaction N and the force of friction F . Note that F prevents the ladder from sliding away from the wall and is therefore directed toward the wall.

For translational equilibrium, taking the forces in the vertical direction,

$$N - W = 0 \quad (i)$$

Taking the forces in the horizontal direction,

$$F - F_1 = 0 \quad (ii)$$

For rotational equilibrium, taking the moments of the forces about A ,

$$2\sqrt{2} F_1 - (1/2) W = 0 \quad (iii)$$

$$\text{Now } W = 20 \text{ g} = 20 \times 9.8 \text{ N} = 196.0 \text{ N}$$

$$\text{From (i) } N = 196.0 \text{ N}$$

$$\text{From (iii) } F_1 = W/4\sqrt{2} = 196.0/4\sqrt{2} = 34.6 \text{ N}$$

$$\text{From (ii) } F = F_1 = 34.6 \text{ N}$$

$$F_2 = \sqrt{F^2 + N^2} = 199.0 \text{ N}$$

The force F_2 makes an angle α with the horizontal,

$$\tan \alpha = N/F = 4\sqrt{2}, \quad \alpha = \tan^{-1}(4\sqrt{2}) \approx 80^\circ$$

7.9 MOMENT OF INERTIA

We have already mentioned that we are developing the study of rotational motion parallel to the study of translational motion with which we are familiar. We have yet to answer one major question in this connection. **What is the analogue of mass in rotational motion?** We shall attempt to answer this question in the present section. To keep the discussion simple, we shall consider rotation about a fixed axis only. Let us try to get an expression for the **kinetic energy of a rotating body**. We know that for a body rotating about a fixed axis, each particle of the body moves in a circle with linear velocity given by Eq. (7.19). (Refer to Fig. 7.16). For a particle at a distance from the axis, the

linear velocity is $v_i = r_i \omega$. The kinetic energy of motion of this particle is

$$k_i = \frac{1}{2} m_i v_i^2 = \frac{1}{2} m_i r_i^2 \omega^2$$

where m_i is the mass of the particle. The total kinetic energy K of the body is then given by the sum of the kinetic energies of individual particles,

$$K = \sum_{i=1}^n k_i = \frac{1}{2} \sum_{i=1}^n (m_i r_i^2 \omega^2)$$

Here n is the number of particles in the body. Note ω is the same for all particles. Hence, taking ω out of the sum,

$$K = \frac{1}{2} \omega^2 \left(\sum_{i=1}^n m_i r_i^2 \right)$$

We define a new parameter characterising the rigid body, called the moment of inertia I , given by

$$I = \sum_{i=1}^n m_i r_i^2 \quad (7.34)$$

With this definition,

$$K = \frac{1}{2} I \omega^2 \quad (7.35)$$

Note that the parameter I is independent of the magnitude of the angular velocity. It is a characteristic of the rigid body and the axis about which it rotates.

Compare Eq. (7.35) for the kinetic energy of a rotating body with the expression for the kinetic energy of a body in linear (translational) motion,

$$K = \frac{1}{2} m v^2$$

Here, m is the mass of the body and v is its velocity. We have already noted the analogy between angular velocity ω (in respect of rotational motion about a fixed axis) and linear velocity v (in respect of linear motion). It is then evident that the parameter, moment of inertia I , is the desired rotational analogue of mass in linear motion. In rotation (about a fixed axis), the moment of inertia plays a similar role as mass does in linear motion.

We now apply the definition Eq. (7.34), to calculate the moment of inertia in two simple cases.

(a) Consider a thin ring of radius R and mass M , rotating in its own plane around its centre

with angular velocity ω . Each mass element of the ring is at a distance R from the axis, and moves with a speed $R\omega$. The kinetic energy is therefore,

$$K = \frac{1}{2} M v^2 = \frac{1}{2} M R^2 \omega^2$$

Comparing with Eq. (7.35) we get $I = MR^2$ for the ring.

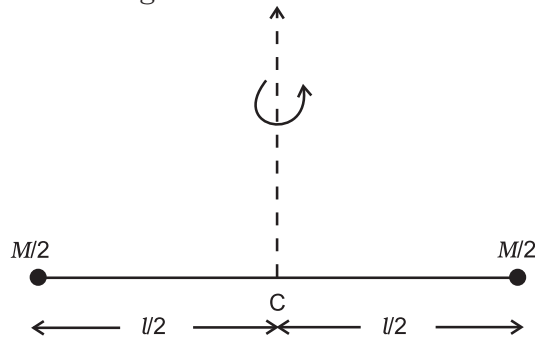


Fig. 7.28 A light rod of length l with a pair of masses rotating about an axis through the centre of mass of the system and perpendicular to the rod. The total mass of the system is M .

- (b) Next, take a rigid rod of negligible mass of length l with a pair of small masses, rotating about an axis through the centre of mass perpendicular to the rod (Fig. 7.28). Each mass $M/2$ is at a distance $l/2$ from the axis. The moment of inertia of the masses is therefore given by

$$(M/2)(l/2)^2 + (M/2)(l/2)^2$$

Thus, for the pair of masses, rotating about the axis through the centre of mass perpendicular to the rod

$$I = Ml^2 / 4$$

Table 7.1 simply gives the moment of inertia of various familiar regular shaped bodies about specific axes. (The derivations of these expressions are beyond the scope of this textbook and you will study them in higher classes.)

As the mass of a body resists a change in its state of linear motion, it is a measure of its inertia in linear motion. Similarly, as the moment of inertia about a given axis of rotation resists a change in its rotational motion, it can be regarded as a measure of rotational inertia of the body; it is a measure of the way in which different parts of the body are distributed at different distances from the axis. Unlike the mass of a body, the moment of inertia is not a fixed quantity but depends on distribution of mass about the axis of rotation, and the orientation and position of the axis of rotation

with respect to the body as a whole. As a measure of the way in which the mass of a rotating rigid body is distributed with respect to the axis of rotation, we can define a new parameter, the **radius of gyration**. It is related to the moment of inertia and the total mass of the body.

Notice from the Table 7.1 that in all cases, we can write $I = Mk^2$, where k has the dimension of length. For a rod, about the perpendicular axis at its midpoint, $k^2 = L^2/12$, i.e. $k = L/\sqrt{12}$. Similarly, $k = R/2$ for the circular disc about its diameter. The length k is a geometric property of the body and axis of rotation. It is called the **radius of gyration**. **The radius of gyration of a body about an axis** may be defined as the distance from the axis of a mass point whose mass is equal to the mass of the whole body and whose moment of inertia is equal to the moment of inertia of the body about the axis.

Thus, the moment of inertia of a rigid body depends on the mass of the body, its shape and size; distribution of mass about the axis of rotation, and the position and orientation of the axis of rotation.

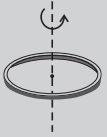
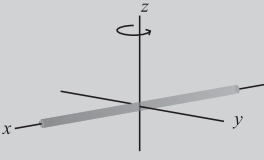
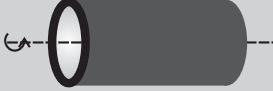
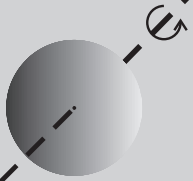
From the definition, Eq. (7.34), we can infer that the dimensions of moments of inertia are ML^2 and its SI units are kg m^2 .

The property of this extremely important quantity I , as a measure of rotational inertia of the body, has been put to a great practical use. The machines, such as steam engine and the automobile engine, etc., that produce rotational motion have a disc with a large moment of inertia, called a **flywheel**. Because of its large moment of inertia, the flywheel resists the sudden increase or decrease of the speed of the vehicle. It allows a gradual change in the speed and prevents jerky motions, thereby ensuring a smooth ride for the passengers on the vehicle.

7.10 THEOREMS OF PERPENDICULAR AND PARALLEL AXES

These are two useful theorems relating to moment of inertia. We shall first discuss the theorem of perpendicular axes and its simple yet instructive application in working out the moments of inertia of some regular-shaped bodies.

Table 7.1 Moments of inertia of some regular shaped bodies about specific axes

Z	Body	Axis	Figure	I
(1)	Thin circular ring, radius R	Perpendicular to plane, at centre		$M R^2$
(2)	Thin circular ring, radius R	Diameter		$M R^2/2$
(3)	Thin rod, length L	Perpendicular to rod, at mid point		$M L^2/12$
(4)	Circular disc, radius R	Perpendicular to disc at centre		$M R^2/2$
(5)	Circular disc, radius R	Diameter		$M R^2/4$
(6)	Hollow cylinder, radius R	Axis of cylinder		$M R^2$
(7)	Solid cylinder, radius R	Axis of cylinder		$M R^2/2$
(8)	Solid sphere, radius R	Diameter		$2 M R^2/5$

Theorem of perpendicular axes

This theorem is applicable to bodies which are planar. In practice this means the theorem applies to flat bodies whose thickness is very small compared to their other dimensions (e.g. length, breadth or radius). Fig. 7.29 illustrates

the theorem. It states that **the moment of inertia of a planar body (lamina) about an axis perpendicular to its plane is equal to the sum of its moments of inertia about two perpendicular axes concurrent with perpendicular axis and lying in the plane of the body.**

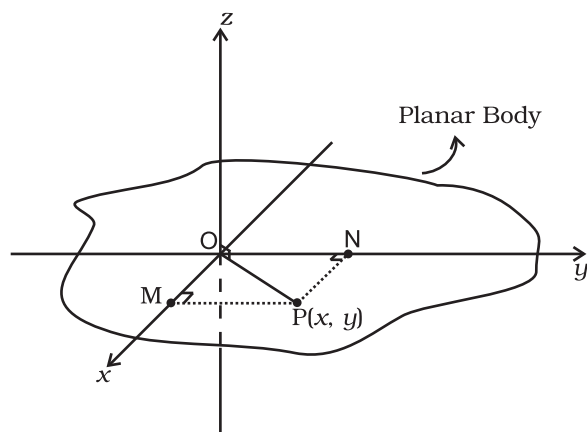


Fig. 7.29 Theorem of perpendicular axes applicable to a planar body; x and y axes are two perpendicular axes in the plane and the z -axis is perpendicular to the plane.

The figure shows a planar body. An axis perpendicular to the body through a point O is taken as the z -axis. Two mutually perpendicular axes lying in the plane of the body and concurrent with z -axis, i.e., passing through O , are taken as the x and y -axes. The theorem states that

$$I_z = I_x + I_y \quad (7.36)$$

Let us look at the usefulness of the theorem through an example.

► **Example 7.10** What is the moment of inertia of a disc about one of its diameters?

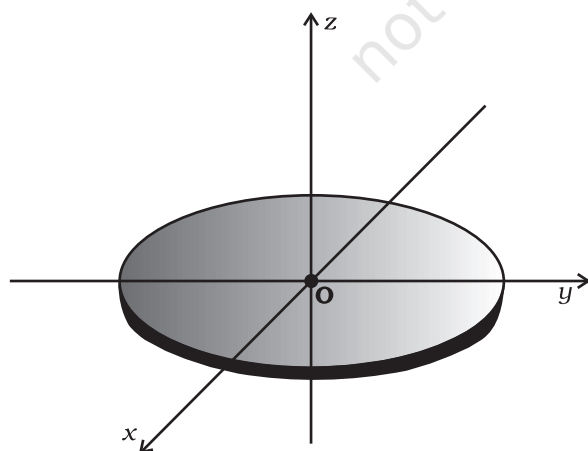


Fig. 7.30 Moment of inertia of a disc about a diameter, given its moment of inertia about the perpendicular axis through its centre.

Answer We assume the moment of inertia of the disc about an axis perpendicular to it and through its centre to be known; it is $MR^2/2$, where M is the mass of the disc and R is its radius (Table 7.1)

The disc can be considered to be a planar body. Hence the theorem of perpendicular axes is applicable to it. As shown in Fig. 7.30, we take three concurrent axes through the centre of the disc, O , as the x -, y - and z -axes; x - and y -axes lie in the plane of the disc and z -axis is perpendicular to it. By the theorem of perpendicular axes,

$$I_z = I_x + I_y$$

Now, x and y axes are along two diameters of the disc, and by symmetry the moment of inertia of the disc is the same about any diameter. Hence

$$\begin{aligned} I_x &= I_y \\ \text{and} \quad I_z &= 2I_x \\ \text{But} \quad I_z &= MR^2/2 \\ \text{So finally,} \quad I_x &= I_z/2 = MR^2/4 \end{aligned}$$

Thus the moment of inertia of a disc about any of its diameter is $MR^2/4$.

Find similarly the moment of inertia of a ring about any of its diameters. Will the theorem be applicable to a solid cylinder?

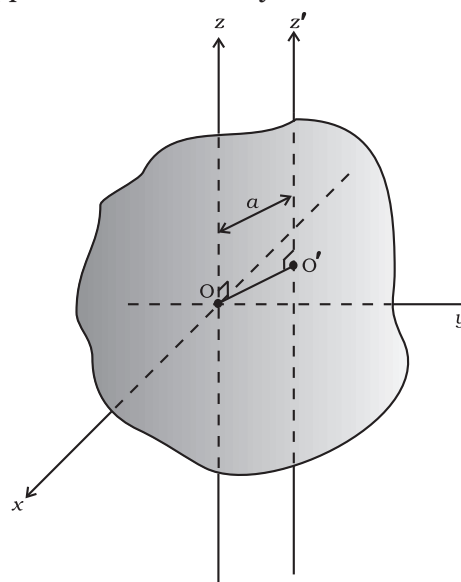


Fig. 7.31 The theorem of parallel axes The z and z' axes are two parallel axes separated by a distance a ; O is the centre of mass of the body, $OO' = a$.

7.10.1 Theorem of parallel axes

This theorem is applicable to a body of any shape. It allows to find the moment of inertia of a body about any axis, given the moment of inertia of the body about a parallel axis through the centre of mass of the body. We shall only state this theorem and not give its proof. We shall, however, apply it to a few simple situations which will be enough to convince us about the usefulness of the theorem. The theorem may be stated as follows:

The moment of inertia of a body about any axis is equal to the sum of the moment of inertia of the body about a parallel axis passing through its centre of mass and the product of its mass and the square of the distance between the two parallel axes. As shown in the Fig. 7.31, z and z' are two parallel axes, separated by a distance a . The z -axis passes through the centre of mass O of the rigid body. Then according to the theorem of parallel axes

$$I_{z'} = I_z + Ma^2 \quad (7.37)$$

where I_z and $I_{z'}$ are the moments of inertia of the body about the z and z' axes respectively, M is the total mass of the body and a is the perpendicular distance between the two parallel axes.

► **Example 7.11** What is the moment of inertia of a rod of mass M , length l about an axis perpendicular to it through one end?

Answer For the rod of mass M and length l , $I = Ml^2/12$. Using the parallel axes theorem, $I' = I + Ma^2$ with $a = l/2$ we get,

$$I' = M \frac{l^2}{12} + M \left(\frac{l}{2} \right)^2 = \frac{Ml^2}{3}$$

We can check this independently since I is half the moment of inertia of a rod of mass $2M$ and length $2l$ about its midpoint,

$$I' = 2M \cdot \frac{4l^2}{12} \times \frac{1}{2} = \frac{Ml^2}{3}$$

► **Example 7.12** What is the moment of inertia of a ring about a tangent to the circle of the ring?

Answer

The tangent to the ring in the plane of the ring is parallel to one of the diameters of the ring.

The distance between these two parallel axes is R , the radius of the ring. Using the parallel axes theorem,

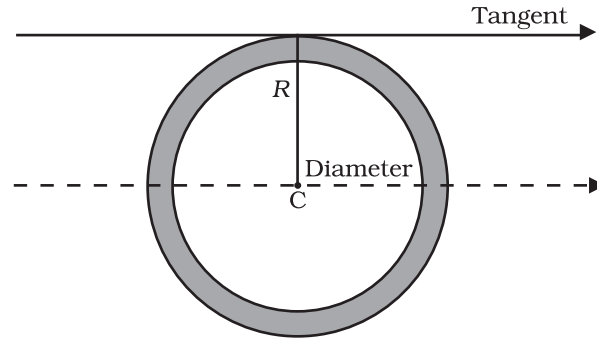


Fig. 7.32

$$I_{\text{tangent}} = I_{\text{dia}} + MR^2 = \frac{MR^2}{2} + MR^2 = \frac{3}{2}MR^2. \quad \blacktriangleleft$$

7.11 KINEMATICS OF ROTATIONAL MOTION ABOUT A FIXED AXIS

We have already indicated the analogy between rotational motion and translational motion. For example, the angular velocity ω plays the same role in rotation as the linear velocity $\mathbf{v}$ in translation. We wish to take this analogy further. In doing so we shall restrict the discussion only to rotation about fixed axis. This case of motion involves only one degree of freedom, i.e., needs only one independent variable to describe the motion. This in translation corresponds to linear motion. This section is limited only to kinematics. We shall turn to dynamics in later sections.

We recall that for specifying the angular displacement of the rotating body we take any particle like P (Fig. 7.33) of the body. Its angular displacement θ in the plane it moves is the angular displacement of the whole body; θ is measured from a fixed direction in the plane of motion of P , which we take to be the x' -axis, chosen parallel to the x -axis. Note, as shown, the axis of rotation is the z -axis and the plane of the motion of the particle is the $x-y$ plane. Fig. 7.33 also shows θ_0 , the angular displacement at $t = 0$.

We also recall that the angular velocity is the time rate of change of angular displacement, $\omega = d\theta/dt$. Note since the axis of rotation is fixed,

there is no need to treat angular velocity as a vector. Further, the angular acceleration, $\alpha = d\omega/dt$.

The kinematical quantities in rotational motion, angular displacement (θ), angular velocity (ω) and angular acceleration (α) respectively are analogous to kinematic quantities in linear motion, displacement (x), velocity (v) and acceleration (a). We know the kinematical equations of linear motion with uniform (i.e. constant) acceleration:

$$v = v_0 + at \quad (a)$$

$$x = x_0 + v_0 t + \frac{1}{2} at^2 \quad (b)$$

$$v^2 = v_0^2 + 2ax \quad (c)$$

where x_0 = initial displacement and v_0 = initial velocity. The word 'initial' refers to values of the quantities at $t = 0$

The corresponding kinematic equations for rotational motion with uniform angular acceleration are:

$$\omega = \omega_0 + \alpha t \quad (7.38)$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2 \quad (7.39)$$

$$\text{and } \omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0) \quad (7.40)$$

where θ_0 = initial angular displacement of the rotating body, and ω_0 = initial angular velocity of the body.

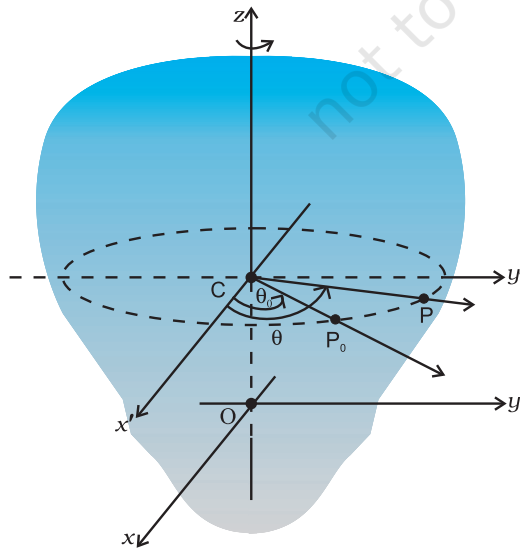


Fig. 7.33 Specifying the angular position of a rigid body.

► **Example 7.13** Obtain Eq. (7.38) from first principles.

Answer The angular acceleration is uniform, hence

$$\frac{d\omega}{dt} = \alpha = \text{constant} \quad (i)$$

Integrating this equation,

$$\omega = \int \alpha dt + c$$

$$= \alpha t + c \quad (\text{as } \alpha \text{ is constant})$$

At $t = 0$, $\omega = \omega_0$ (given)

From (i) we get at $t = 0$, $\omega = c = \omega_0$

Thus, $\omega = \alpha t + \omega_0$ as required.

With the definition of $\omega = d\theta/dt$ we may integrate Eq. (7.38) to get Eq. (7.39). This derivation and the derivation of Eq. (7.40) is left as an exercise.

► **Example 7.14** The angular speed of a motor wheel is increased from 1200 rpm to 3120 rpm in 16 seconds. (i) What is its angular acceleration, assuming the acceleration to be uniform? (ii) How many revolutions does the engine make during this time?

Answer

(i) We shall use $\omega = \omega_0 + \alpha t$

ω_0 = initial angular speed in rad/s

$$= 2\pi \times \text{angular speed in rev/s}$$

$$= \frac{2\pi \times \text{angular speed in rev/min}}{60 \text{ s/min}}$$

$$= \frac{2\pi \times 1200}{60} \text{ rad/s}$$

$$= 40\pi \text{ rad/s}$$

Similarly ω = final angular speed in rad/s

$$= \frac{2\pi \times 3120}{60} \text{ rad/s}$$

$$= 2\pi \times 52 \text{ rad/s}$$

$$= 104\pi \text{ rad/s}$$

$\therefore$ Angular acceleration

$$\alpha = \frac{\omega - \omega_0}{t} = 4\pi \text{ rad/s}^2$$

The angular acceleration of the engine
 $= 4\pi \text{ rad/s}^2$

(ii) The angular displacement in time t is given by

$$\begin{aligned}\theta &= \omega_0 t + \frac{1}{2} \alpha t^2 \\ &= (40\pi \times 16 + \frac{1}{2} \times 4\pi \times 16^2) \text{ rad} \\ &= (640\pi + 512\pi) \text{ rad} \\ &= 1152\pi \text{ rad}\end{aligned}$$

$$\text{Number of revolutions} = \frac{1152\pi}{2\pi} = 576$$

7.12 DYNAMICS OF ROTATIONAL MOTION ABOUT A FIXED AXIS

Table 7.2 lists quantities associated with linear motion and their analogues in rotational motion. We have already compared kinematics of the two motions. Also, we know that in rotational motion moment of inertia and torque play the same role as mass and force respectively in linear motion. Given this we should be able to guess what the other analogues indicated in the table are. For example, we know that in linear motion, work done is given by $F dx$, in rotational motion about a fixed axis it should be $\tau d\theta$, since we already know the correspondence $dx \rightarrow d\theta$ and $F \rightarrow \tau$. It is, however, necessary that these correspondences are established on sound dynamical considerations. This is what we now turn to.

Before we begin, we note a **simplification that arises in the case of rotational motion about a fixed axis**. Since the axis is fixed, only those components of torques, which are along the direction of the fixed axis need to be considered in our discussion. Only these components can cause the body to rotate about the axis. A component of the torque perpendicular to the axis of rotation will tend to turn the axis from its position. We specifically assume that there will arise necessary forces of constraint to cancel the effect of the perpendicular components of the (external) torques, so that the fixed position of the axis will be maintained. The perpendicular components of the torques, therefore need not be taken into account. This means that for our calculation of torques on a rigid body:

- (1) We need to consider only those forces that lie in planes perpendicular to the axis. Forces which are parallel to the axis will give torques perpendicular to the axis and need not be taken into account.
- (2) We need to consider only those components of the position vectors which are perpendicular to the axis. Components of position vectors along the axis will result in torques perpendicular to the axis and need not be taken into account.

Work done by a torque

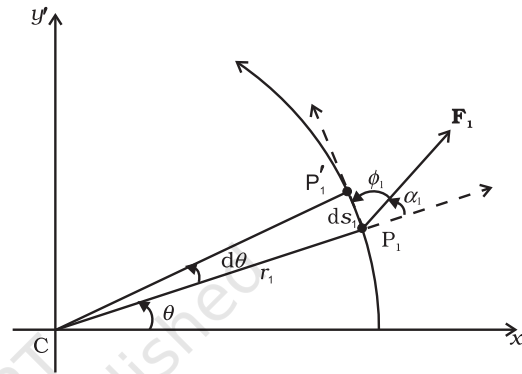


Fig. 7.34 Work done by a force $\mathbf{F}_1$ acting on a particle of a body rotating about a fixed axis; the particle describes a circular path with centre C on the axis; arc $P_1P'_1(ds_1)$ gives the displacement of the particle.

Figure 7.34 shows a cross-section of a rigid body rotating about a fixed axis, which is taken as the z -axis (perpendicular to the plane of the page; see Fig. 7.33). As said above we need to consider only those forces which lie in planes perpendicular to the axis. Let $\mathbf{F}_1$ be one such typical force acting as shown on a particle of the body at point P_1 with its line of action in a plane perpendicular to the axis. For convenience we call this to be the $x'-y'$ plane (coincident with the plane of the page). The particle at P_1 describes a circular path of radius r_1 with centre C on the axis; $CP_1 = r_1$.

In time Δt , the point moves to the position P'_1 . The displacement of the particle ds_1 , therefore, has magnitude $ds_1 = r_1 d\theta$ and direction tangential at P_1 to the circular path as shown. Here $d\theta$ is the angular displacement of the particle, $d\theta = \angle P_1CP'_1$. The work done by the force on the particle is

$$dW_1 = \mathbf{F}_1 \cdot d\mathbf{s}_1 = F_1 ds_1 \cos \phi_1 = F_1 (r_1 d\theta) \sin \alpha_1$$

where ϕ_1 is the angle between $\mathbf{F}_1$ and the tangent

Table 7.2 Comparison of Translational and Rotational Motion

Linear Motion	Rotational Motion about a Fixed Axis
1 Displacement x	Angular displacement θ
2 Velocity $v = dx/dt$	Angular velocity $\omega = d\theta/dt$
3 Acceleration $a = dv/dt$	Angular acceleration $\alpha = d\omega/dt$
4 Mass M	Moment of inertia I
5 Force $F = Ma$	Torque $\tau = I \alpha$
6 Work $dW = F ds$	Work $W = \tau d\theta$
7 Kinetic energy $K = Mv^2/2$	Kinetic energy $K = I\omega^2/2$
8 Power $P = F v$	Power $P = \tau \omega$
9 Linear momentum $p = Mv$	Angular momentum $L = I\omega$

at P_1 , and α_1 is the angle between $\mathbf{F}_1$ and the radius vector $\mathbf{OP}_1$; $\phi_1 + \alpha_1 = 90^\circ$.

The torque due to $\mathbf{F}_1$ about the origin is $\mathbf{OP}_1 \times \mathbf{F}_1$. Now $\mathbf{OP}_1 = \mathbf{OC} + \mathbf{CP}_1$. [Refer to Fig. 7.17(b).] Since $\mathbf{OC}$ is along the axis, the torque resulting from it is excluded from our consideration. The effective torque due to $\mathbf{F}_1$ is $\boldsymbol{\tau}_1 = \mathbf{CP}_1 \times \mathbf{F}_1$; it is directed along the axis of rotation and has a magnitude $\tau_1 = r_1 F_1 \sin \alpha$. Therefore,

$$dW_1 = \tau_1 d\theta$$

If there are more than one forces acting on the body, the work done by all of them can be added to give the total work done on the body. Denoting the magnitudes of the torques due to the different forces as $\tau_1, \tau_2, \dots$ etc,

$$dW = (\tau_1 + \tau_2 + \dots) d\theta$$

Remember, the forces giving rise to the torques act on different particles, but the angular displacement $d\theta$ is the same for all particles. Since all the torques considered are parallel to the fixed axis, the magnitude τ of the total torque is just the algebraic sum of the magnitudes of the torques, i.e., $\tau = \tau_1 + \tau_2 + \dots$. We, therefore, have

$$dW = \tau d\theta \quad (7.41)$$

This expression gives the work done by the total (external) torque τ which acts on the body rotating about a fixed axis. Its similarity with the corresponding expression

$$dW = F ds$$

for linear (translational) motion is obvious.

Dividing both sides of Eq. (7.41) by dt gives

$$P = \frac{dW}{dt} = \tau \frac{d\theta}{dt} = \tau \omega$$

$$\text{or } P = \tau \omega \quad (7.42)$$

This is the instantaneous power. Compare this expression for power in the case of rotational motion about a fixed axis with that of power in the case of linear motion,

$$P = Fv$$

In a perfectly rigid body there is no internal motion. The work done by external torques is therefore, not dissipated and goes on to increase the kinetic energy of the body. The rate at which work is done on the body is given by Eq. (7.42). This is to be equated to the rate at which kinetic energy increases. The rate of increase of kinetic energy is

$$\frac{d}{dt} \left(\frac{I\omega^2}{2} \right) = I \frac{(2\omega)}{2} \frac{d\omega}{dt}$$

We assume that the moment of inertia does not change with time. This means that the mass of the body does not change, the body remains rigid and also the axis does not change its position with respect to the body.

Since $\alpha = d\omega/dt$, we get

$$\frac{d}{dt} \left(\frac{I\omega^2}{2} \right) = I \omega \alpha$$

Equating rates of work done and of increase in kinetic energy,

$$\tau \omega = I \omega \alpha$$

$$\tau = I\alpha \quad (7.43)$$

Eq. (7.43) is similar to Newton's second law for linear motion expressed symbolically as

$$F = ma$$

Just as force produces acceleration, torque produces angular acceleration in a body. The angular acceleration is directly proportional to the applied torque and is inversely proportional to the moment of inertia of the body. In this respect, Eq. (7.43) can be called Newton's second law for rotational motion about a fixed axis.

► **Example 7.15** A cord of negligible mass is wound round the rim of a fly wheel of mass 20 kg and radius 20 cm. A steady pull of 25 N is applied on the cord as shown in Fig. 7.35. The flywheel is mounted on a horizontal axle with frictionless bearings.

- Compute the angular acceleration of the wheel.
- Find the work done by the pull, when 2m of the cord is unwound.
- Find also the kinetic energy of the wheel at this point. Assume that the wheel starts from rest.
- Compare answers to parts (b) and (c).

Answer

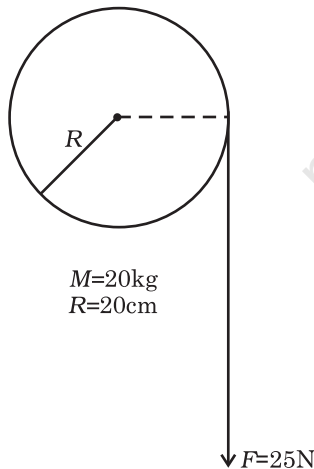


Fig. 7.35

- We use $I\alpha = \tau$
the torque $\tau = FR$
 $= 25 \times 0.20 \text{ Nm (as } R = 0.20\text{m)}$
 $= 5.0 \text{ Nm}$

I = Moment of inertia of flywheel about its

$$\text{axis} = \frac{MR^2}{2}$$

$$= \frac{20.0 \times (0.2)^2}{2} = 0.4 \text{ kg m}^2$$

$$\alpha = \text{angular acceleration} \\ = 5.0 \text{ N m} / 0.4 \text{ kg m}^2 = 12.5 \text{ s}^{-2}$$

- Work done by the pull unwinding 2m of the cord

$$= 25 \text{ N} \times 2\text{m} = 50 \text{ J}$$

- Let ω be the final angular velocity. The

$$\text{kinetic energy gained} = \frac{1}{2} I \omega^2,$$

since the wheel starts from rest. Now,

$$\omega^2 = \omega_0^2 + 2\alpha\theta, \quad \omega_0 = 0$$

The angular displacement θ = length of unwound string / radius of wheel
 $= 2\text{m} / 0.2 \text{ m} = 10 \text{ rad}$

$$\omega^2 = 2 \times 12.5 \times 10.0 = 250 (\text{rad/s})^2$$

$$\therefore \text{K.E. gained} = \frac{1}{2} \times 0.4 \times 250 = 50 \text{ J}$$

- The answers are the same, i.e. the kinetic energy gained by the wheel = work done by the force. There is no loss of energy due to friction. ◀

7.13 ANGULAR MOMENTUM IN CASE OF ROTATION ABOUT A FIXED AXIS

We have studied in section 7.7, the angular momentum of a system of particles. We already know from there that the time rate of total angular momentum of a system of particles about a point is equal to the total external torque on the system taken about the same point. When the total external torque is zero, the total angular momentum of the system is conserved.

We now wish to study the angular momentum in the special case of rotation about a fixed axis. The general expression for the total angular momentum of the system of n particles is

$$\mathbf{L} = \sum_{i=1}^N \mathbf{r}_i \times \mathbf{p}_i \quad (7.25b)$$

We first consider the angular momentum of a typical particle of the rotating rigid body. We then sum up the contributions of individual particles to get $\mathbf{L}$ of the whole body.

For a typical particle $\mathbf{l} = \mathbf{r} \times \mathbf{p}$. As seen in the last section $\mathbf{r} = \mathbf{OP} = \mathbf{OC} + \mathbf{CP}$ [Fig. 7.17(b)]. With

$$\mathbf{p} = m \mathbf{v},$$

$$\mathbf{l} = (\mathbf{OC} \times m \mathbf{v}) + (\mathbf{CP} \times m \mathbf{v})$$

The magnitude of the linear velocity $\mathbf{v}$ of the particle at P is given by $v = \omega r_{\perp}$ where $r_{\perp}$ is the length of CP or the perpendicular distance of P from the axis of rotation. Further, $\mathbf{v}$ is tangential at P to the circle which the particle describes. Using the right-hand rule one can check that $\mathbf{CP} \times \mathbf{v}$ is parallel to the fixed axis. The unit vector along the fixed axis (chosen as the z-axis) is $\hat{\mathbf{k}}$. Hence

$$\begin{aligned} \mathbf{CP} \times m \mathbf{v} &= r_{\perp} (m v) \hat{\mathbf{k}} \\ &= m r_{\perp}^2 \omega \hat{\mathbf{k}} \quad (\text{since } v = \omega r_{\perp}) \end{aligned}$$

Similarly, we can check that $\mathbf{OC} \times \mathbf{v}$ is perpendicular to the fixed axis. Let us denote the part of $\mathbf{l}$ along the fixed axis (i.e. the z-axis) by $\mathbf{l}_z$, then

$$\mathbf{l}_z = \mathbf{CP} \times m \mathbf{v} = m r_{\perp}^2 \omega \hat{\mathbf{k}}$$

$$\text{and } \mathbf{l} = \mathbf{l}_z + \mathbf{OC} \times m \mathbf{v}$$

We note that $\mathbf{l}_z$ is parallel to the fixed axis, but $\mathbf{l}$ is not. In general, for a particle, the angular momentum $\mathbf{l}$ is not along the axis of rotation, i.e. for a particle, $\mathbf{l}$ and $\boldsymbol{\omega}$ are not necessarily parallel. Compare this with the corresponding fact in translation. For a particle, $\mathbf{p}$ and $\mathbf{v}$ are always parallel to each other.

For computing the total angular momentum of the whole rigid body, we add up the contribution of each particle of the body.

$$\text{Thus } \mathbf{L} = \sum \mathbf{l}_i = \sum \mathbf{l}_{iz} + \sum \mathbf{OC}_i \times m_i \mathbf{v}_i$$

We denote by $\mathbf{L}_{\perp}$ and $\mathbf{L}_z$ the components of $\mathbf{L}$ respectively perpendicular to the z-axis and along the z-axis;

$$\mathbf{L}_{\perp} = \sum \mathbf{OC}_i \times m_i \mathbf{v}_i \quad (7.44a)$$

where m_i and $\mathbf{v}_i$ are respectively the mass and the velocity of the i^{th} particle and C_i is the centre of the circle described by the particle;

$$\text{and } \mathbf{L}_z = \sum \mathbf{l}_{iz} = \left(\sum_i m_i r_i^2 \right) \omega \hat{\mathbf{k}}$$

$$\text{or } \mathbf{L}_z = I \omega \hat{\mathbf{k}} \quad (7.44b)$$

The last step follows since the perpendicular distance of the i^{th} particle from the axis is r_i ; and by definition the moment of inertia of the body about the axis of rotation is $I = \sum m_i r_i^2$.

$$\text{Note } \mathbf{L} = \mathbf{L}_z + \mathbf{L}_{\perp} \quad (7.44c)$$

The rigid bodies which we have mainly considered in this chapter are symmetric about the axis of rotation, i.e. the axis of rotation is one of their symmetry axes. For such bodies, for a given $\mathbf{OC}_i$, for every particle which has a velocity $\mathbf{v}_i$, there is another particle of velocity $-\mathbf{v}_i$ located diametrically opposite on the circle with centre C_i described by the particle. Together such pairs will contribute zero to $\mathbf{L}_{\perp}$ and as a result for symmetric bodies $\mathbf{L}_{\perp}$ is zero, and hence

$$\mathbf{L} = \mathbf{L}_z = I \omega \hat{\mathbf{k}} \quad (7.44d)$$

For bodies, which are not symmetric about the axis of rotation, $\mathbf{L}$ is not equal to $\mathbf{L}_z$ and hence $\mathbf{L}$ does not lie along the axis of rotation.

Referring to Table 7.1, can you tell in which cases $\mathbf{L} = \mathbf{L}_z$ will not apply?

Let us differentiate Eq. (7.44b). Since $\hat{\mathbf{k}}$ is a fixed (constant) vector, we get

$$\frac{d}{dt}(\mathbf{L}_z) = \left(\frac{d}{dt}(I \omega) \right) \hat{\mathbf{k}}$$

Now, Eq. (7.28b) states

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}$$

As we have seen in the last section, only those components of the external torques which are along the axis of rotation, need to be taken into account, when we discuss rotation about a fixed axis. This means we can take $\boldsymbol{\tau} = \tau \hat{\mathbf{k}}$. Since $\mathbf{L} = \mathbf{L}_z + \mathbf{L}_{\perp}$ and the direction of $\mathbf{L}_z$ (vector $\hat{\mathbf{k}}$) is fixed, it follows that for rotation about a fixed axis,

$$\frac{d\mathbf{L}_z}{dt} = \tau \hat{\mathbf{k}} \quad (7.45a)$$

$$\text{and } \frac{d\mathbf{L}_{\perp}}{dt} = 0 \quad (7.45b)$$

Thus, for rotation about a fixed axis, the component of angular momentum perpendicular to the fixed axis is constant. As

$\mathbf{L}_z = I \omega \hat{\mathbf{k}}$, we get from Eq. (7.45a),

$$\frac{d}{dt}(I \omega) = \tau \quad (7.45c)$$

If the moment of inertia I does not change with time,

$$\frac{d}{dt}(I\omega) = I \frac{d\omega}{dt} = I\alpha$$

and we get from Eq. (7.45c),

$$\tau = I\alpha \quad (7.43)$$

We have already derived this equation using the work - kinetic energy route.

7.13.1 Conservation of angular momentum

We are now in a position to revisit the principle of conservation of angular momentum in the context of rotation about a fixed axis. From Eq. (7.45c), if the external torque is zero,

$$L_z = I\omega = \text{constant} \quad (7.46)$$

For symmetric bodies, from Eq. (7.44d), L_z may be replaced by L . (L and L_z are respectively the magnitudes of $\mathbf{L}$ and $\mathbf{L}_z$.)

This then is the required form, for fixed axis rotation, of Eq. (7.29a), which expresses the general law of conservation of angular momentum of a system of particles. Eq. (7.46) applies to many situations that we come across in daily life. You may do this experiment with your friend. Sit on a swivel chair (a chair with a seat, free to rotate about a pivot) with your arms folded and feet not resting on, i.e., away from, the ground. Ask your friend to rotate the chair rapidly. While the chair is rotating with

considerable angular speed stretch your arms horizontally. What happens? Your angular speed is reduced. If you bring back your arms closer to your body, the angular speed increases again. This is a situation where the principle of conservation of angular momentum is applicable. If friction in the rotational mechanism is neglected, there is no external torque about the axis of rotation of the chair and hence $I\omega$ is constant. Stretching the arms increases I about the axis of rotation, resulting in decreasing the angular speed ω . Bringing the arms closer to the body has the opposite effect.

A circus acrobat and a diver take advantage of this principle. Also, skaters and classical, Indian or western, dancers performing a pirouette (a spinning about a tip-top) on the toes of one foot display 'mastery' over this principle. Can you explain?

7.14 ROLLING MOTION

One of the most common motions observed in daily life is the rolling motion. All wheels used in transportation have rolling motion. For specificity we shall begin with the case of a disc, but the result will apply to any rolling body rolling on a level surface. We shall assume that the disc rolls without slipping. This means that at any instant of time the bottom of the disc



Fig 7.36 (a) A demonstration of conservation of angular momentum. A girl sits on a swivel chair and stretches her arms/ brings her arms closer to the body.

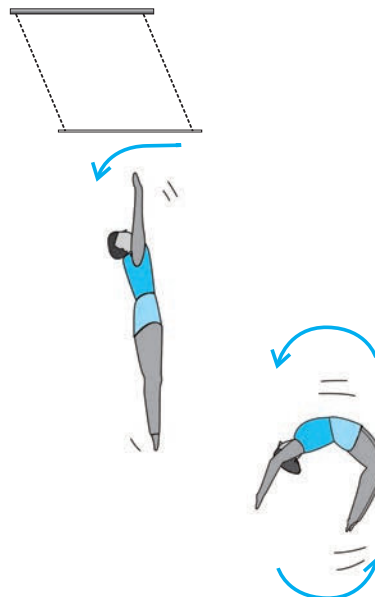


Fig 7.36 (b) An acrobat employing the principle of conservation of angular momentum in her performance.

which is in contact with the surface is at rest on the surface.

We have remarked earlier that rolling motion is a combination of rotation and translation. We know that the translational motion of a system of particles is the motion of its centre of mass.

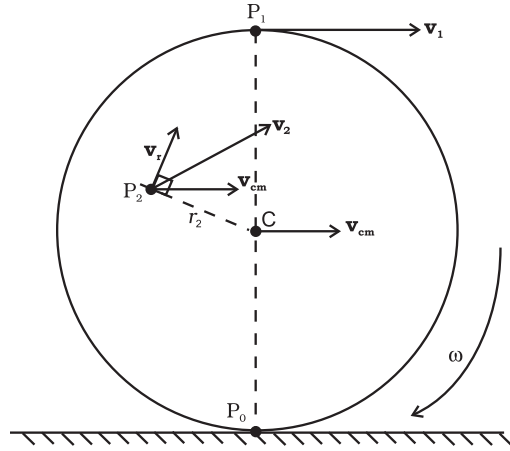


Fig. 7.37 The rolling motion (without slipping) of a disc on a level surface. Note at any instant, the point of contact P_0 of the disc with the surface is at rest; the centre of mass of the disc moves with velocity, v_{cm} . The disc rotates with angular velocity ω about its axis which passes through C; $v_{cm} = R\omega$, where R is the radius of the disc.

Let $\mathbf{v}_{cm}$ be the velocity of the centre of mass and therefore the translational velocity of the disc. Since the centre of mass of the rolling disc is at its geometric centre C (Fig. 7.37), $\mathbf{v}_{cm}$ is the velocity of C. It is parallel to the level surface. The rotational motion of the disc is about its symmetry axis, which passes through C. Thus, the velocity of any point of the disc, like P_0 , P_1 or P_2 , consists of two parts, one is the translational velocity $\mathbf{v}_{cm}$ and the other is the linear velocity $\mathbf{v}_r$ on account of rotation. The magnitude of $\mathbf{v}_r$ is $v_r = r\omega$, where ω is the angular velocity of the rotation of the disc about the axis and r is the distance of the point from the axis (i.e. from C). The velocity $\mathbf{v}_r$ is directed perpendicular to the radius vector of the given point with respect to C. In Fig. 7.37, the velocity of the point P_2 ($\mathbf{v}_2$) and its components $\mathbf{v}_r$ and $\mathbf{v}_{cm}$ are shown; $\mathbf{v}_r$ here is perpendicular to CP_2 . It is easy to show that $\mathbf{v}_2$ is perpendicular to the line P_0P_2 . Therefore the line passing through P_0 and parallel to ω is called the instantaneous axis of rotation.

At P_0 , the linear velocity, $\mathbf{v}_r$, due to rotation is directed exactly opposite to the translational velocity $\mathbf{v}_{cm}$. Further the magnitude of $\mathbf{v}_r$ here is $R\omega$, where R is the radius of the disc. The condition that P_0 is instantaneously at rest requires $v_{cm} = R\omega$. Thus for the disc the condition for rolling without slipping is

$$v_{cm} = R\omega \quad (7.47)$$

Incidentally, this means that the velocity of point P_1 at the top of the disc ($\mathbf{v}_1$) has a magnitude $v_{cm} + R\omega$ or $2v_{cm}$ and is directed parallel to the level surface. The condition (7.47) applies to all rolling bodies.

7.14.1 Kinetic Energy of Rolling Motion

Our next task will be to obtain an expression for the kinetic energy of a rolling body. The kinetic energy of a rolling body can be separated into kinetic energy of translation and kinetic energy of rotation. This is a special case of a general result for a system of particles, according to which the kinetic energy of a system of particles (K) can be separated into the kinetic energy of translational motion of the centre of mass ($MV^2/2$) and kinetic energy of rotational motion about the centre of mass of the system of particles (K'). Thus,

$$K = K' + MV^2/2 \quad (7.48)$$

We assume this general result (see Exercise 7.31), and apply it to the case of rolling motion. In our notation, the kinetic energy of the centre of mass, i.e., the kinetic energy of translation, of the rolling body is $mv_{cm}^2/2$, where m is the mass of the body and v_{cm} is the centre of the mass velocity. Since the motion of the rolling body about the centre of mass is rotation, K' represents the kinetic energy of rotation of the body; $K' = I\omega^2/2$, where I is the moment of inertia about the appropriate axis, which is the symmetry axis of the rolling body. The kinetic energy of a rolling body, therefore, is given by

$$K = \frac{1}{2} I\omega^2 + \frac{1}{2} mv_{cm}^2 \quad (7.49a)$$

Substituting $I = mk^2$ where k = the corresponding radius of gyration of the body and $v_{cm} = R\omega$, we get

$$K = \frac{1}{2} \frac{mk^2 v_{cm}^2}{R^2} + \frac{1}{2} mv_{cm}^2$$

$$\text{or } K = \frac{1}{2} mv_{cm}^2 \left(1 + \frac{k^2}{R^2} \right) \quad (7.49b)$$

Equation (7.49b) applies to any rolling body: a disc, a cylinder, a ring or a sphere.

► **Example 7.16** Three bodies, a ring, a solid cylinder and a solid sphere roll down the same inclined plane without slipping. They start from rest. The radii of the bodies are identical. Which of the bodies reaches the ground with maximum velocity?

Answer We assume conservation of energy of the rolling body, i.e. there is no loss of energy due to friction etc. The potential energy lost by the body in rolling down the inclined plane ($= mgh$) must, therefore, be equal to kinetic energy gained. (See Fig. 7.38) Since the bodies start from rest the kinetic energy gained is equal to the final kinetic energy of the bodies. From

Eq. (7.49b), $K = \frac{1}{2}mv^2 \left(1 + \frac{k^2}{R^2}\right)$, where v is the final velocity of (the centre of mass of) the body. Equating K and mgh ,

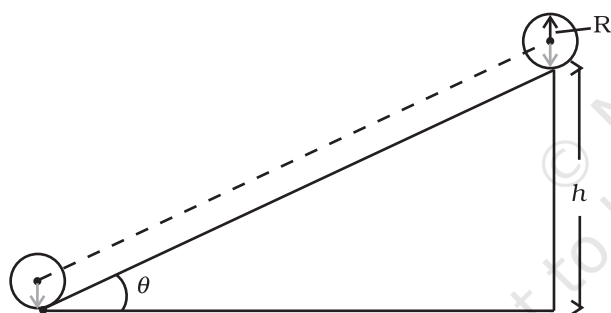


Fig. 7.38

$$mgh = \frac{1}{2}mv^2 \left(1 + \frac{k^2}{R^2}\right)$$

$$\text{or } v^2 = \left(\frac{2gh}{1 + k^2/R^2}\right)$$

Note v is independent of the mass of the rolling body;

For a ring, $k^2 = R^2$

$$v_{\text{ring}} = \sqrt{\frac{2gh}{1+1}}$$

$$= \sqrt{gh}$$

For a solid cylinder $k^2 = R^2/2$

$$v_{\text{disc}} = \sqrt{\frac{2gh}{1+1/2}}$$

$$= \sqrt{\frac{4gh}{3}}$$

For a solid sphere $k^2 = 2R^2/5$

$$v_{\text{sphere}} = \sqrt{\frac{2gh}{1+2/5}}$$

$$= \sqrt{\frac{10gh}{7}}$$

From the results obtained it is clear that among the three bodies the sphere has the greatest and the ring has the least velocity of the centre of mass at the bottom of the inclined plane.

Suppose the bodies have the same mass. Which body has the greatest rotational kinetic energy while reaching the bottom of the inclined plane? ◀

SUMMARY

1. Ideally, a rigid body is one for which the distances between different particles of the body do not change, even though there are forces on them.
2. A rigid body fixed at one point or along a line can have only rotational motion. A rigid body not fixed in some way can have either pure translational motion or a combination of translational and rotational motions.
3. In rotation about a fixed axis, every particle of the rigid body moves in a circle which lies in a plane perpendicular to the axis and has its centre on the axis. Every Point in the rotating rigid body has the same angular velocity at any instant of time.
4. In pure translation, every particle of the body moves with the same velocity at any instant of time.
5. Angular velocity is a vector. Its magnitude is $\omega = d\theta/dt$ and it is directed along the axis of rotation. For rotation about a fixed axis, this vector ω has a fixed direction.

6. The vector or cross product of two vector **a** and **b** is a vector written as **a**×**b**. The magnitude of this vector is $ab\sin\theta$ and its direction is given by the right handed screw or the right hand rule.
7. The linear velocity of a particle of a rigid body rotating about a fixed axis is given by $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, where **r** is the position vector of the particle with respect to an origin along the fixed axis. The relation applies even to more general rotation of a rigid body with one point fixed. In that case **r** is the position vector of the particle with respect to the fixed point taken as the origin.
8. The centre of mass of a system of n particles is defined as the point whose position vector is

$$\mathbf{R} = \frac{\sum m_i \mathbf{r}_i}{M}$$

9. Velocity of the centre of mass of a system of particles is given by $\mathbf{V} = \mathbf{P}/M$, where **P** is the linear momentum of the system. The centre of mass moves as if all the mass of the system is concentrated at this point and all the external forces act at it. If the total external force on the system is zero, then the total linear momentum of the system is constant.
10. The angular momentum of a system of n particles about the origin is

$$\mathbf{L} = \sum_{i=1}^n \mathbf{r}_i \times \mathbf{p}_i$$

The torque or moment of force on a system of n particles about the origin is

$$\boldsymbol{\tau} = \sum_i \mathbf{r}_i \times \mathbf{F}_i$$

The force $\mathbf{F}_i$ acting on the i^{th} particle includes the external as well as internal forces. Assuming Newton's third law of motion and that forces between any two particles act along the line joining the particles, we can show $\boldsymbol{\tau}_{\text{int}} = \mathbf{0}$ and

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}_{\text{ext}}$$

11. A rigid body is in mechanical equilibrium if

- (1) it is in translational equilibrium, i.e., the total external force on it is zero : $\sum \mathbf{F}_i = \mathbf{0}$,
and
- (2) it is in rotational equilibrium, i.e. the total external torque on it is zero :

$$\sum \boldsymbol{\tau}_i = \sum \mathbf{r}_i \times \mathbf{F}_i = \mathbf{0}.$$

12. The centre of gravity of an extended body is that point where the total gravitational torque on the body is zero.

13. The moment of inertia of a rigid body about an axis is defined by the formula $I = \sum m_i r_i^2$ where r_i is the perpendicular distance of the i^{th} point of the body from the axis. The

kinetic energy of rotation is $K = \frac{1}{2} I \omega^2$.

14. The theorem of parallel axes: $I'_z = I_z + M\alpha^2$, allows us to determine the moment of inertia of a rigid body about an axis as the sum of the moment of inertia of the body about a parallel axis through its centre of mass and the product of mass and square of the perpendicular distance between these two axes.

15. Rotation about a fixed axis is directly analogous to linear motion in respect of kinematics and dynamics.
16. For a rigid body rotating about a fixed axis (say, z-axis) of rotation, $L_z = I\omega$, where I is the moment of inertia about z-axis. In general, the angular momentum $\mathbf{L}$ for such a body is not along the axis of rotation. Only if the body is symmetric about the axis of rotation, $\mathbf{L}$ is along the axis of rotation. In that case, $|\mathbf{L}| = L_z = I\omega$. The angular acceleration of a rigid body rotating about a fixed axis is given by $I\alpha = \tau$. If the external torque τ acting on the body is zero, the component of angular momentum about the fixed axis (say, z-axis), $L_z (=I\omega)$ of such a rotating body is constant.
17. For rolling motion without slipping $v_{cm} = R\omega$, where v_{cm} is the velocity of translation (i.e. of the centre of mass), R is the radius and m is the mass of the body. The kinetic energy of such a rolling body is the sum of kinetic energies of translation and rotation:

$$K = \frac{1}{2} m v_{cm}^2 + \frac{1}{2} I \omega^2.$$

Quantity	Symbols	Dimensions	Units	Remarks
Angular velocity	ω	$[T^{-1}]$	rad s ⁻¹	$\mathbf{v} = \omega \times \mathbf{r}$
Angular momentum	$\mathbf{L}$	$[ML^2 T^{-1}]$	J s	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$
Torque	$\boldsymbol{\tau}$	$[ML^2 T^{-2}]$	N m	$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$
Moment of Inertia	I	$[ML^2]$	kg m ²	$I = \sum m_i r_i^2$

POINTS TO PONDER

- To determine the motion of the centre of mass of a system no knowledge of internal forces of the system is required. For this purpose we need to know only the external forces on the body.
- Separating the motion of a system of particles as the motion of the centre of mass, (i.e., the translational motion of the system) and motion about (i.e. relative to) the centre of mass of the system is a useful technique in dynamics of a system of particles. One example of this technique is separating the kinetic energy of a system of particles K as the kinetic energy of the system about its centre of mass K' and the kinetic energy of the centre of mass $MV^2/2$,

$$K = K' + MV^2/2$$
- Newton's Second Law for finite sized bodies (or systems of particles) is based in Newton's Second Law and also Newton's Third Law for particles.
- To establish that the time rate of change of the total angular momentum of a system of particles is the total external torque in the system, we need not only Newton's second law for particles, but also Newton's third law with the provision that the forces between any two particles act along the line joining the particles.
- The vanishing of the total external force and the vanishing of the total external torque are independent conditions. We can have one without the other. In a couple, total external force is zero, but total torque is non-zero.
- The total torque on a system is independent of the origin if the total external force is zero.
- The centre of gravity of a body coincides with its centre of mass only if the gravitational field does not vary from one part of the body to the other.
- The angular momentum $\mathbf{L}$ and the angular velocity $\boldsymbol{\omega}$ are not necessarily parallel vectors. However, for the simpler situations discussed in this chapter when rotation is about a fixed axis which is an axis of symmetry of the rigid body, the relation $\mathbf{L} = I\boldsymbol{\omega}$ holds good, where I is the moment of the inertia of the body about the rotation axis.

EXERCISES

- 7.1** Give the location of the centre of mass of a (i) sphere, (ii) cylinder, (iii) ring, and (iv) cube, each of uniform mass density. Does the centre of mass of a body necessarily lie inside the body ?
- 7.2** In the HCl molecule, the separation between the nuclei of the two atoms is about $1.27 \text{ \AA}$ ($1 \text{ \AA} = 10^{-10} \text{ m}$). Find the approximate location of the CM of the molecule, given that a chlorine atom is about 35.5 times as massive as a hydrogen atom and nearly all the mass of an atom is concentrated in its nucleus.
- 7.3** A child sits stationary at one end of a long trolley moving uniformly with a speed V on a smooth horizontal floor. If the child gets up and runs about on the trolley in any manner, what is the speed of the CM of the (trolley + child) system ?
- 7.4** Show that the area of the triangle contained between the vectors $\mathbf{a}$ and $\mathbf{b}$ is one half of the magnitude of $\mathbf{a} \times \mathbf{b}$.
- 7.5** Show that $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ is equal in magnitude to the volume of the parallelepiped formed on the three vectors, $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.
- 7.6** Find the components along the x , y , z axes of the angular momentum $\mathbf{l}$ of a particle, whose position vector is $\mathbf{r}$ with components x , y , z and momentum is $\mathbf{p}$ with components p_x , p_y and p_z . Show that if the particle moves only in the x - y plane the angular momentum has only a z -component.
- 7.7** Two particles, each of mass m and speed v , travel in opposite directions along parallel lines separated by a distance d . Show that the angular momentum vector of the two particle system is the same whatever be the point about which the angular momentum is taken.
- 7.8** A non-uniform bar of weight W is suspended at rest by two strings of negligible weight as shown in Fig. 7.39. The angles made by the strings with the vertical are 36.9° and 53.1° respectively. The bar is 2 m long. Calculate the distance d of the centre of gravity of the bar from its left end.

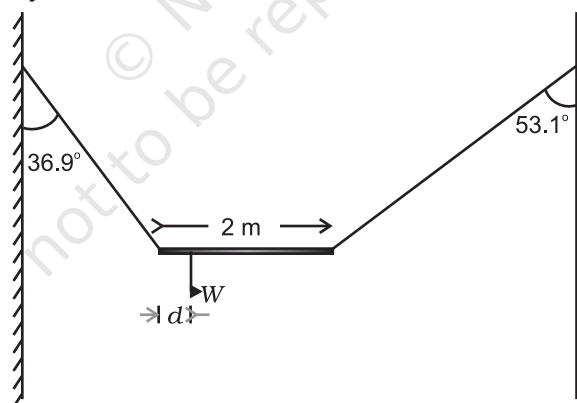


Fig. 7.39

- 7.9** A car weighs 1800 kg . The distance between its front and back axles is 1.8 m . Its centre of gravity is 1.05 m behind the front axle. Determine the force exerted by the level ground on each front wheel and each back wheel.
- 7.10** (a) Find the moment of inertia of a sphere about a tangent to the sphere, given the moment of inertia of the sphere about any of its diameters to be $\frac{2MR^2}{5}$, where M is the mass of the sphere and R is the radius of the sphere.
 (b) Given the moment of inertia of a disc of mass M and radius R about any of its diameters to be $\frac{MR^2}{4}$, find its moment of inertia about an axis normal to the disc and passing through a point on its edge.

- 7.11** Torques of equal magnitude are applied to a hollow cylinder and a solid sphere, both having the same mass and radius. The cylinder is free to rotate about its standard axis of symmetry, and the sphere is free to rotate about an axis passing through its centre. Which of the two will acquire a greater angular speed after a given time.
- 7.12** A solid cylinder of mass 20 kg rotates about its axis with angular speed 100 rad s^{-1} . The radius of the cylinder is 0.25 m. What is the kinetic energy associated with the rotation of the cylinder? What is the magnitude of angular momentum of the cylinder about its axis?
- 7.13** (a) A child stands at the centre of a turntable with his two arms outstretched. The turntable is set rotating with an angular speed of 40 rev/min. How much is the angular speed of the child if he folds his hands back and thereby reduces his moment of inertia to $2/5$ times the initial value? Assume that the turntable rotates without friction.
- (b) Show that the child's new kinetic energy of rotation is more than the initial kinetic energy of rotation. How do you account for this increase in kinetic energy?
- 7.14** A rope of negligible mass is wound round a hollow cylinder of mass 3 kg and radius 40 cm. What is the angular acceleration of the cylinder if the rope is pulled with a force of 30 N? What is the linear acceleration of the rope? Assume that there is no slipping.
- 7.15** To maintain a rotor at a uniform angular speed of 200 rad s^{-1} , an engine needs to transmit a torque of 180 N m. What is the power required by the engine? (Note: uniform angular velocity in the absence of friction implies zero torque. In practice, applied torque is needed to counter frictional torque). Assume that the engine is 100% efficient.
- 7.16** From a uniform disk of radius R , a circular hole of radius $R/2$ is cut out. The centre of the hole is at $R/2$ from the centre of the original disc. Locate the centre of gravity of the resulting flat body.
- 7.17** A metre stick is balanced on a knife edge at its centre. When two coins, each of mass 5 g are put one on top of the other at the 12.0 cm mark, the stick is found to be balanced at 45.0 cm. What is the mass of the metre stick?
- 7.18** A solid sphere rolls down two different inclined planes of the same heights but different angles of inclination. (a) Will it reach the bottom with the same speed in each case? (b) Will it take longer to roll down one plane than the other? (c) If so, which one and why?
- 7.19** A hoop of radius 2 m weighs 100 kg. It rolls along a horizontal floor so that its centre of mass has a speed of 20 cm/s. How much work has to be done to stop it?
- 7.20** The oxygen molecule has a mass of $5.30 \times 10^{-26} \text{ kg}$ and a moment of inertia of $1.94 \times 10^{-46} \text{ kg m}^2$ about an axis through its centre perpendicular to the lines joining the two atoms. Suppose the mean speed of such a molecule in a gas is 500 m/s and that its kinetic energy of rotation is two thirds of its kinetic energy of translation. Find the average angular velocity of the molecule.
- 7.21** A solid cylinder rolls up an inclined plane of angle of inclination 30° . At the bottom of the inclined plane the centre of mass of the cylinder has a speed of 5 m/s.
- (a) How far will the cylinder go up the plane?
- (b) How long will it take to return to the bottom?

Additional Exercises

- 7.22** As shown in Fig.7.40, the two sides of a step ladder BA and CA are 1.6 m long and hinged at A. A rope DE, 0.5 m is tied half way up. A weight 40 kg is suspended from a point F, 1.2 m from B along the ladder BA. Assuming the floor to be frictionless and neglecting the weight of the ladder, find the tension in the rope and forces exerted by the floor on the ladder. (Take $g = 9.8 \text{ m/s}^2$)
(Hint: Consider the equilibrium of each side of the ladder separately.)

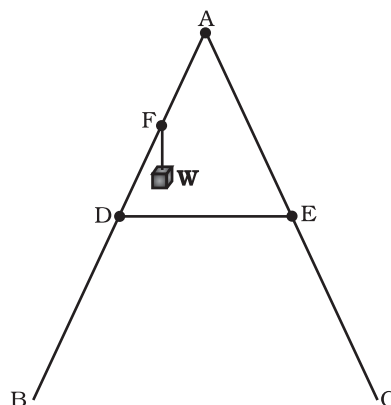


Fig.7.40

- 7.23** A man stands on a rotating platform, with his arms stretched horizontally holding a 5 kg weight in each hand. The angular speed of the platform is 30 revolutions per minute. The man then brings his arms close to his body with the distance of each weight from the axis changing from 90cm to 20cm. The moment of inertia of the man together with the platform may be taken to be constant and equal to 7.6 kg m^2 .
 (a) What is his new angular speed? (Neglect friction.)
 (b) Is kinetic energy conserved in the process? If not, from where does the change come about?
- 7.24** A bullet of mass 10 g and speed 500 m/s is fired into a door and gets embedded exactly at the centre of the door. The door is 1.0 m wide and weighs 12 kg. It is hinged at one end and rotates about a vertical axis practically without friction. Find the angular speed of the door just after the bullet embeds into it.
 (Hint: The moment of inertia of the door about the vertical axis at one end is $ML^2/3$.)
- 7.25** Two discs of moments of inertia I_1 and I_2 about their respective axes (normal to the disc and passing through the centre), and rotating with angular speeds ω_1 and ω_2 are brought into contact face to face with their axes of rotation coincident. (a) What is the angular speed of the two-disc system? (b) Show that the kinetic energy of the combined system is less than the sum of the initial kinetic energies of the two discs. How do you account for this loss in energy? Take $\omega_1 \neq \omega_2$.
- 7.26** (a) Prove the theorem of perpendicular axes.
 (Hint : Square of the distance of a point (x, y) in the x - y plane from an axis through the origin and perpendicular to the plane is $x^2 + y^2$).
 (b) Prove the theorem of parallel axes.
 (Hint : If the centre of mass of a system of n particles is chosen to be the origin $\sum m_i \mathbf{r}_i = 0$).
- 7.27** Prove the result that the velocity v of translation of a rolling body (like a ring, disc, cylinder or sphere) at the bottom of an inclined plane of a height h is given by
- $$v^2 = \frac{2gh}{(1 + k^2 / R^2)}$$
- using dynamical consideration (i.e. by consideration of forces and torques). Note k is the radius of gyration of the body about its symmetry axis, and R is the radius of the body. The body starts from rest at the top of the plane.
- 7.28** A disc rotating about its axis with angular speed ω_0 is placed lightly (without any translational push) on a perfectly frictionless table. The radius of the disc is R . What

are the linear velocities of the points A, B and C on the disc shown in Fig. 7.41? Will the disc roll in the direction indicated ?

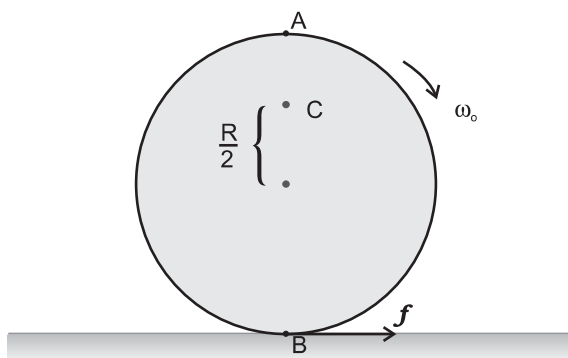


Fig. 7.41

- 7.29** Explain why friction is necessary to make the disc in Fig. 7.41 roll in the direction indicated.
- Give the direction of frictional force at B, and the sense of frictional torque, before perfect rolling begins.
 - What is the force of friction after perfect rolling begins ?
- 7.30** A solid disc and a ring, both of radius 10 cm are placed on a horizontal table simultaneously, with initial angular speed equal to $10\pi \text{ rad s}^{-1}$. Which of the two will start to roll earlier ? The co-efficient of kinetic friction is $\mu_k = 0.2$.
- 7.31** A cylinder of mass 10 kg and radius 15 cm is rolling perfectly on a plane of inclination 30° . The co-efficient of static friction $\mu_s = 0.25$.
- How much is the force of friction acting on the cylinder ?
 - What is the work done against friction during rolling ?
 - If the inclination θ of the plane is increased, at what value of θ does the cylinder begin to skid, and not roll perfectly ?
- 7.32** Read each statement below carefully, and state, with reasons, if it is true or false;
- During rolling, the force of friction acts in the same direction as the direction of motion of the CM of the body.
 - The instantaneous speed of the point of contact during rolling is zero.
 - The instantaneous acceleration of the point of contact during rolling is zero.
 - For perfect rolling motion, work done against friction is zero.
 - A wheel moving down a perfectly *frictionless* inclined plane will undergo slipping (not rolling) motion.
- 7.33** Separation of Motion of a system of particles into motion of the centre of mass and motion about the centre of mass :
- Show $\mathbf{p} = \mathbf{p}'_i + m_i \mathbf{V}$
 where $\mathbf{p}_i$ is the momentum of the i th particle (of mass m_i) and $\mathbf{p}'_i = m_i \mathbf{v}'_i$. Note $\mathbf{v}'_i$ is the velocity of the i th particle relative to the centre of mass.
 Also, prove using the definition of the centre of mass $\sum \mathbf{p}'_i = 0$
 - Show $K = K' + \frac{1}{2}MV^2$
 where K is the total kinetic energy of the system of particles, K' is the total kinetic energy of the system when the particle velocities are taken with respect to the centre of mass and $MV^2/2$ is the kinetic energy of the translation of the system as a whole (i.e. of the centre of mass motion of the system). The result has been used in Sec. 7.14.
 - Show $\mathbf{L} = \mathbf{L}' + \mathbf{R} \times M\mathbf{V}$
 where $\mathbf{L}' = \sum \mathbf{r}'_i \times \mathbf{p}'_i$ is the angular momentum of the system about the centre of mass with

velocities taken relative to the centre of mass. Remember $\mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}$; rest of the notation is the standard notation used in the chapter. Note $\mathbf{L}'$ and $M\mathbf{R} \times \mathbf{V}$ can be said to be angular momenta, respectively, about and of the centre of mass of the system of particles.

(d) Show $\frac{d\mathbf{L}'}{dt} = \sum \mathbf{r}'_i \times \frac{d\mathbf{p}'}{dt}$

Further, show that

$$\frac{d\mathbf{L}'}{dt} = \boldsymbol{\tau}'_{ext}$$

where $\boldsymbol{\tau}'_{ext}$ is the sum of all external torques acting on the system about the centre of mass.

(Hint : Use the definition of centre of mass and third law of motion. Assume the internal forces between any two particles act along the line joining the particles.)

Pluto - A Dwarf Planet

The International Astronomical Union (IAU) at the IAU 2006 General Assembly held on August 24, 2006, in Prague in Czech Republic, adopted a new definition of planets in our Solar System. According to the new definition, Pluto is no longer a planet. This means that the Solar System consists of eight planets: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune. According to the IAU usage, the 'planet' and 'other bodies' in our Solar System, except satellites, are to be defined into three distinct categories of celestial objects in the following way:

1. A 'planet' is a celestial body that (a) is in orbit around the Sun, (b) has sufficient mass for its self-gravity to overcome rigid body forces so that it assumes a hydrostatic equilibrium (nearly round) shape, and (c) has cleared the neighbourhood around its orbit.
2. A 'dwarf planet' is a celestial body that (a) is in orbit around the Sun, (b) has sufficient mass for its self-gravity to overcome rigid body forces so that it assumes a hydrostatic equilibrium (nearly round) shape, (c) has not cleared the neighbourhood around its orbit, and (d) is not a satellite.
3. All 'other objects', except satellites, orbiting the Sun, shall be referred to collectively as 'Small Solar-System Bodies'.

Unlike other eight planets in the Solar System, Pluto's orbital path overlaps with 'other objects' and the planet Neptune. The 'other objects' currently include most of the Solar System asteroids, most of the Trans-Neptunian Objects (TNOs), comets, and other small bodies.

Pluto is a 'dwarf planet' by the above definition and is recognised as the prototype of a new category of Trans-Neptunian Objects.

CHAPTER EIGHT



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GRAVITATION

- 8.1 Introduction
- 8.2 Kepler's laws
- 8.3 Universal law of gravitation
- 8.4 The gravitational constant
- 8.5 Acceleration due to gravity of the earth
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8.1 INTRODUCTION

Early in our lives, we become aware of the tendency of all material objects to be attracted towards the earth. Anything thrown up falls down towards the earth, going uphill is lot more tiring than going downhill, raindrops from the clouds above fall towards the earth and there are many other such phenomena. Historically it was the Italian Physicist Galileo (1564-1642) who recognised the fact that all bodies, irrespective of their masses, are accelerated towards the earth with a constant acceleration. It is said that he made a public demonstration of this fact. To find the truth, he certainly did experiments with bodies rolling down inclined planes and arrived at a value of the acceleration due to gravity which is close to the more accurate value obtained later.

A seemingly unrelated phenomenon, observation of stars, planets and their motion has been the subject of attention in many countries since the earliest of times. Observations since early times recognised stars which appeared in the sky with positions unchanged year after year. The more interesting objects are the planets which seem to have regular motions against the background of stars. The earliest recorded model for planetary motions proposed by Ptolemy about 2000 years ago was a 'geocentric' model in which all celestial objects, stars, the sun and the planets, all revolved around the earth. The only motion that was thought to be possible for celestial objects was motion in a circle. Complicated schemes of motion were put forward by Ptolemy in order to describe the observed motion of the planets. The planets were described as moving in circles with the centre of the circles themselves moving in larger circles. Similar theories were also advanced by Indian astronomers some 400 years later. However a more elegant model in which the Sun was the centre around which the planets revolved – the 'heliocentric' model – was already mentioned by Aryabhata (5th century A.D.) in his treatise. A thousand years later, a Polish monk named Nicolas

Copernicus (1473-1543) proposed a definitive model in which the planets moved in circles around a fixed central sun. His theory was discredited by the church, but notable amongst its supporters was Galileo who had to face prosecution from the state for his beliefs.

It was around the same time as Galileo, a nobleman called Tycho Brahe (1546-1601) hailing from Denmark, spent his entire lifetime recording observations of the planets with the naked eye. His compiled data were analysed later by his assistant Johannes Kepler (1571-1640). He could extract from the data three elegant laws that now go by the name of Kepler's laws. These laws were known to Newton and enabled him to make a great scientific leap in proposing his universal law of gravitation.

8.2 KEPLER'S LAWS

The three laws of Kepler can be stated as follows:

1. Law of orbits : All planets move in elliptical orbits with the Sun situated at one of the foci

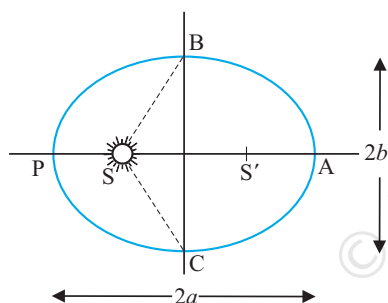


Fig. 8.1(a) An ellipse traced out by a planet around the sun. The closest point is P and the farthest point is A, P is called the perihelion and A the aphelion. The semimajor axis is half the distance AP.

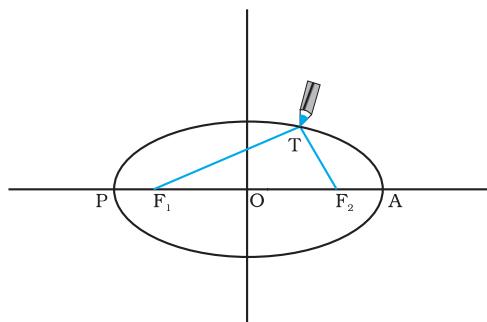


Fig. 8.1(b) Drawing an ellipse. A string has its ends fixed at F_1 and F_2 . The tip of a pencil holds the string taut and is moved around.

of the ellipse (Fig. 8.1a). This law was a deviation from the Copernican model which allowed only circular orbits. The ellipse, of which the circle is a special case, is a closed curve which can be drawn very simply as follows.

Select two points F_1 and F_2 . Take a length of a string and fix its ends at F_1 and F_2 by pins. With the tip of a pencil stretch the string taut and then draw a curve by moving the pencil keeping the string taut throughout. (Fig. 8.1(b)) The closed curve you get is called an ellipse. Clearly for any point T on the ellipse, the sum of the distances from F_1 and F_2 is a constant. F_1 , F_2 are called the foci. Join the points F_1 and F_2 and extend the line to intersect the ellipse at points P and A as shown in Fig. 8.1(b). The midpoint of the line PA is the centre of the ellipse O and the length $PO = AO$ is called the semi-major axis of the ellipse. For a circle, the two foci merge into one and the semi-major axis becomes the radius of the circle.

2. Law of areas : The line that joins any planet to the sun sweeps equal areas in equal intervals of time (Fig. 8.2). This law comes from the observations that planets appear to move slower when they are farther from the sun than when they are nearer.

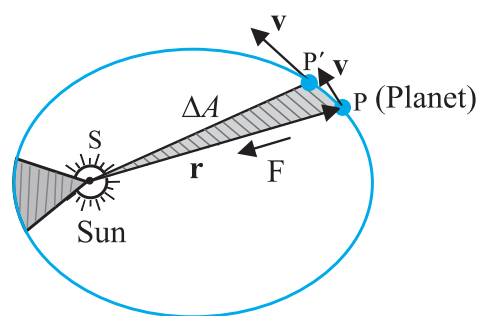


Fig. 8.2 The planet P moves around the sun in an elliptical orbit. The shaded area is the area ΔA swept out in a small interval of time Δt .

3. Law of periods : The square of the time period of revolution of a planet is proportional to the cube of the semi-major axis of the ellipse traced out by the planet.

Table 8.1 gives the approximate time periods of revolution of eight* planets around the sun along with values of their semi-major axes.

* Refer to information given in the Box on Page 182

Table 8.1 Data from measurement of planetary motions given below confirm Kepler's Law of Periods

(a = Semi-major axis in units of 10^{10} m.
 T = Time period of revolution of the planet in years(y).
 Q = The quotient (T^2/a^3) in units of $10^{-34} \text{ y}^2 \text{ m}^{-3}$.)

Planet	a	T	Q
Mercury	5.79	0.24	2.95
Venus	10.8	0.615	3.00
Earth	15.0	1	2.96
Mars	22.8	1.88	2.98
Jupiter	77.8	11.9	3.01
Saturn	143	29.5	2.98
Uranus	287	84	2.98
Neptune	450	165	2.99
Pluto*	590	248	2.99

The law of areas can be understood as a consequence of conservation of angular momentum which is valid for any central force. A central force is such that the force on the planet is along the vector joining the Sun and the planet. Let the Sun be at the origin and let the position and momentum of the planet be denoted by $\mathbf{r}$ and $\mathbf{p}$ respectively. Then the area swept out by the planet of mass m in time interval Δt is (Fig. 8.2) $\Delta \mathbf{A}$ given by

$$\Delta \mathbf{A} = \frac{1}{2} (\mathbf{r} \times \mathbf{v} \Delta t) \quad (8.1)$$

Hence

$$\begin{aligned} \Delta \mathbf{A} / \Delta t &= \frac{1}{2} (\mathbf{r} \times \mathbf{p}) / m, \text{ (since } \mathbf{v} = \mathbf{p} / m) \\ &= \mathbf{L} / (2m) \end{aligned} \quad (8.2)$$

where $\mathbf{v}$ is the velocity, $\mathbf{L}$ is the angular momentum equal to $(\mathbf{r} \times \mathbf{p})$. For a central force, which is directed along $\mathbf{r}$, $\mathbf{L}$ is a constant



Johannes Kepler (1571–1630) was a scientist of German origin. He formulated the three laws of planetary motion based on the painstaking observations of Tycho Brahe and coworkers. Kepler himself was an assistant to Brahe and it took him sixteen long years to arrive at the three planetary laws. He is also known as the founder of geometrical optics, being the first to describe what happens to light after it enters a telescope.

as the planet goes around. Hence, $\Delta \mathbf{A} / \Delta t$ is a constant according to the last equation. This is the law of areas. Gravitation is a central force and hence the law of areas follows.

Example 8.1 Let the speed of the planet at the perihelion P in Fig. 8.1(a) be v_p and the Sun-planet distance SP be r_p . Relate $\{r_p, v_p\}$ to the corresponding quantities at the aphelion $\{r_A, v_A\}$. Will the planet take equal times to traverse BAC and CPB ?

Answer The magnitude of the angular momentum at P is $L_p = m_p r_p v_p$, since inspection tells us that $\mathbf{r}_p$ and $\mathbf{v}_p$ are mutually perpendicular. Similarly, $L_A = m_p r_A v_A$. From angular momentum conservation

$$m_p r_p v_p = m_p r_A v_A$$

$$\text{or } \frac{v_p}{v_A} = \frac{r_A}{r_p}$$

Since $r_A > r_p$, $v_p > v_A$.

The area $SBAC$ bounded by the ellipse and the radius vectors SB and SC is larger than $SBPC$ in Fig. 8.1. From Kepler's second law, equal areas are swept in equal times. Hence the planet will take a longer time to traverse BAC than CPB .

8.3 UNIVERSAL LAW OF GRAVITATION

Legend has it that observing an apple falling from a tree, Newton was inspired to arrive at an universal law of gravitation that led to an explanation of terrestrial gravitation as well as of Kepler's laws. Newton's reasoning was that the moon revolving in an orbit of radius R_m was subject to a centripetal acceleration due to earth's gravity of magnitude

$$a_m = \frac{V^2}{R_m} = \frac{4\pi^2 R_m}{T^2} \quad (8.3)$$

where V is the speed of the moon related to the time period T by the relation $V = 2\pi R_m / T$. The time period T is about 27.3 days and R_m was already known then to be about $3.84 \times 10^8 \text{ m}$. If we substitute these numbers in Eq. (8.3), we get a value of a_m much smaller than the value of acceleration due to gravity g on the surface of the earth, arising also due to earth's gravitational attraction.

* Refer to information given in the Box on Page 182

Central Forces

We know the time rate of change of the angular momentum of a single particle about the origin is

$$\frac{d\mathbf{l}}{dt} = \mathbf{r} \times \mathbf{F}$$

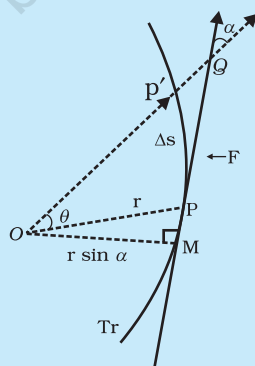
The angular momentum of the particle is conserved, if the torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ due to the force $\mathbf{F}$ on it vanishes. This happens either when $\mathbf{F}$ is zero or when $\mathbf{F}$ is along $\mathbf{r}$. We are interested in forces which satisfy the latter condition. Central forces satisfy this condition. A 'central' force is always directed towards or away from a fixed point, i.e., along the position vector of the point of application of the force with respect to the fixed point. (See Figure below.) Further, the magnitude of a central force F depends on r , the distance of the point of application of the force from the fixed point; $F = F(r)$.

In the motion under a central force the angular momentum is always conserved. Two important results follow from this:

- (1) The motion of a particle under the central force is always confined to a plane.
- (2) The position vector of the particle with respect to the centre of the force (i.e. the fixed point) has a constant areal velocity. In other words the position vector sweeps out equal areas in equal times as the particle moves under the influence of the central force.

Try to prove both these results. You may need to know that the areal velocity is given by : $dA/dt = \frac{1}{2} r v \sin \alpha$.

An immediate application of the above discussion can be made to the motion of a planet under the gravitational force of the sun. For convenience the sun may be taken to be so heavy that it is at rest. The gravitational force of the sun on the planet is directed towards the sun. This force also satisfies the requirement $F = F(r)$, since $F = G m_1 m_2 / r^2$ where m_1 and m_2 are respectively the masses of the planet and the sun and G is the universal constant of gravitation. The two results (1) and (2) described above, therefore, apply to the motion of the planet. In fact, the result (2) is the well-known second law of Kepler.



Tr is the trajectory of the particle under the central force. At a position P, the force is directed along $\overrightarrow{OP}$, O is the centre of the force taken as the origin. In time Δt , the particle moves from P to P', arc $PP' = \Delta s = v \Delta t$. The tangent PQ at P to the trajectory gives the direction of the velocity at P. The area swept in Δt is the area of sector $POP' \approx (r \sin \alpha) PP'/2 = (r v \sin \alpha) \Delta t/2$.

This clearly shows that the force due to earth's gravity decreases with distance. If one assumes that the gravitational force due to the earth decreases in proportion to the inverse square of the distance from the centre of the earth, we will have $a_m \propto R_m^{-2}$; $g \propto R_E^{-2}$ and we get

$$\frac{g}{a_m} = \frac{R_m^2}{R_E^2} \approx 3600 \quad (8.4)$$

in agreement with a value of $g \approx 9.8 \text{ m s}^{-2}$ and the value of a_m from Eq. (8.3). These observations led Newton to propose the following Universal Law of Gravitation :

Every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

The quotation is essentially from Newton's famous treatise called 'Mathematical Principles of Natural Philosophy' (Principia for short).

Stated Mathematically, Newton's gravitation law reads : The force $\mathbf{F}$ on a point mass m_2 due to another point mass m_1 has the magnitude

$$|\mathbf{F}| = G \frac{m_1 m_2}{r^2} \quad (8.5)$$

Equation (8.5) can be expressed in vector form as

$$\begin{aligned} \mathbf{F} &= G \frac{m_1 m_2}{r^2} (-\hat{\mathbf{r}}) = -G \frac{m_1 m_2}{r^2} \hat{\mathbf{r}} \\ &= -G \frac{m_1 m_2}{|\mathbf{r}|^3} \mathbf{r} \end{aligned}$$

where G is the universal gravitational constant, $\hat{\mathbf{r}}$ is the unit vector from m_1 to m_2 and $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$ as shown in Fig. 8.3.

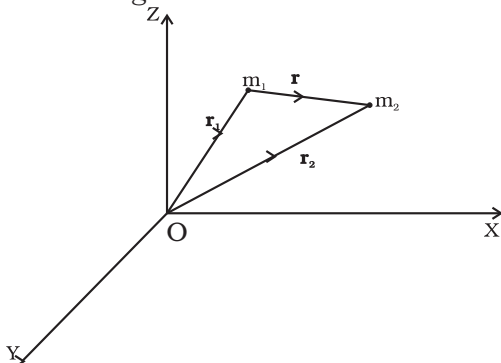


Fig. 8.3 Gravitational force on m_1 due to m_2 is along $\mathbf{r}$ where the vector $\mathbf{r}$ is $(\mathbf{r}_2 - \mathbf{r}_1)$.

The gravitational force is attractive, i.e., the force $\mathbf{F}$ is along $-\mathbf{r}$. The force on point mass m_1 due to m_2 is of course $-\mathbf{F}$ by Newton's third law. Thus, the gravitational force $\mathbf{F}_{12}$ on the body 1 due to 2 and $\mathbf{F}_{21}$ on the body 2 due to 1 are related as $\mathbf{F}_{12} = -\mathbf{F}_{21}$.

Before we can apply Eq. (8.5) to objects under consideration, we have to be careful since the law refers to **point** masses whereas we deal with extended objects which have finite size. If we have a collection of point masses, the force on any one of them is the vector sum of the gravitational forces exerted by the other point masses as shown in Fig 8.4.

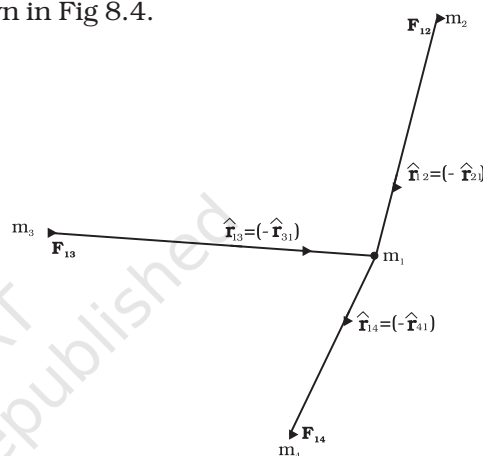


Fig. 8.4 Gravitational force on point mass m_1 is the vector sum of the gravitational forces exerted by m_2 , m_3 and m_4 .

The total force on m_1 is

$$\mathbf{F}_1 = \frac{Gm_2 m_1}{r_{21}^2} \hat{\mathbf{r}}_{21} + \frac{Gm_3 m_1}{r_{31}^2} \hat{\mathbf{r}}_{31} + \frac{Gm_4 m_1}{r_{41}^2} \hat{\mathbf{r}}_{41}$$

► **Example 8.2** Three equal masses of m kg each are fixed at the vertices of an equilateral triangle ABC.

- (a) What is the force acting on a mass $2m$ placed at the centroid G of the triangle?
 (b) What is the force if the mass at the vertex A is doubled?

Take $AG = BG = CG = 1 \text{ m}$ (see Fig. 8.5)

Answer (a) The angle between GC and the positive x -axis is 30° and so is the angle between GB and the negative x -axis. The individual forces in vector notation are

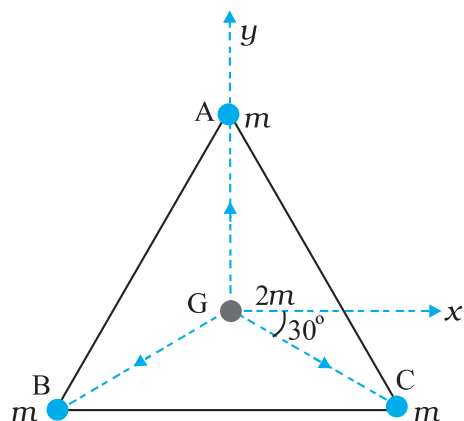


Fig. 8.5 Three equal masses are placed at the three vertices of the ΔABC . A mass $2m$ is placed at the centroid G .

$$\mathbf{F}_{GA} = \frac{Gm(2m)}{1} \hat{\mathbf{j}}$$

$$\mathbf{F}_{GB} = \frac{Gm(2m)}{1} (-\hat{\mathbf{i}} \cos 30^\circ - \hat{\mathbf{j}} \sin 30^\circ)$$

$$\mathbf{F}_{GC} = \frac{Gm(2m)}{1} (+\hat{\mathbf{i}} \cos 30^\circ - \hat{\mathbf{j}} \sin 30^\circ)$$

From the principle of superposition and the law of vector addition, the resultant gravitational force $\mathbf{F}_R$ on $(2m)$ is

$$\mathbf{F}_R = \mathbf{F}_{GA} + \mathbf{F}_{GB} + \mathbf{F}_{GC}$$

$$\mathbf{F}_R = 2Gm^2 \hat{\mathbf{j}} + 2Gm^2 (-\hat{\mathbf{i}} \cos 30^\circ - \hat{\mathbf{j}} \sin 30^\circ) + 2Gm^2 (\hat{\mathbf{i}} \cos 30^\circ - \hat{\mathbf{j}} \sin 30^\circ) = 0$$

Alternatively, one expects on the basis of symmetry that the resultant force ought to be zero.

(b) Now if the mass at vertex A is doubled then

$$\mathbf{F}'_{GA} = \frac{G2m \cdot 2m}{1} \hat{\mathbf{j}} = 4Gm^2 \hat{\mathbf{j}}$$

$$\mathbf{F}'_{GB} = \mathbf{F}_{GB} \text{ and } \mathbf{F}'_{GC} = \mathbf{F}_{GC}$$

$$\mathbf{F}'_R = \mathbf{F}'_{GA} + \mathbf{F}'_{GB} + \mathbf{F}'_{GC}$$

$$\mathbf{F}'_R = 2Gm^2 \hat{\mathbf{j}}$$

For the gravitational force between an extended object (like the earth) and a point mass, Eq. (8.5) is not directly applicable. Each point mass in the extended object will exert a force on the given point mass and these force will not all be in the same direction. We have to add up these forces vectorially for all the point masses in the extended object to get the total force. This is easily done using calculus. For two special cases, a simple law results when you do that :

- (1) **The force of attraction between a hollow spherical shell of uniform density and a point mass situated outside is just as if the entire mass of the shell is concentrated at the centre of the shell.**

Qualitatively this can be understood as follows: Gravitational forces caused by the various regions of the shell have components along the line joining the point mass to the centre as well as along a direction perpendicular to this line. The components perpendicular to this line cancel out when summing over all regions of the shell leaving only a resultant force along the line joining the point to the centre. The magnitude of this force works out to be as stated above.

Newton's Principia

Kepler had formulated his third law by 1619. The announcement of the underlying universal law of gravitation came about seventy years later with the publication in 1687 of Newton's masterpiece **Philosophiae Naturalis Principia Mathematica**, often simply called the **Principia**.

Around 1685, Edmund Halley (after whom the famous Halley's comet is named), came to visit Newton at Cambridge and asked him about the nature of the trajectory of a body moving under the influence of an inverse square law. Without hesitation Newton replied that it had to be an ellipse, and further that he had worked it out long ago around 1665 when he was forced to retire to his farm house from Cambridge on account of a plague outbreak. Unfortunately, Newton had lost his papers. Halley prevailed upon Newton to produce his work in book form and agreed to bear the cost of publication. Newton accomplished this feat in eighteen months of superhuman effort. The **Principia** is a singular scientific masterpiece and in the words of Lagrange it is "the greatest production of the human mind." The Indian born astrophysicist and Nobel laureate S. Chandrasekhar spent ten years writing a treatise on the **Principia**. His book, Newton's **Principia for the Common Reader** brings into sharp focus the beauty, clarity and breath taking economy of Newton's methods.

- (2) **The force of attraction due to a hollow spherical shell of uniform density, on a point mass situated inside it is zero.**

Qualitatively, we can again understand this result. Various regions of the spherical shell attract the point mass inside it in various directions. These forces cancel each other completely.

8.4 THE GRAVITATIONAL CONSTANT

The value of the gravitational constant G entering the Universal law of gravitation can be determined experimentally and this was first done by English scientist Henry Cavendish in 1798. The apparatus used by him is schematically shown in figure.8.6

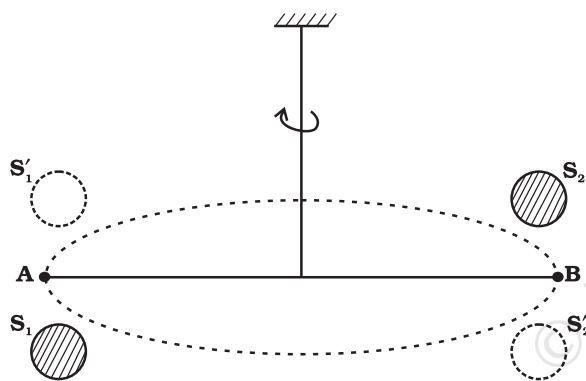


Fig. 8.6 Schematic drawing of Cavendish's experiment. S_1 and S_2 are large spheres which are kept on either side (shown shaded) of the masses at A and B. When the big spheres are taken to the other side of the masses (shown by dotted circles), the bar AB rotates a little since the torque reverses direction. The angle of rotation can be measured experimentally.

The bar AB has two small lead spheres attached at its ends. The bar is suspended from a rigid support by a fine wire. Two large lead spheres are brought close to the small ones but on opposite sides as shown. The big spheres attract the nearby small ones by equal and opposite force as shown. There is no net force on the bar but only a torque which is clearly equal to F times the length of the bar, where F is the force of attraction between a big sphere and

its neighbouring small sphere. Due to this torque, the suspended wire gets twisted till such time as the restoring torque of the wire equals the gravitational torque. If θ is the angle of twist of the suspended wire, the restoring torque is proportional to θ , equal to $\tau\theta$. Where τ is the restoring couple per unit angle of twist. τ can be measured independently e.g. by applying a known torque and measuring the angle of twist. The gravitational force between the spherical balls is the same as if their masses are concentrated at their centres. Thus if d is the separation between the centres of the big and its neighbouring small ball, M and m their masses, the gravitational force between the big sphere and its neighbouring small ball is.

$$F = G \frac{Mm}{d^2} \quad (8.6)$$

If L is the length of the bar AB, then the torque arising out of F is F multiplied by L . At equilibrium, this is equal to the restoring torque and hence

$$G \frac{Mm}{d^2} L = \tau \theta \quad (8.7)$$

Observation of θ thus enables one to calculate G from this equation.

Since Cavendish's experiment, the measurement of G has been refined and the currently accepted value is

$$G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2 \quad (8.8)$$

8.5 ACCELERATION DUE TO GRAVITY OF THE EARTH

The earth can be imagined to be a sphere made of a large number of concentric spherical shells with the smallest one at the centre and the largest one at its surface. A point outside the earth is obviously outside all the shells. Thus, all the shells exert a gravitational force at the point outside just as if their masses are concentrated at their common centre according to the result stated in section 8.3. The total mass of all the shells combined is just the mass of the earth. Hence, at a point outside the earth, the gravitational force is just as if its entire mass of the earth is concentrated at its centre.

For a point inside the earth, the situation is different. This is illustrated in Fig. 8.7.

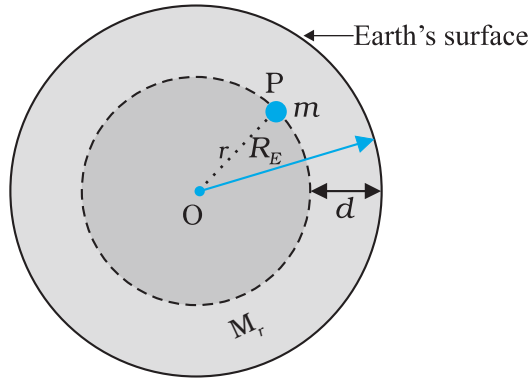


Fig. 8.7 The mass m is in a mine located at a depth d below the surface of the Earth of mass M_E and radius R_E . We treat the Earth to be spherically symmetric.

Again consider the earth to be made up of concentric shells as before and a point mass m situated at a distance r from the centre. The point P lies outside the sphere of radius r . For the shells of radius greater than r , the point P lies inside. Hence according to result stated in the last section, they exert no gravitational force on mass m kept at P . The shells with radius $\leq r$ make up a sphere of radius r for which the point P lies on the surface. This smaller sphere therefore exerts a force on a mass m at P as if its mass M_r is concentrated at the centre. Thus the force on the mass m at P has a magnitude

$$F = \frac{Gm(M_r)}{r^2} \quad (8.9)$$

We assume that the entire earth is of uniform density and hence its mass is $M_E = \frac{4\pi}{3} R_E^3 \rho$ where M_E is the mass of the earth R_E is its radius and ρ is the density. On the other hand the

mass of the sphere M_r of radius r is $\frac{4\pi}{3} \rho r^3$ and hence

$$F = Gm \left(\frac{4\pi}{3} r \right) \frac{r^3}{r^2} = Gm \left(\frac{M_E}{R_E^3} \right) r^3 = \frac{GmM_E}{R_E^3} r \quad (8.10)$$

If the mass m is situated on the surface of earth, then $r = R_E$ and the gravitational force on it is, from Eq. (8.10)

$$F = G \frac{M_E m}{R_E^2} \quad (8.11)$$

The acceleration experienced by the mass m , which is usually denoted by the symbol g is related to F by Newton's 2nd law by relation $F = mg$. Thus

$$g = \frac{F}{m} = \frac{GM_E}{R_E^2} \quad (8.12)$$

Acceleration g is readily measurable. R_E is a known quantity. The measurement of G by Cavendish's experiment (or otherwise), combined with knowledge of g and R_E enables one to estimate M_E from Eq. (8.12). This is the reason why there is a popular statement regarding Cavendish : "Cavendish weighed the earth".

8.6 ACCELERATION DUE TO GRAVITY BELOW AND ABOVE THE SURFACE OF EARTH

Consider a point mass m at a height h above the surface of the earth as shown in Fig. 8.8(a). The radius of the earth is denoted by R_E . Since this point is outside the earth,

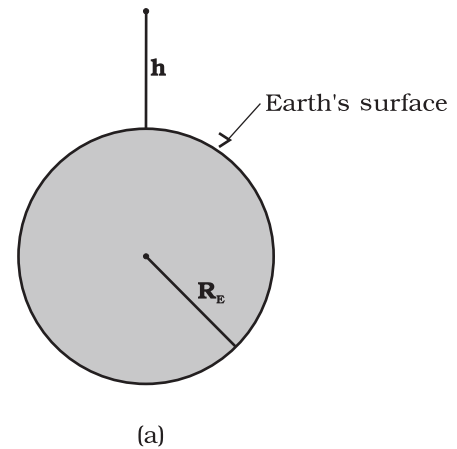


Fig. 8.8 (a) g at a height h above the surface of the earth.

its distance from the centre of the earth is $(R_E + h)$. If $F(h)$ denoted the magnitude of the force on the point mass m , we get from Eq. (8.5) :

$$F(h) = \frac{GM_E m}{(R_E + h)^2} \quad (8.13)$$

The acceleration experienced by the point mass is $F(h)/m \equiv g(h)$ and we get

$$g(h) = \frac{F(h)}{m} = \frac{GM_E}{(R_E + h)^2} \quad (8.14)$$

This is clearly less than the value of g on the surface of earth : $g = \frac{GM_E}{R_E^2}$. For $h \ll R_E$, we can expand the RHS of Eq. (8.14) :

$$g(h) = \frac{GM_E}{R_E^2(1 + h/R_E)^2} = g(1 + h/R_E)^{-2}$$

For $\frac{h}{R_E} \ll 1$, using binomial expression,

$$g(h) \approx g \left(1 - \frac{2h}{R_E} \right) \quad (8.15)$$

Equation (8.15) thus tells us that for small heights h above the value of g decreases by a factor $(1 - 2h/R_E)$.

Now, consider a point mass m at a depth d below the surface of the earth (Fig. 8.8(b)), so that its distance from the centre of the earth is $(R_E - d)$ as shown in the figure. The earth can be thought of as being composed of a smaller sphere of radius $(R_E - d)$ and a spherical shell of thickness d . The force on m due to the outer shell of thickness d is zero because the result quoted in the previous section. As far as the smaller sphere of radius $(R_E - d)$ is concerned, the point mass is outside it and hence according to the result quoted earlier, the force due to this smaller sphere is just as if the entire mass of the smaller sphere is concentrated at the centre. If M_s is the mass of the smaller sphere, then,

$$M_s/M_E = (R_E - d)^3 / R_E^3 \quad (8.16)$$

Since mass of a sphere is proportional to be cube of its radius.

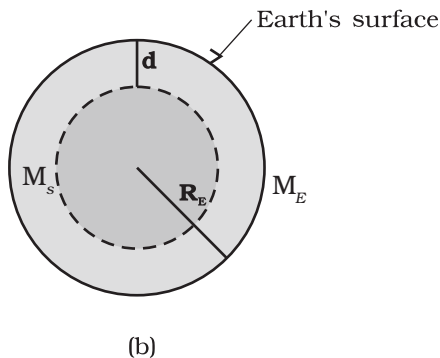


Fig. 8.8 (b) g at a depth d . In this case only the smaller sphere of radius $(R_E - d)$ contributes to g .

Thus the force on the point mass is

$$F(d) = G M_s m / (R_E - d)^2 \quad (8.17)$$

Substituting for M_s from above, we get

$$F(d) = G M_E m (R_E - d) / R_E^3 \quad (8.18)$$

and hence the acceleration due to gravity at a depth d ,

$$g(d) = \frac{F(d)}{m} \text{ is}$$

$$g(d) = \frac{F(d)}{m} = \frac{GM_E}{R_E^3} (R_E - d)$$

$$= g \frac{R_E - d}{R_E} = g(1 - d/R_E) \quad (8.19)$$

Thus, as we go down below earth's surface, the acceleration due gravity decreases by a factor $(1 - d/R_E)$. The remarkable thing about acceleration due to earth's gravity is that it is maximum on its surface decreasing whether you go up or down.

8.7 GRAVITATIONAL POTENTIAL ENERGY

We had discussed earlier the notion of potential energy as being the energy stored in the body at its given position. If the position of the particle changes on account of forces acting on it, then the change in its potential energy is just the amount of work done on the body by the force. As we had discussed earlier, forces for which the work done is independent of the path are the conservative forces.

The force of gravity is a conservative force and we can calculate the potential energy of a body arising out of this force, called the gravitational potential energy. Consider points close to the surface of earth, at distances from the surface much smaller than the radius of the earth. In such cases, the force of gravity is practically a constant equal to mg , directed towards the centre of the earth. If we consider a point at a height h_1 from the surface of the earth and another point vertically above it at a height h_2 from the surface, the work done in lifting the particle of mass m from the first to the second position is denoted by W_{12}

$$W_{12} = \text{Force} \times \text{displacement}$$

$$= mg(h_2 - h_1) \quad (8.20)$$

If we associate a potential energy $W(h)$ at a point at a height h above the surface such that

$$W(h) = mgh + W_0 \quad (8.21)$$

(where W_0 = constant) ;

then it is clear that

$$W_{12} = W(h_2) - W(h_1) \quad (8.22)$$

The work done in moving the particle is just the difference of potential energy between its final and initial positions. Observe that the constant W_0 cancels out in Eq. (8.22). Setting $h = 0$ in the last equation, we get $W(h = 0) = W_0$. $h = 0$ means points on the surface of the earth. Thus, W_0 is the potential energy on the surface of the earth.

If we consider points at arbitrary distance from the surface of the earth, the result just derived is not valid since the assumption that the gravitational force mg is a constant is no longer valid. However, from our discussion we know that a point outside the earth, the force of gravitation on a particle directed towards the centre of the earth is

$$F = \frac{GM_E m}{r^2} \quad (8.23)$$

where M_E = mass of earth, m = mass of the particle and r its distance from the centre of the earth. If we now calculate the work done in lifting a particle from $r = r_1$ to $r = r_2$ ($r_2 > r_1$) along a vertical path, we get instead of Eq. (8.20)

$$\begin{aligned} W_{12} &= \int_{r_1}^{r_2} \frac{GM m}{r^2} dr \\ &= -GM_E m \left(\frac{1}{r_2} - \frac{1}{r_1} \right) \end{aligned} \quad (8.24)$$

In place of Eq. (8.21), we can thus associate a potential energy $W(r)$ at a distance r , such that

$$W(r) = -\frac{GM_E m}{r} + W_1, \quad (8.25)$$

valid for $r > R$,

so that once again $W_{12} = W(r_2) - W(r_1)$. Setting $r = \text{infinity}$ in the last equation, we get $W(r = \text{infinity}) = W_1$. Thus, W_1 is the potential energy at infinity. One should note that only the difference of potential energy between two points has a definite meaning from Eqs. (8.22) and (8.24). One conventionally sets W_1 equal to zero, so that the potential energy at a point is just the amount of work done in displacing the particle from infinity to that point.

We have calculated the potential energy at a point of a particle due to gravitational forces on it due to the earth and it is proportional to the mass of the particle. The gravitational potential due to the gravitational force of the earth is defined as the potential energy of a particle of unit mass at that point. From the earlier discussion, we learn that the gravitational potential energy associated with two particles of masses m_1 and m_2 separated by distance by a distance r is given by

$$V = -\frac{Gm_1 m_2}{r} \quad (\text{if we choose } V = 0 \text{ as } r \rightarrow \infty)$$

It should be noted that an isolated system of particles will have the total potential energy that equals the sum of energies (given by the above equation) for all possible pairs of its constituent particles. This is an example of the application of the superposition principle.

► **Example 8.3** Find the potential energy of a system of four particles placed at the vertices of a square of side l . Also obtain the potential at the centre of the square.

Answer Consider four masses each of mass m at the corners of a square of side l ; See Fig. 8.9. We have four mass pairs at distance l and two diagonal pairs at distance $\sqrt{2}l$

Hence,

$$W(r) = -4 \frac{Gm^2}{l} - 2 \frac{Gm^2}{\sqrt{2}l}$$

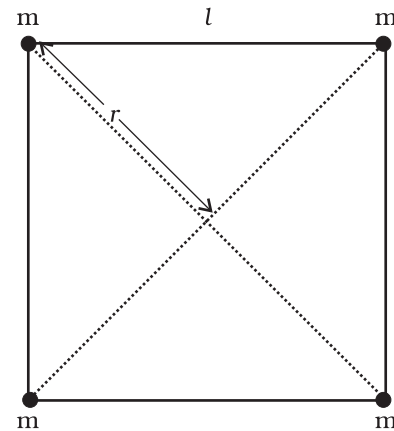


Fig. 8.9

$$= -\frac{2Gm^2}{l} \left(2 + \frac{1}{\sqrt{2}} \right) = -5.41 \frac{Gm^2}{l}$$

The gravitational potential at the centre of the square ($r = \sqrt{2}l/2$) is

$$U(r) = -4\sqrt{2} \frac{Gm}{l}$$

8.8 ESCAPE SPEED

If a stone is thrown by hand, we see it falls back to the earth. Of course using machines we can shoot an object with much greater speeds and with greater and greater initial speed, the object scales higher and higher heights. A natural query that arises in our mind is the following: 'can we throw an object with such high initial speeds that it does not fall back to the earth?'

The principle of conservation of energy helps us to answer this question. Suppose the object did reach infinity and that its speed there was V_f . The energy of an object is the sum of potential and kinetic energy. As before W_i denotes that gravitational potential energy of the object at infinity. The total energy of the projectile at infinity then is

$$E(\infty) = W_i + \frac{mV_f^2}{2} \quad (8.26)$$

If the object was thrown initially with a speed V_i from a point at a distance $(h+R_E)$ from the centre of the earth (R_E = radius of the earth), its energy initially was

$$E(h+R_E) = \frac{1}{2}mV_i^2 - \frac{GmM_E}{(h+R_E)} + W_i \quad (8.27)$$

By the principle of energy conservation Eqs. (8.26) and (8.27) must be equal. Hence

$$\frac{mV_i^2}{2} - \frac{GmM_E}{(h+R_E)} = \frac{mV_f^2}{2} \quad (8.28)$$

The R.H.S. is a positive quantity with a minimum value zero hence so must be the L.H.S. Thus, an object can reach infinity as long as V_i is such that

$$\frac{mV_i^2}{2} - \frac{GmM_E}{(h+R_E)} \geq 0 \quad (8.29)$$

The minimum value of V_i corresponds to the case when the L.H.S. of Eq. (8.29) equals zero.

Thus, the minimum speed required for an object to reach infinity (i.e. escape from the earth) corresponds to

$$\frac{1}{2}m(V_i)_{\min}^2 = \frac{GmM_E}{h+R_E} \quad (8.30)$$

If the object is thrown from the surface of the earth, $h = 0$, and we get

$$(V_i)_{\min} = \sqrt{\frac{2GM_E}{R_E}} \quad (8.31)$$

Using the relation $g = GM_E / R_E^2$, we get

$$(V_i)_{\min} = \sqrt{2gR_E} \quad (8.32)$$

Using the value of g and R_E , numerically $(V_i)_{\min} \approx 11.2$ km/s. This is called the escape speed, sometimes loosely called the escape velocity.

Equation (8.32) applies equally well to an object thrown from the surface of the moon with g replaced by the acceleration due to Moon's gravity on its surface and R_E replaced by the radius of the moon. Both are smaller than their values on earth and the escape speed for the moon turns out to be 2.3 km/s, about five times smaller. This is the reason that moon has no atmosphere. Gas molecules if formed on the surface of the moon having velocities larger than this will escape the gravitational pull of the moon.

Example 8.4 Two uniform solid spheres of equal radii R , but mass M and $4M$ have a centre to centre separation $6R$, as shown in Fig. 8.10. The two spheres are held fixed. A projectile of mass m is projected from the surface of the sphere of mass M directly towards the centre of the second sphere. Obtain an expression for the minimum speed v of the projectile so that it reaches the surface of the second sphere.

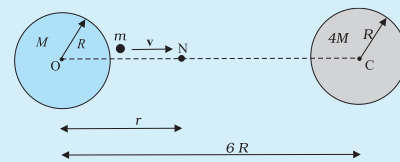


Fig. 8.10

Answer The projectile is acted upon by two mutually opposing gravitational forces of the two

spheres. The neutral point N (see Fig. 8.10) is defined as the position where the two forces cancel each other exactly. If $ON = r$, we have

$$\frac{G M m}{r^2} = \frac{4 G M m}{(6R-r)^2}$$

$$(6R-r)^2 = 4r^2$$

$$6R-r = \pm 2r$$

$$r = 2R \text{ or } -6R.$$

The neutral point $r = -6R$ does not concern us in this example. Thus $ON = r = 2R$. It is sufficient to project the particle with a speed which would enable it to reach N. Thereafter, the greater gravitational pull of $4M$ would suffice. The mechanical energy at the surface of M is

$$E_i = \frac{1}{2} m v^2 - \frac{G M m}{R} - \frac{4 G M m}{5 R}.$$

At the neutral point N, the speed approaches zero. The mechanical energy at N is purely potential.

$$E_N = -\frac{G M m}{2 R} - \frac{4 G M m}{4 R}.$$

From the principle of conservation of mechanical energy

$$\frac{1}{2} v^2 - \frac{GM}{R} - \frac{4GM}{5R} = -\frac{GM}{2R} - \frac{GM}{R}$$

or

$$v^2 = \frac{2 G M}{R} \left(\frac{4}{5} - \frac{1}{2} \right)$$

$$v = \left(\frac{3 G M}{5 R} \right)^{1/2}$$

A point to note is that the speed of the projectile is zero at N, but is nonzero when it strikes the heavier sphere $4 M$. The calculation of this speed is left as an exercise to the students.

8.9 EARTH SATELLITES

Earth satellites are objects which revolve around the earth. Their motion is very similar to the motion of planets around the Sun and hence Kepler's laws of planetary motion are equally applicable to them. In particular, their orbits around the earth are circular or elliptic. Moon is the only natural satellite of the earth with a near circular orbit with a time period of approximately 27.3 days which is also roughly equal to the rotational period of the moon about

its own axis. Since, 1957, advances in technology have enabled many countries including India to launch artificial earth satellites for practical use in fields like telecommunication, geophysics and meteorology.

We will consider a satellite in a circular orbit of a distance $(R_E + h)$ from the centre of the earth, where R_E = radius of the earth. If m is the mass of the satellite and V its speed, the centripetal force required for this orbit is

$$F(\text{centripetal}) = \frac{mV^2}{(R_E + h)} \quad (8.33)$$

directed towards the centre. This centripetal force is provided by the gravitational force, which is

$$F(\text{gravitation}) = \frac{G m M_E}{(R_E + h)^2} \quad (8.34)$$

where M_E is the mass of the earth.

Equating R.H.S of Eqs. (8.33) and (8.34) and cancelling out m , we get

$$V^2 = \frac{G M_E}{(R_E + h)} \quad (8.35)$$

Thus V decreases as h increases. From equation (8.35), the speed V for $h = 0$ is

$$V^2 (h = 0) = GM / R_E = gR_E \quad (8.36)$$

where we have used the relation $g = GM / R_E^2$. In every orbit, the satellite traverses a distance $2\pi(R_E + h)$ with speed V . Its time period T therefore is

$$T = \frac{2\pi(R_E + h)}{V} = \frac{2\pi(R_E + h)^{3/2}}{\sqrt{G M_E}} \quad (8.37)$$

on substitution of value of V from Eq. (8.35). Squaring both sides of Eq. (8.37), we get

$$T^2 = k (R_E + h)^3 \text{ (where } k = 4 \pi^2 / G M_E \text{)} \quad (8.38)$$

which is Kepler's law of periods, as applied to motion of satellites around the earth. For a satellite very close to the surface of earth h can be neglected in comparison to R_E in Eq. (8.38). Hence, for such satellites, T is T_0 , where

$$T_0 = 2\pi\sqrt{R_E / g} \quad (8.39)$$

If we substitute the numerical values $g \approx 9.8 \text{ m s}^{-2}$ and $R_E = 6400 \text{ km.}$, we get

$$T_0 = 2\pi\sqrt{\frac{6.4 \times 10^6}{9.8}} \text{ s}$$

Which is approximately 85 minutes.

► **Example 8.5** The planet Mars has two moons, phobos and delmos. (i) phobos has a period 7 hours, 39 minutes and an orbital radius of 9.4×10^3 km. Calculate the mass of mars. (ii) Assume that earth and mars move in circular orbits around the sun, with the martian orbit being 1.52 times the orbital radius of the earth. What is the length of the martian year in days ?

Answer (i) We employ Eq. (8.38) with the sun's mass replaced by the martian mass M_m

$$T^2 = \frac{4\pi^2}{GM_m} R^3$$

$$M_m = \frac{4\pi^2}{G} \frac{R^3}{T^2}$$

$$= \frac{4 \times (3.14)^2 \times (9.4)^3 \times 10^{18}}{6.67 \times 10^{-11} \times (459 \times 60)^2}$$

$$M_m = \frac{4 \times (3.14)^2 \times (9.4)^3 \times 10^{18}}{6.67 \times (4.59 \times 6)^2 \times 10^{-5}}$$

$$= 6.48 \times 10^{23} \text{ kg.}$$

(ii) Once again Kepler's third law comes to our aid,

$$\frac{T_M^2}{T_E^2} = \frac{R_{MS}^3}{R_{ES}^3}$$

where R_{MS} is the mars-sun distance and R_{ES} is the earth-sun distance.

$$\therefore T_M = (1.52)^{3/2} \times 365$$

$$= 684 \text{ days}$$

We note that the orbits of all planets except Mercury, Mars and Pluto* are very close to being circular. For example, the ratio of the semi-minor to semi-major axis for our Earth is, $b/a = 0.99986$.

► **Example 8.6 Weighing the Earth :** You are given the following data: $g = 9.81 \text{ ms}^{-2}$, $R_E = 6.37 \times 10^6 \text{ m}$, the distance to the moon $R = 3.84 \times 10^8 \text{ m}$ and the time period of the moon's revolution is 27.3 days. Obtain the mass of the Earth M_E in two different ways.

Answer From Eq. (8.12) we have

$$M_E = \frac{g R_E^2}{G}$$

$$= \frac{9.81 \times (6.37 \times 10^6)^2}{6.67 \times 10^{-11}}$$

$$= 5.97 \times 10^{24} \text{ kg.}$$

The moon is a satellite of the Earth. From the derivation of Kepler's third law [see Eq. (8.38)]

$$T^2 = \frac{4\pi^2 R^3}{G M_E}$$

$$M_E = \frac{4\pi^2 R^3}{G T^2}$$

$$= \frac{4 \times 3.14 \times 3.14 \times (3.84)^3 \times 10^{24}}{6.67 \times 10^{-11} \times (27.3 \times 24 \times 60 \times 60)^2}$$

$$= 6.02 \times 10^{24} \text{ kg}$$

Both methods yield almost the same answer, the difference between them being less than 1%.

► **Example 8.7** Express the constant k of Eq. (8.38) in days and kilometres. Given $k = 10^{-13} \text{ s}^2 \text{ m}^{-3}$. The moon is at a distance of $3.84 \times 10^5 \text{ km}$ from the earth. Obtain its time-period of revolution in days.

Answer Given

$$k = 10^{-13} \text{ s}^2 \text{ m}^{-3}$$

$$= 10^{-13} \left[\frac{1}{(24 \times 60 \times 60)^2} \text{ d}^2 \right] \left[\frac{1}{(1/1000)^3 \text{ km}^3} \right]$$

$$= 1.33 \times 10^{-14} \text{ d}^2 \text{ km}^{-3}$$

Using Eq. (8.38) and the given value of k , the time period of the moon is

$$T^2 = (1.33 \times 10^{-14}) (3.84 \times 10^5)^3$$

$$T = 27.3 \text{ d}$$

Note that Eq. (8.38) also holds for elliptical orbits if we replace $(R_E + h)$ by the semi-major axis of the ellipse. The earth will then be at one of the foci of this ellipse.

8.10 ENERGY OF AN ORBITING SATELLITE

Using Eq. (8.35), the kinetic energy of the satellite in a circular orbit with speed v is

$$K.E = \frac{1}{2} m v^2$$

$$= \frac{G m M_E}{2(R_E + h)} \quad (8.40)$$

* Refer to information given in the Box on Page 182

Considering gravitational potential energy at infinity to be zero, the potential energy at distance $(R_E + h)$ from the centre of the earth is

$$P.E = -\frac{G m M_E}{(R_E + h)} \quad (8.41)$$

The K.E is positive whereas the P.E is negative. However, in magnitude the K.E. is half the P.E, so that the total E is

$$E = K.E + P.E = -\frac{G m M_E}{2(R_E + h)} \quad (8.42)$$

The total energy of an circularly orbiting satellite is thus negative, with the potential energy being negative but twice is magnitude of the positive kinetic energy.

When the orbit of a satellite becomes elliptic, both the K.E. and P.E. vary from point to point. The total energy which remains constant is negative as in the circular orbit case. This is what we expect, since as we have discussed before if the total energy is positive or zero, the object escapes to infinity. Satellites are always at finite distance from the earth and hence their energies cannot be positive or zero.

Example 8.8 A 400 kg satellite is in a circular orbit of radius $2R_E$ about the Earth. How much energy is required to transfer it to a circular orbit of radius $4R_E$? What are the changes in the kinetic and potential energies?

Answer Initially,

$$E_i = -\frac{G M_E m}{4 R_E}$$

While finally

$$E_f = -\frac{G M_E m}{8 R_E}$$

The change in the total energy is

$$\Delta E = E_f - E_i$$

$$\begin{aligned} &= \frac{G M_E m}{8 R_E} = \left(\frac{G M_E}{R_E^2} \right) \frac{m R_E}{8} \\ \Delta E &= \frac{g m R_E}{8} = \frac{9.81 \times 400 \times 6.37 \times 10^6}{8} = 3.13 \times 10^9 \text{ J} \end{aligned}$$

The kinetic energy is reduced and it mimics ΔE , namely, $\Delta K = K_f - K_i = -3.13 \times 10^9 \text{ J}$.

The change in potential energy is twice the change in the total energy, namely

$$\Delta V = V_f - V_i = -6.25 \times 10^9 \text{ J}$$

8.11 GEOSTATIONARY AND POLAR SATELLITES

An interesting phenomenon arises if we arrange the value of $(R_E + h)$ such that T in Eq. (8.37) becomes equal to 24 hours. If the circular orbit is in the equatorial plane of the earth, such a satellite, having the same period as the period of rotation of the earth about its own axis would appear stationary viewed from a point on earth. The $(R_E + h)$ for this purpose works out to be large as compared to R_E :

$$R_E + h = \left(\frac{T^2 G M_E}{4\pi^2} \right)^{1/3} \quad (8.43)$$

and for $T = 24$ hours, h works out to be 35800 km. which is much larger than R_E . Satellites in a circular orbits around the earth in the equatorial plane with $T = 24$ hours are called Geostationary Satellites. Clearly, since the earth rotates with the same period, the satellite would appear fixed from any point on earth. It takes very powerful rockets to throw up a satellite to such large heights above the earth but this has been done in view of the several benefits of many practical applications.

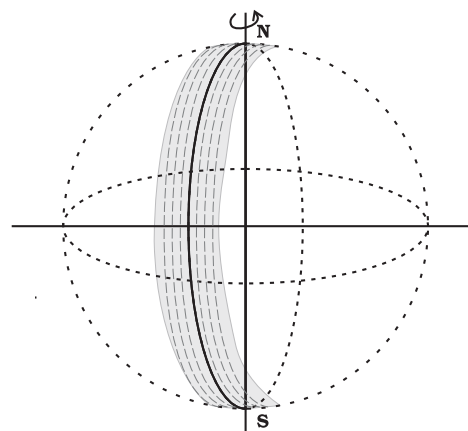


Fig. 8.11 A Polar satellite. A strip on earth's surface (shown shaded) is visible from the satellite during one cycle. For the next revolution of the satellite, the earth has rotated a little on its axis so that an adjacent strip becomes visible.

It is known that electromagnetic waves above a certain frequency are not reflected from ionosphere. Radio waves used for radio broadcast which are in the frequency range 2 MHz to 10 MHz, are below the critical frequency. They are therefore reflected by the ionosphere.

India's Leap into Space

India started its space programme in 1962 when Indian National Committee for Space Research was set up by the Government of India which was superseded by the Indian Space Research Organisation (ISRO) in 1969. ISRO identified the role and importance of space technology in nation's development and bringing space to the service of the common man. India launched its first low orbit satellite Aryabhata in 1975, for which the launch vehicle was provided by the erstwhile Soviet Union. ISRO started employing its indigenous launching vehicle in 1979 by sending Rohini series of satellites into space from its main launch site at Satish Dhawan Space Center, Sriharikota, Andhra Pradesh. The tremendous progress in India's space programme has made ISRO one of the six largest space agencies in the world. ISRO develops and delivers application specific satellite products and tools for broadcasts, communication, weather forecasts, disaster management tools, Geographic Information System, cartography, navigation, telemedicine, dedicated distance education satellite etc. In order to achieve complete self-reliance in these applications, cost effective and reliable Polar Satellite Launch Vehicle (PSLV) was developed in early 1990s. PSLV has thus become a favoured carrier for satellites of various countries, promoting unprecedented international collaboration. In 2001, the Geosynchronous Satellite Launch Vehicle (GSLV) was developed for launching heavier and more demanding Geosynchronous communication satellites. Various research centers and autonomous institutions for remote sensing, astronomy and astrophysics, atmospheric sciences and space research are functioning under the aegis of the Department of Space, Government of India. Success of lunar (Chandrayaan) and inter planetary (Mangalyaan) missions along with other scientific projects has been landmark achievements of ISRO. Future endeavors of ISRO include human space flight projects, the development of heavy lift launchers, reusable launch vehicles, semi-cryogenic engines, single and two stage to orbit (SSTO and TSTO) vehicles, development and use of composite materials for space application etc. In 1984 Rakesh Sharma became the first Indian to go into outer space aboard in a USSR spaceship. (www.isro.gov.in)

Thus radio waves broadcast from an antenna can be received at points far away where the direct wave fail to reach on account of the curvature of the earth. Waves used in television broadcast or other forms of communication have much higher frequencies and thus cannot be received beyond the line of sight. A Geostationary satellite, appearing fixed above the broadcasting station can however receive these signals and broadcast them back to a wide area on earth. The INSAT group of satellites sent up by India are one such group of Geostationary satellites widely used for telecommunications in India.

Another class of satellites are called the Polar satellites (Fig. 8.11). These are low altitude ($h \approx 500$ to 800 km) satellites, but they go around the poles of the earth in a north-south direction whereas the earth rotates around its axis in an east-west direction. Since its time period is around 100 minutes it crosses any altitude many times a day. However, since its height h above the earth is about 500 - 800 km, a camera fixed on it can view only small strips of the earth in one orbit. Adjacent strips are viewed in the next orbit, so that in effect the whole earth can be viewed strip by strip during the entire day. These satellites can view polar and equatorial regions

at close distances with good resolution. Information gathered from such satellites is extremely useful for remote sensing, meteorology as well as for environmental studies of the earth.

8.12 WEIGHTLESSNESS

Weight of an object is the force with which the earth attracts it. We are conscious of our own weight when we stand on a surface, since the surface exerts a force opposite to our weight to keep us at rest. The same principle holds good when we measure the weight of an object by a spring balance hung from a fixed point e.g. the ceiling. The object would fall down unless it is subject to a force opposite to gravity. This is exactly what the spring exerts on the object. This is because the spring is pulled down a little by the gravitational pull of the object and in turn the spring exerts a force on the object vertically upwards.

Now, imagine that the top end of the balance is no longer held fixed to the top ceiling of the room. Both ends of the spring as well as the object move with identical acceleration g . The spring is not stretched and does not exert any upward force on the object which is moving down with acceleration g due to gravity. The reading

recorded in the spring balance is zero since the spring is not stretched at all. If the object were a human being, he or she will not feel his weight since there is no upward force on him. Thus, when an object is in free fall, it is weightless and this phenomenon is usually called the phenomenon of weightlessness.

In a satellite around the earth, every part and parcel of the satellite has an acceleration towards the centre of the earth which is exactly

the value of earth's acceleration due to gravity at that position. Thus in the satellite everything inside it is in a state of free fall. This is just as if we were falling towards the earth from a height. Thus, in a manned satellite, people inside experience no gravity. Gravity for us defines the vertical direction and thus for them there are no horizontal or vertical directions, all directions are the same. Pictures of astronauts floating in a satellite show this fact.

SUMMARY

1. Newton's law of universal gravitation states that the gravitational force of attraction between any two particles of masses m_1 and m_2 separated by a distance r has the magnitude

$$F = G \frac{m_1 m_2}{r^2}$$

where G is the universal gravitational constant, which has the value $6.672 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$.

2. If we have to find the resultant gravitational force acting on the particle m due to a number of masses $M_1, M_2, \dots, M_n$ etc. we use the principle of superposition. Let $F_1, F_2, \dots, F_n$ be the individual forces due to $M_1, M_2, \dots, M_n$ each given by the law of gravitation. From the principle of superposition each force acts independently and uninfluenced by the other bodies. The resultant force F_R is then found by vector addition

$$F_R = F_1 + F_2 + \dots + F_n = \sum_{i=1}^n F_i$$

where the symbol ' Σ ' stands for summation.

3. Kepler's laws of planetary motion state that
 - (a) All planets move in elliptical orbits with the Sun at one of the focal points
 - (b) The radius vector drawn from the Sun to a planet sweeps out equal areas in equal time intervals. This follows from the fact that the force of gravitation on the planet is central and hence angular momentum is conserved.
 - (c) The square of the orbital period of a planet is proportional to the cube of the semi-major axis of the elliptical orbit of the planet
 The period T and radius R of the circular orbit of a planet about the Sun are related by

$$T^2 = \left(\frac{4\pi^2}{G M_s} \right) R^3$$

where M_s is the mass of the Sun. Most planets have nearly circular orbits about the Sun. For elliptical orbits, the above equation is valid if R is replaced by the semi-major axis, a .

4. The acceleration due to gravity.
 - (a) at a height h above the earth's surface

$$g(h) = \frac{G M_E}{(R_E + h)^2}$$

$$\approx \frac{G M_E}{R_E^2} \left(1 - \frac{2h}{R_E} \right) \quad \text{for } h \ll R_E$$

$$g(h) = g(0) \left(1 - \frac{2h}{R_E} \right) \quad \text{where } g(0) = \frac{G M_E}{R_E^2}$$

(b) at depth d below the earth's surface is

$$g(d) = \frac{G M_E}{R_E^2} \left(1 - \frac{d}{R_E}\right) = g(0) \left(1 - \frac{d}{R_E}\right)$$

5. The gravitational force is a conservative force, and therefore a potential energy function can be defined. The *gravitational potential energy* associated with two particles separated by a distance r is given by

$$V = -\frac{G m_1 m_2}{r}$$

where V is taken to be zero at $r \rightarrow \infty$. The total potential energy for a system of particles is the sum of energies for all pairs of particles, with each pair represented by a term of the form given by above equation. This prescription follows from the principle of superposition.

6. If an isolated system consists of a particle of mass m moving with a speed v in the vicinity of a massive body of mass M , the total mechanical energy of the particle is given by

$$E = \frac{1}{2} m v^2 - \frac{G M m}{r}$$

That is, the total mechanical energy is the sum of the kinetic and potential energies. The total energy is a constant of motion.

7. If m moves in a circular orbit of radius a about M , where $M \gg m$, the total energy of the system is

$$E = -\frac{G M m}{2a}$$

with the choice of the arbitrary constant in the potential energy given in the point 5., above. The total energy is negative for any bound system, that is, one in which the orbit is closed, such as an elliptical orbit. The kinetic and potential energies are

$$K = \frac{G M m}{2a}$$

$$V = -\frac{G M m}{a}$$

8. The escape speed from the surface of the earth is

$$v_e = \sqrt{\frac{2 G M_E}{R_E}} = \sqrt{2 g R_E}$$

and has a value of 11.2 km s^{-1} .

9. If a particle is outside a uniform spherical shell or solid sphere with a spherically symmetric internal mass distribution, the sphere attracts the particle as though the mass of the sphere or shell were concentrated at the centre of the sphere.
10. If a particle is inside a uniform spherical shell, the gravitational force on the particle is zero. If a particle is inside a homogeneous solid sphere, the force on the particle acts toward the centre of the sphere. This force is exerted by the spherical mass interior to the particle.
11. A geostationary (geosynchronous communication) satellite moves in a circular orbit in the equatorial plane at a approximate distance of $4.22 \times 10^4 \text{ km}$ from the earth's centre.

Physical Quantity	Symbol	Dimensions	Unit	Remarks
Gravitational Constant	G	$[M^{-1} L^3 T^{-2}]$	$N m^2 kg^{-2}$	6.67×10^{-11}
Gravitational Potential Energy	$V(r)$	$[M L^2 T^{-2}]$	J	$-\frac{GMm}{r}$ (scalar)
Gravitational Potential	$U(r)$	$[L^2 T^{-2}]$	$J kg^{-1}$	$-\frac{GM}{r}$ (scalar)
Gravitational Intensity	$\mathbf{E}$ or $\mathbf{g}$	$[L T^{-2}]$	$m s^{-2}$	$\frac{GM}{r^2} \hat{r}$ (vector)

POINTS TO PONDER

1. In considering motion of an object under the gravitational influence of another object the following quantities are conserved:
 - (a) Angular momentum
 - (b) Total mechanical energy
 Linear momentum is **not** conserved
2. Angular momentum conservation leads to Kepler's second law. However, it is not special to the inverse square law of gravitation. It holds for any central force.
3. In Kepler's third law (see Eq. (8.1) and $T^2 = K_s R^3$. The constant K_s is the same for all planets in circular orbits. This applies to satellites orbiting the Earth [(Eq. (8.38))].
4. An astronaut experiences weightlessness in a space satellite. This is not because the gravitational force is small at that location in space. It is because both the astronaut and the satellite are in "free fall" towards the Earth.
5. The *gravitational potential energy* associated with two particles separated by a distance r is given by

$$V = -\frac{G m_1 m_2}{r} + \text{constant}$$

The constant can be given any value. The simplest choice is to take it to be zero. With this choice

$$V = -\frac{G m_1 m_2}{r}$$

This choice implies that $V \rightarrow 0$ as $r \rightarrow \infty$. Choosing location of zero of the gravitational energy is the same as choosing the arbitrary constant in the potential energy. Note that the gravitational force is not altered by the choice of this constant.

6. The total mechanical energy of an object is the sum of its kinetic energy (which is always positive) and the potential energy. Relative to infinity (i.e. if we presume that the potential energy of the object at infinity is zero), the gravitational potential energy of an object is negative. The total energy of a satellite is negative.
7. The commonly encountered expression mgh for the potential energy is actually an approximation to the difference in the gravitational potential energy discussed in the point 6, above.
8. Although the gravitational force between two particles is central, the force between two finite rigid bodies is not necessarily along the line joining their centre of mass. For a spherically symmetric body however the force on a particle external to the body is as if the mass is concentrated at the centre and this force is therefore central.
9. The gravitational force on a particle inside a spherical shell is zero. However, (unlike a metallic shell which shields electrical forces) the shell does not shield other bodies outside it from exerting gravitational forces on a particle inside. *Gravitational shielding is not possible.*

EXERCISES**8.1** Answer the following :

- (a) You can shield a charge from electrical forces by putting it inside a hollow conductor. Can you shield a body from the gravitational influence of nearby matter by putting it inside a hollow sphere or by some other means ?
- (b) An astronaut inside a small space ship orbiting around the earth cannot detect gravity. If the space station orbiting around the earth has a large size, can he hope to detect gravity ?
- (c) If you compare the gravitational force on the earth due to the sun to that due to the moon, you would find that the Sun's pull is greater than the moon's pull. (you can check this yourself using the data available in the succeeding exercises). However, the tidal effect of the moon's pull is greater than the tidal effect of sun. Why ?

- 8.2** Choose the correct alternative :
- Acceleration due to gravity increases/decreases with increasing altitude.
 - Acceleration due to gravity increases/decreases with increasing depth (assume the earth to be a sphere of uniform density).
 - Acceleration due to gravity is independent of mass of the earth/mass of the body.
 - The formula $-GMm(1/r_2 - 1/r_1)$ is more/less accurate than the formula $mg(r_2 - r_1)$ for the difference of potential energy between two points r_2 and r_1 distance away from the centre of the earth.
- 8.3** Suppose there existed a planet that went around the sun twice as fast as the earth. What would be its orbital size as compared to that of the earth ?
- 8.4** Io, one of the satellites of Jupiter, has an orbital period of 1.769 days and the radius of the orbit is 4.22×10^8 m. Show that the mass of Jupiter is about one-thousandth that of the sun.
- 8.5** Let us assume that our galaxy consists of 2.5×10^{11} stars each of one solar mass. How long will a star at a distance of 50,000 ly from the galactic centre take to complete one revolution ? Take the diameter of the Milky Way to be 10^5 ly.
- 8.6** Choose the correct alternative:
- If the zero of potential energy is at infinity, the total energy of an orbiting satellite is negative of its kinetic/potential energy.
 - The energy required to launch an orbiting satellite out of earth's gravitational influence is more/less than the energy required to project a stationary object at the same height (as the satellite) out of earth's influence.
- 8.7** Does the escape speed of a body from the earth depend on (a) the mass of the body, (b) the location from where it is projected, (c) the direction of projection, (d) the height of the location from where the body is launched?
- 8.8** A comet orbits the sun in a highly elliptical orbit. Does the comet have a constant (a) linear speed, (b) angular speed, (c) angular momentum, (d) kinetic energy, (e) potential energy, (f) total energy throughout its orbit? Neglect any mass loss of the comet when it comes very close to the Sun.
- 8.9** Which of the following symptoms is likely to afflict an astronaut in space (a) swollen feet, (b) swollen face, (c) headache, (d) orientational problem.
- 8.10** In the following two exercises, choose the correct answer from among the given ones: The gravitational intensity at the centre of a hemispherical shell of uniform mass density has the direction indicated by the arrow (see Fig 8.12) (i) a, (ii) b, (iii) c, (iv) O.

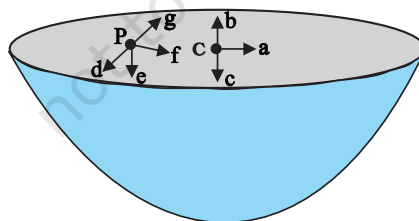


Fig. 8.12

- 8.11** For the above problem, the direction of the gravitational intensity at an arbitrary point P is indicated by the arrow (i) d, (ii) e, (iii) f, (iv) g.
- 8.12** A rocket is fired from the earth towards the sun. At what distance from the earth's centre is the gravitational force on the rocket zero ? Mass of the sun = 2×10^{30} kg, mass of the earth = 6×10^{24} kg. Neglect the effect of other planets etc. (orbital radius = 1.5×10^{11} m).
- 8.13** How will you 'weigh the sun', that is estimate its mass? The mean orbital radius of the earth around the sun is 1.5×10^8 km.
- 8.14** A saturn year is 29.5 times the earth year. How far is the saturn from the sun if the earth is 1.50×10^8 km away from the sun ?
- 8.15** A body weighs 63 N on the surface of the earth. What is the gravitational force on it due to the earth at a height equal to half the radius of the earth ?
- 8.16** Assuming the earth to be a sphere of uniform mass density, how much would a body

- weigh half way down to the centre of the earth if it weighed 250 N on the surface ?
- 8.17** A rocket is fired vertically with a speed of 5 km s^{-1} from the earth's surface. How far from the earth does the rocket go before returning to the earth ? Mass of the earth = $6.0 \times 10^{24} \text{ kg}$; mean radius of the earth = $6.4 \times 10^6 \text{ m}$; $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$.
- 8.18** The escape speed of a projectile on the earth's surface is 11.2 km s^{-1} . A body is projected out with thrice this speed. What is the speed of the body far away from the earth? Ignore the presence of the sun and other planets.
- 8.19** A satellite orbits the earth at a height of 400 km above the surface. How much energy must be expended to rocket the satellite out of the earth's gravitational influence? Mass of the satellite = 200 kg; mass of the earth = $6.0 \times 10^{24} \text{ kg}$; radius of the earth = $6.4 \times 10^6 \text{ m}$; $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$.
- 8.20** Two stars each of one solar mass ($= 2 \times 10^{30} \text{ kg}$) are approaching each other for a head on collision. When they are at a distance 10^9 km , their speeds are negligible. What is the speed with which they collide ? The radius of each star is 10^4 km . Assume the stars to remain undistorted until they collide. (Use the known value of G).
- 8.21** Two heavy spheres each of mass 100 kg and radius 0.10 m are placed 1.0 m apart on a horizontal table. What is the gravitational force and potential at the mid point of the line joining the centres of the spheres ? Is an object placed at that point in equilibrium? If so, is the equilibrium stable or unstable ?

Additional Exercises

- 8.22** As you have learnt in the text, a geostationary satellite orbits the earth at a height of nearly 36,000 km from the surface of the earth. What is the potential due to earth's gravity at the site of this satellite ? (Take the potential energy at infinity to be zero). Mass of the earth = $6.0 \times 10^{24} \text{ kg}$, radius = 6400 km.
- 8.23** A star 2.5 times the mass of the sun and collapsed to a size of 12 km rotates with a speed of 1.2 rev. per second. (Extremely compact stars of this kind are known as neutron stars. Certain stellar objects called pulsars belong to this category). Will an object placed on its equator remain stuck to its surface due to gravity ? (mass of the sun = $2 \times 10^{30} \text{ kg}$).
- 8.24** A spaceship is stationed on Mars. How much energy must be expended on the spaceship to launch it out of the solar system ? Mass of the space ship = 1000 kg; mass of the sun = $2 \times 10^{30} \text{ kg}$; mass of mars = $6.4 \times 10^{23} \text{ kg}$; radius of mars = 3395 km; radius of the orbit of mars = $2.28 \times 10^8 \text{ km}$; $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$.
- 8.25** A rocket is fired 'vertically' from the surface of mars with a speed of 2 km s^{-1} . If 20% of its initial energy is lost due to martian atmospheric resistance, how far will the rocket go from the surface of mars before returning to it ? Mass of mars = $6.4 \times 10^{23} \text{ kg}$; radius of mars = 3395 km; $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$.

APPENDIX 8.1 : LIST OF INDIAN SATELLITES

S.No.	Name	Launch Date	Launch Vehicle	Application
1.	Aryabhata	Apr. 19, 1975	C-1 Intercosmos ^a	Experimental
2.	Bhaskara-I	Jun. 07, 1979	C-1 Intercosmos ^a	Earth Observation, Experimental
3.	Rohini Technology Payload (RTP)	Aug. 10, 1979	SLV-3E1 ^b	Experimental
4.	Rohini Satellite RS-1	Jul. 18, 1980	SLV-3E2 ^b	Experimental
5.	Rohini Satellite RS-D1	May 31, 1981	SLV-3D1 ^b	Earth Observation
6.	APPLE	Jun. 19, 1981	Ariane -1(V-3) ^c	Communication, Experimental
7.	Bhaskara-II	Nov. 20, 1981	C-1 Intercosmos ^a	Earth Observation, Experimental
8.	INSAT-1A	Apr. 10, 1982	Delta ^d	Communication
9.	Rohini Satellite RS-D2	Apr. 17, 1983	SLV-3 ^b	Earth Observation
10.	INSAT-1B	Aug. 30, 1983	Shuttle [PAM-D] ^d	Communication
11.	SROSS-1	Mar. 24, 1987	ASLV-D1 ^b	Experimental
12.	IRS-1A	Mar. 17, 1988	Vostok ^e	Earth Observation
13.	SROSS-2	Jul. 13, 1988	ASLV-D2 ^b	Earth Observation, Experimental
14.	INSAT-1C	Jul. 22, 1988	Ariane-3 ^c	Communication
15.	INSAT-1D	Jun. 12, 1990	Delta 4925 ^d	Communication
16.	IRS-1B	Aug. 29, 1991	Vostok ^e	Earth Observation
17.	SROSS-C	May 20, 1992	ASLV-D3 ^b	Experimental
18.	INSAT-2A	Jul. 10, 1992	Ariane-44L H10 ^c	Communication
19.	INSAT-2B	Jul. 23, 1993	Ariane-44L H10 ^{+c}	Communication
20.	IRS-1E	Sep. 20, 1993	PSLV-D1 ^b	Earth Observation
21.	SROSS-C2	May 04, 1994	ASLV-D4 ^b	Experimental
22.	IRS-P2	Oct. 15, 1994	PSLV-D2 ^b	Earth Observation
23.	INSAT-2C	Dec. 07, 1995	Ariane-44L H10-3 ^c	Communication
24.	IRS-1C	Dec. 28, 1995	Molniya ^e	Earth Observation
25.	IRS-P3	Mar. 21, 1996	PSLV-D3/IRS-P3 ^b	Earth Observation
26.	INSAT-2D	Jun. 04, 1997	Ariane-44L H10-3 ^c	Communication
27.	IRS-1D	Sep. 29, 1997	PSLV-C1/IRS-1D ^b	Earth Observation
28.	INSAT-2E	Apr. 03, 1999	Ariane-42P H10-3 ^c	Communication
29.	Oceansat (IRS-P4)	May 26, 1999	PSLV-C2/IRS-P4 ^b	Earth Observation
30.	INSAT-3B	Mar. 22, 2000	Ariane-5G ^c	Communication
31.	GSAT-1	Apr. 18, 2001	GSLV-D1/GSAT-1 ^b	Communication
32.	The Technology Experiment Satellite (TES)	Oct. 22, 2001	PSLV-C3/TES ^b	Earth Observation
33.	INSAT-3C	Jan. 24, 2002	Ariane5-V147 ^c	Climate & Environment, Communication

34.	KALPANA-1	Sep. 12, 2002	PSLV-C4/ KALPANA-1 ^b	Climate & Environment, Communication
35.	INSAT-3A	Apr. 10, 2003	Ariane5-V160 ^c	Climate & Environment, Communication
36.	GSAT-2	May 08, 2003	GSLV-D2/GSAT-2 ^b	Communication
37.	INSAT-3E	Sep. 28, 2003	Ariane5-V162 ^c	Communication
38.	IRS-P6 / RESOURCESAT-1	Oct. 17, 2003	PSLV-C5/ RESOURCESAT-1 ^b	Earth Observation
39.	EDUSAT	Sep. 20, 2004	GSLV-F01/ EDUSAT(GSAT-3) ^b	Communication
40.	HAMSAT	May 05, 2005	PSLV-C6/ CARTOSAT-1/HAMSAT ^b	Communication
41.	CARTOSAT-1	May 05, 2005	PSLV-C6/ CARTOSAT-1/HAMSAT ^b	Earth Observation
42.	INSAT-4A	Dec. 22, 2005	Ariane5-V169c	Communication
43.	INSAT-4C	Jul. 10, 2006	GSLV-F02/INSAT-4C ^b	Communication
44.	CARTOSAT-2	Jan. 10, 2007	PSLV-C7/CARTOSAT-2 /SRE-1 ^b	Earth Observation
45.	SRE-1	Jan. 10, 2007	PSLV-C7/CARTOSAT-2 /SRE-1 ^b	Experimental
46.	INSAT-4B	Mar. 12, 2007	Ariane5 ^c	Communication
47.	INSAT-4CR	Sep. 02, 2007	GSLV-F04/INSAT-4 CR ^b	Communication
48.	IMS-1	Apr. 28, 2008	PSLV-C9/ CARTOSAT-2A ^b	Earth Observation
49.	CARTOSAT - 2A	Apr. 28, 2008	PSLV-C9/ CARTOSAT-2A ^b	Earth Observation
50.	Chandrayaan-1	Oct. 22, 2008	PSLV-C11 ^b	Planetary Observation
51.	RISAT-2	Apr. 20, 2009	PSLV-C12/RISAT-2 ^b	Earth Observation
52.	ANUSAT	Apr. 20, 2009	PSLV-C12/RISAT-2 ^b	University/ Academic Institute
53.	Oceansat-2	Sep. 23, 2009	PSLV-C14/ OCEANSAT-2 ^b	Climate & Environment, Earth Observation
54.	GSAT-4	Apr. 15, 2010	GSLV-D3 / GSAT-4 ^b	Communication
55.	CARTOSAT-2B	Jul. 12, 2010	PSLV-C15/ CARTOSAT-2B ^b	Earth Observation
56.	STUDSAT	Jul. 12, 2010	PSLV-C15/ CARTOSAT-2B ^b	University/ Academic Institute
57.	GSAT-5P	Dec. 25, 2010	GSLV-F06/GSAT-5P ^b	Communication
58.	RESOURCESAT-2	Apr. 20, 2011	PSLV-C16/ RESOURCESAT-2 ^b	Earth Observation
59.	YOUTHSAT	Apr. 20, 2011	PSLV-C16/ RESOURCESAT-2 ^b	Student Satellite
60.	GSAT-8	May 21, 2011	Ariane-5 VA-202 ^c	Communication

61.	GSAT-12	Jul. 15, 2011	PSLV-C17/GSAT-12 ^b	Communication
62.	Megha-Tropiques	Oct. 12, 2011	PSLV-C18/Megha-Tropiques ^b	Climate & Environment, Earth Observation
63.	SRMSat	Oct. 12, 2011	PSLV-C18/Megha-Tropiques ^b	University/Academic Institute
64.	Jugnu	Oct. 12, 2011	PSLV-C18/Megha-Tropiques ^b	University/Academic Institute
65.	RISAT-1	Apr. 26, 2012	PSLV-C19/RISAT-1 ^b	Earth Observation
66.	GSAT-10	Sep. 29, 2012	Ariane-5 VA-209 ^c	Communication, Navigation
67.	SARAL	Feb. 25, 2013	PSLV-C20/SARAL ^b	Climate & Environment, Earth Observation
68.	IRNSS-1A	Jul. 01, 2013	PSLV-C22/IRNSS-1A ^b	Navigation
69.	INSAT-3D	Jul. 26, 2013	Ariane-5 VA-214 ^c	Climate & Environment, Disaster Management System
70.	GSAT-7	Aug. 30, 2013	Ariane-5 VA-215 ^c	Communication
71.	Mars Orbiter Mission Spacecraft (Mangalyaan-1)	Nov. 05, 2013	PSLV-C25 ^b	Planetary Observation
72.	GSAT-14	Jan. 05, 2014	GSLV-D5/GSAT-14 ^b	Communication
73.	IRNSS-1B	Apr. 04, 2014	PSLV-C24/IRNSS-1B ^b	Navigation
74.	IRNSS-1C	Oct. 16, 2014	PSLV-C26/IRNSS-1C ^b	Navigation
75.	GSAT-16	Dec. 07, 2014	Ariane-5 VA-221 ^c	Communication
76.	Crew module Atmospheric Reentry Experiment	Dec. 18, 2014	LVM-3/CARE Mission ^b	Experimental
77.	IRNSS-1D	Mar. 28, 2015	PSLV-C27/IRNSS-1D ^b	Navigation
78.	GSAT-6 (INSAT-4E)	Aug. 27, 2015	GSLV-D6 ^b	Communication
79.	Astrosat	Sep. 28, 2015	PSLV-C30 ^b	Space Sciences
80.	GSAT-15	Nov. 11, 2015	Ariane-5 VA-227 ^c	Communication, Navigation
81.	IRNSS-1E	Jan. 20, 2016	PSLV-C31/IRNSS-1E ^b	Navigation
82.	IRNSS-1F	Mar. 10, 2016	PSLV-C32/IRNSS-1F ^b	Navigation
83.	IRNSS-1G	Apr. 28, 2016	PSLV-C33/IRNSS-1G ^b	Navigation
84.	Cartosat-2 Series Satellite	Jun. 22, 2016	PSLV-C34/CARTOSAT-2 Series Satellite ^b	Earth Observation
85.	SathyabamaSat	Jun. 22, 2016	PSLV-C34/CARTOSAT-2 Series Satellite ^b	University/Academic Institute
86.	Swayam	Jun. 22, 2016	PSLV-C34/CARTOSAT-2 Series Satellite ^b	University/Academic Institute

87.	INSAT-3DR	Sep. 08, 2016	GSLV-F05/ INSAT-3DR ^b	Climate & Environment, Disaster Management System
88.	ScatSat-1	Sep. 26, 2016	PSLV-C35/ SCATSAT-1 ^b	Climate & Environment
89.	Pratham	Sep. 26, 2016	PSLV-C35/ SCATSAT-1 ^b	University/ Academic Institute
90.	PiSat	Sep. 26, 2016	PSLV-C35/ SCATSAT-1 ^b	University/ Academic Institute
91.	GSAT-18	Oct. 06, 2016	Ariane-5 VA-231 ^c	Communication
92.	ResourceSat-2A	Dec. 07, 2016	PSLV-C36/ RESOURCESAT-2A ^b	Earth Observation
93.	Cartosat -2 Series Satellite	Feb. 15, 2017	PSLV-C37/Cartosat -2 Series Satellite ^b	Earth Observation
94.	INS-1A	Feb. 15, 2017	PSLV-C37/Cartosat -2 Series Satellite ^b	Experimental
95.	INS-1B	Feb. 15, 2017	PSLV-C37/Cartosat -2 Series Satellite ^b	Experimental
96.	GSAT-9	May 05, 2017	GSLV-F09/GSAT-9 ^b	Communication
97.	GSAT-19	Jun. 05, 2017	GSLV Mk III-D1/ GSAT-19 Mission ^b	Communication
98.	Cartosat-2 Series Satellite	Jun. 23, 2017	PSLV-C38/Cartosat-2 Series Satellite ^b	Earth Observation
99.	NIUSAT	Jun. 23, 2017	PSLV-C38/Cartosat-2 Series Satellite ^b	University/ Academic Institute
100.	GSAT-17	Jun. 29, 2017	Ariane-5 VA-238 ^c	Communication
101.	IRNSS-1H	Aug. 31, 2017	PSLV-C39/IRNSS- 1H Mission ^b	Navigation
102.	INS-1C	Jan. 12, 2018	PSLV-C40/Cartosat-2 Series Satellite Mission ^b	Experimental
103.	Mircosat	Jan. 12, 2018	PSLV-C40/Cartosat-2 Series Satellite Mission ^b	Experimental
104.	Cartosat-2 Series Satellite	Jan. 12, 2018	PSLV-C40/Cartosat-2 Series Satellite Mission ^b	Earth Observation
105.	GSAT-6A	Mar. 29, 2018	GSLV-F08/GSAT-6A Mission ^b	Communication
106.	IRNSS-1I	Apr. 12, 2018	PSLV-C41/IRNSS-1I ^b	Navigation

India has so far also launched 239 foreign satellites of 28 countries from Satish Dhawan Space Center, Sriharikota, Andhra Pradesh: May 26, 1999 (02); Oct. 22, 2001 (02); Jan. 10, 2007 (02); Apr. 23, 2007 (01); Jan. 21, 2008 (01); Apr. 28, 2008 (08); Sep. 23, 2009 (06); July 12, 2010 (03); Jan. 12, 2011 (01); Apr. 20, 2011 (01) Sep. 09, 2012 (02); Feb. 25, 2013 (06); June 30, 2014 (05); July 10, 2015 (05); Sep. 28, 2015 (06); Dec. 16, 2015 (06); June 22, 2016 (17); Sep. 26, 2016 (05); Feb. 15, 2017 (101) and thus setting a world record; June 23, 2017 (29). Jan 12, 2018 (28); Sep. 16, 2018 (02).

Details can be seen at www.isro.gov.in.

- a Launched from Kapustin Yar Missile and Space Complex, Soviet Union (now Russia)
- b Launched from Satish Dhawan Space Centre, Sriharikota, Andhra Pradesh
- c Launched from Centre Spatial Guyanais, Kourou, French Guiana
- d Launched from Air Force Eastern Test Range, Florida
- e Launched from Baikonur Cosmodrome, Kazakhstan

ANSWERS

Chapter 2

- 2.1** (a) 10^{-6} ; (b) 1.5×10^4 ; (c) 5; (d) 11.3, 1.13×10^4 .
- 2.2** (a) 10^7 ; (b) 10^{-16} ; (c) 3.9×10^4 ; (d) 6.67×10^{-8} .
- 2.5** 500
- 2.6** (c)
- 2.7** 0.035 mm
- 2.9** 94.1
- 2.10** (a) 1; (b) 3; (c) 4; (d) 4; (e) 4; (f) 4.
- 2.11** 8.72 m^2 ; 0.0855 m^3
- 2.12** (a) 2.3 kg; (b) 0.02 g
- 2.13** 13%; 3.8
- 2.14** (b) and (c) are wrong on dimensional grounds. Hint: The argument of a trigonometric function must always be dimensionless.
- 2.15** The correct formula is $m = m_0(1 - v^2/c^2)^{-1/2}$
- 2.16** $\cong 3 \times 10^{-7} \text{ m}^3$
- 2.17** $\cong 10^4$; intermolecular separation in a gas is much larger than the size of a molecule.
- 2.18** Near objects make greater angle than distant (far off) objects at the eye of the observer. When you are moving, the angular change is less for distant objects than nearer objects. So, these distant objects seem to move along with you, but the nearer objects in opposite direction.
- 2.19** $\cong 3 \times 10^{16} \text{ m}$; as a unit of length 1 parsec is defined to be equal to $3.084 \times 10^{16} \text{ m}$.
- 2.20** 1.32 parsec; $2.64''$ (second of arc)
- 2.23** $1.4 \times 10^3 \text{ kg m}^{-3}$; the mass density of the Sun is in the range of densities of liquids / solids and *not* gases. This high density arises due to inward gravitational attraction on outer layers due to inner layers of the Sun.
- 2.24** $1.429 \times 10^5 \text{ km}$

- 2.25** Hint: $\tan \theta$ must be dimensionless. The correct formula is $\tan \theta = v/v'$ where v' is the speed of rainfall.
- 2.26** Accuracy of 1 part in 10^{11} to 10^{12}
- 2.27** $\cong 0.7 \times 10^3 \text{ kg m}^{-3}$. In the solid phase atoms are tightly packed, so the atomic mass density is close to the mass density of the solid.
- 2.28** $\cong 0.3 \times 10^{18} \text{ kg m}^{-3}$ – Nuclear density is typically 10^{15} times atomic density of matter.
- 2.29** $3.84 \times 10^8 \text{ m}$
- 2.30** 55.8 km
- 2.31** $2.8 \times 10^{22} \text{ km}$
- 2.32** 3,581 km
- 2.33** Hint: the quantity $e^4 / (16 \pi^2 \epsilon_0^2 m_p m_e^2 c^3 G)$ has the dimension of time.

Chapter 3

- 3.1** (a), (b)
- 3.2** (a) A....B, (b) A....B, (c) B....A, (d) Same, (e) B....A....once.
- 3.4** 37 s
- 3.5** 1000 km/h
- 3.6** 3.06 m s^{-2} ; 11.4 s
- 3.7** 1250 m (Hint: view the motion of B relative to A)
- 3.8** 1 m s^{-2} (Hint: view the motion of B and C relative to A)
- 3.9** $T = 9 \text{ min}$, speed = 40 km/h. Hint: $vT / (v - 20) = 18$; $vT / (v + 20) = 6$
- 3.10** (a) Vertically downwards; (b) zero velocity, acceleration of 9.8 m s^{-2} downwards; (c) $x > 0$ (upward and downward motion); $v < 0$ (upward), $v > 0$ (downward), $a > 0$ throughout; (d) 44.1 m, 6 s.
- 3.11** (a) True; (b) False; (c) True (if the particle rebounds instantly with the same speed, it implies infinite acceleration which is unphysical); (d) False (true only when the chosen positive direction is along the direction of motion)
- 3.14** (a) 5 km h^{-1} , 5 km h^{-1} ; (b) 0, 6 km h^{-1} ; (c) $\frac{15}{8} \text{ km h}^{-1}$, $\frac{45}{8} \text{ km h}^{-1}$
- 3.15** Because, for an arbitrarily small interval of time, the magnitude of displacement is equal to the length of the path.
- 3.16** All the four graphs are impossible. (a) a particle cannot have two different positions at the same time; (b) a particle cannot have velocity in opposite directions at the same time; (c) speed is always non-negative; (d) total path length of a particle can never decrease with time. (Note, the arrows on the graphs are meaningless).
- 3.17** No, wrong. $x-t$ plot does not show the trajectory of a particle. Context: A body is dropped from a tower ($x = 0$) at $t = 0$.
- 3.18** 105 m s^{-1}

- 3.19** (a) A ball at rest on a smooth floor is kicked, it rebounds from a wall with reduced speed and moves to the opposite wall which stops it; (b) A ball thrown up with some initial velocity rebounding from the floor with reduced speed after each hit; (c) A uniformly moving cricket ball turned back by hitting it with a bat for a very short time-interval.
- 3.20** $x < 0, v < 0, a > 0$; $x > 0, v > 0, a < 0$; $x < 0, v > 0, a > 0$.
- 3.21** Greatest in 3, least in 2; $v > 0$ in 1 and 2, $v < 0$ in 3.
- 3.22** Acceleration magnitude greatest in 2; speed greatest in 3; $v > 0$ in 1, 2 and 3; $a > 0$ in 1 and 3, $a < 0$ in 2; $a = 0$ at A, B, C, D.
- 3.23** A straight line inclined with the time-axis for uniformly accelerated motion; parallel to the time-axis for uniform motion.
- 3.24** 10 s, 10 s
- 3.25** (a) 13 km h^{-1} ; (b) 5 km h^{-1} ; (c) 20 s in either direction, viewed by any one of the parents, the speed of the child is 9 km h^{-1} in either direction; answer to (c) is unaltered.
- 3.26** $x_2 - x_1 = 15 t$ (linear part); $x_2 - x_1 = 200 + 30 t - 5 t^2$ (curved part).
- 3.27** (a) 60 m, 6 m s^{-1} ; (b) 36 m, 9 m s^{-1}
- 3.28** (c), (d), (f)

Chapter 4

- 4.1** Volume, mass, speed, density, number of moles, angular frequency are scalars; the rest are vectors.
- 4.2** Work, current
- 4.3** Impulse
- 4.4** Only (c) and (d) are permissible
- 4.5** (a) T, (b) F, (c) F, (d) T, (e) T
- 4.6** Hint: The sum (difference) of any two sides of a triangle is never less (greater) than the third side. Equality holds for collinear vectors.
- 4.7** All statements except (a) are correct
- 4.8** 400 m for each; B
- 4.9** (a) O; (b) O; (c) 21.4 km h^{-1}
- 4.10** Displacement of magnitude 1 km and direction 60° with the initial direction; total path length = 1.5 km (third turn); null displacement vector; path length = 3 km (sixth turn); 866 m, 30° , 4 km (eighth turn)
- 4.11** (a) 49.3 km h^{-1} ; (b) 21.4 km h^{-1} . No, the average speed equals average velocity magnitude only for a straight path.
- 4.12** About 18° with the vertical, towards the south.
- 4.13** 15 min, 750 m
- 4.14** East (approximately)
- 4.15** 150.5 m
- 4.16** 50 m

- 4.17** 9.9 m s^{-2} , along the radius at every point towards the centre.
- 4.18** 6.4 g
- 4.19** (a) False (true only for uniform circular motion)
(b) True, (c) True.
- 4.20** (a) $\mathbf{v}(t) = (3.0 \hat{\mathbf{i}} - 4.0t \hat{\mathbf{j}})$ $\hat{\mathbf{a}}(t) = -4.0 \hat{\mathbf{j}}$
(b) 8.54 m s^{-1} , 70° with x -axis.
- 4.21** (a) 2 s , 24 m , 21.26 m s^{-1}
- 4.22** $\sqrt{2}$, 45° with the x -axis; $\sqrt{2}$, -45° with the x -axis, $(5/\sqrt{2}, -1/\sqrt{2})$.
- 4.23** (b) and (e)
- 4.24** Only (e) is true
- 4.25** 182 m s^{-1}
- 4.27** No. Rotations in *general* cannot be associated with vectors
- 4.28** A vector can be associated with a plane area
- 4.29** No
- 4.30** At an angle of $\sin^{-1}(1/3) = 19.5^\circ$ with the vertical; 16 km .
- 4.31** 0.86 m s^{-2} , 54.5° with the direction of velocity

Chapter 5

- 5.1** (a) to (d) No net force according to the First Law
(e) No force, since it is far away from all material agencies producing electromagnetic and gravitational forces.
- 5.2** The only force in each case is the force of gravity, (neglecting effects of air) equal to 0.5 N vertically downward. The answers do not change, even if the motion of the pebble is not along the vertical. The pebble is not at rest at the highest point. It has a constant horizontal component of velocity throughout its motion.
- 5.3** (a) 1 N vertically downwards (b) same as in (a)
(c) same as in (a); force at an instant depends on the situation at that instant, not on history.
(d) 0.1 N in the direction of motion of the train.
- 5.4** (i) T
- 5.5** $a = -2.5 \text{ m s}^{-2}$. Using $v = u + at$, $0 = 15 - 2.5t$ i.e., $t = 6.0 \text{ s}$
- 5.6** $a = 1.5/25 = 0.06 \text{ m s}^{-2}$
 $F = 3 \times 0.06 = 0.18 \text{ N}$ in the direction of motion.
- 5.7** Resultant force = 10 N at an angle of $\tan^{-1}(3/4) = 37^\circ$ with the direction of 8 N force.
Acceleration = 2 m s^{-2} in the direction of the resultant force.
- 5.8** $a = -2.5 \text{ m s}^{-2}$, Retarding force = $465 \times 2.5 = 1.2 \times 10^3 \text{ N}$
- 5.9** $F - 20,000 \times 10 = 20000 \times 5.0$, i.e., $F = 3.0 \times 10^5 \text{ N}$
- 5.10** $a = -20 \text{ m s}^{-2}$ $0 \leq t \leq 30 \text{ s}$

$$t = -5 \text{ s} : x = u t = -10 \times 5 = -50 \text{ m}$$

$$t = 25 \text{ s} : x = u t + (\frac{1}{2}) a t^2 = (10 \times 25 - 10 \times 625) \text{ m} = -6 \text{ km}$$

$t = 100 \text{ s}$: First consider motion up to 30 s

$$x_1 = 10 \times 30 - 10 \times 900 = -8700 \text{ m}$$

$$\text{At } t = 30 \text{ s, } v = 10 - 20 \times 30 = -590 \text{ m s}^{-1}$$

$$\text{For motion from 30 s to 100 s : } x_2 = -590 \times 70 = -41300 \text{ m}$$

$$x = x_1 + x_2 = -50 \text{ km}$$

- 5.11** (a) Velocity of car (at $t = 10 \text{ s}$) $= 0 + 2 \times 10 = 20 \text{ m s}^{-1}$

By the First Law, the horizontal component of velocity is 20 m s^{-1} throughout.

Vertical component of velocity (at $t = 11 \text{ s}$) $= 0 + 10 \times 1 = 10 \text{ m s}^{-1}$

Velocity of stone (at $t = 11 \text{ s}$) $= \sqrt{20^2 + 10^2} = \sqrt{500} = 22.4 \text{ m s}^{-1}$ at an angle of $\tan^{-1}(\frac{1}{2})$ with the horizontal.

(b) 10 m s^{-2} vertically downwards.

- 5.12** (a) At the extreme position, the speed of the bob is zero. If the string is cut, it will fall vertically downwards.

(b) At the mean position, the bob has a horizontal velocity. If the string is cut, it will fall along a parabolic path.

- 5.13** The reading on the scale is a measure of the force on the floor by the man. By the Third Law, this is equal and opposite to the normal force N on the man by the floor.

(a) $N = 70 \times 10 = 700 \text{ N}$; Reading is 70 kg

(b) $70 \times 10 - N = 70 \times 5$; Reading is 35 kg

(c) $N - 70 \times 10 = 70 \times 5$; Reading is 105 kg

(d) $70 \times 10 - N = 70 \times 10$; Reading would be zero; the scale would read zero.

- 5.14** (a) In all the three intervals, acceleration and, therefore, force are zero.

(b) 3 kg m s^{-1} at $t = 0$; (c) -3 kg m s^{-1} at $t = 4 \text{ s}$.

- 5.15** If the 20 kg mass is pulled,

$$600 - T = 20 a, \quad T = 10 a$$

$$a = 20 \text{ m s}^{-2}, \quad T = 200 \text{ N}$$

If the 10 kg mass is pulled, $a = 20 \text{ m s}^{-2}$, $T = 400 \text{ N}$

- 5.16** $T - 8 \times 10 = 8 a, 12 \times 10 - T = 12 a$

i.e. $a = 2 \text{ m s}^{-2}$, $T = 96 \text{ N}$

- 5.17** By momentum conservation principle, total final momentum is zero. Two momentum vectors cannot sum to a null momentum unless they are equal and opposite.

- 5.18** Impulse on each ball $= 0.05 \times 12 = 0.6 \text{ kg m s}^{-1}$ in magnitude. The two impulses are opposite in direction.

- 5.19** Use momentum conservation : $100 v = 0.02 \times 80$

$$v = 0.016 \text{ m s}^{-1} = 1.6 \text{ cm s}^{-1}$$

- 5.20** Impulse is directed along the bisector of the initial and final directions. Its magnitude is $0.15 \times 2 \times 15 \times \cos 22.5^\circ = 4.2 \text{ kg m s}^{-1}$

- 5.21** $v = 2\pi \times 1.5 \times \frac{40}{60} = 2\pi \text{ m s}^{-1}$

$$T = \frac{mv^2}{R} = \frac{0.25 \times 4\pi^2}{1.5} = 6.6 \text{ N}$$

$$200 = \frac{mv_{\max}^2}{R}, \text{ which gives } v_{\max} = 35 \text{ m s}^{-1}$$

5.22 Alternative (b) is correct, according to the First Law

5.23 (a) The horse-cart system has no external force in empty space. The mutual forces between the horse and the cart cancel (Third Law). On the ground, the contact force between the system and the ground (friction) causes their motion from rest.

(b) Due to inertia of the body not directly in contact with the seat.

(c) A lawn mower is pulled or pushed by applying force at an angle. When you push, the normal force (N) must be more than its weight, for equilibrium in the vertical direction. This results in greater friction ($f \propto N$) and, therefore, a greater applied force to move. Just the opposite happens while pulling.

(d) To reduce the rate of change of momentum and hence to reduce the force necessary to stop the ball.

5.24 A body with a constant speed of 1 cm s^{-1} receives impulse of magnitude $0.04 \text{ kg} \times 0.02 \text{ m s}^{-1} = 8 \times 10^{-4} \text{ kg m s}^{-1}$ after every 2 s from the walls at $x = 0$ and $x = 2 \text{ cm}$.

5.25 Net force $= 65 \text{ kg} \times 1 \text{ m s}^{-2} = 65 \text{ N}$

$$a_{\max} = \mu_s g = 2 \text{ m s}^{-2}$$

5.26 Alternative (a) is correct. Note $mg + T_2 = mv_2^2/R$; $T_1 - mg = mv_1^2/R$

The moral is : do not confuse the actual material forces on a body (tension, gravitational force, etc) with the effects they produce : centripetal acceleration v_2^2/R or v_1^2/R in this example.

5.27 (a) 'Free body' : crew and passengers

Force on the system by the floor $= F$ upwards; weight of system $= mg$ downwards;

$$\therefore F - mg = ma$$

$$F - 300 \times 10 = 300 \times 15$$

$$F = 7.5 \times 10^3 \text{ N upward}$$

By the Third Law, force on the floor by the crew and passengers $= 7.5 \times 10^3 \text{ N}$ downwards.

(b) 'Free body' : helicopter plus the crew and passengers

Force by air on the system $= R$ upwards; weight of system $= mg$ downwards

$$\therefore R - mg = ma$$

$$R - 1300 \times 10 = 1300 \times 15$$

$$R = 3.25 \times 10^4 \text{ N upwards}$$

By the Third Law, force (action) on the air by the helicopter $= 3.25 \times 10^4 \text{ N}$ downwards.

(c) $3.25 \times 10^4 \text{ N}$ upwards

5.28 Mass of water hitting the wall per second

$$= 10^3 \text{ kg m}^{-3} \times 10^{-2} \text{ m}^2 \times 15 \text{ m s}^{-1} = 150 \text{ kg s}^{-1}$$

$$\text{Force by the wall} = \text{momentum loss of water per second} = 150 \text{ kg s}^{-1} \times 15 \text{ m s}^{-1} = 2.25 \times 10^3 \text{ N}$$

5.29 (a) $3 mg$ (down) (b) $3 mg$ (down) (c) $4 mg$ (up)

5.30 If N is the normal force on the wings,

$$N \cos \theta = mg, \quad N \sin \theta = \frac{mv^2}{R}$$

$$\text{which give } R = \frac{v^2}{g \tan \theta} = \frac{200 \times 200}{10 \times \tan 15^\circ} = 15 \text{ km}$$

- 5.31** The centripetal force is provided by the lateral thrust by the rail on the flanges of the wheels. By the Third Law, the train exerts an equal and opposite thrust on the rail causing its wear and tear.

$$\text{Angle of banking} = \tan^{-1} \left(\frac{v^2}{Rg} \right) = \tan^{-1} \left(\frac{15 \times 15}{30 \times 10} \right) \approx 37^\circ$$

- 5.32** Consider the forces on the man in equilibrium : his weight, force due to the rope and normal force due to the floor.

(a) 750 N (b) 250 N; mode (b) should be adopted.

- 5.33** (a) $T - 400 = 240$, $T = 640$ N

(b) $400 - T = 160$, $T = 240$ N

(c) $T = 400$ N

(d) $T = 0$

The rope will break in case (a).

- 5.34** We assume perfect contact between bodies A and B and the rigid partition. In that case, the self-adjusting normal force on B by the partition (reaction) equals 200 N. There is no impending motion and no friction. The action-reaction forces between A and B are also 200 N. When the partition is removed, kinetic friction comes into play.

$$\text{Acceleration of A + B} = [200 - (150 \times 0.15)] / 15 = 11.8 \text{ m s}^{-2}$$

$$\text{Friction on A} = 0.15 \times 50 = 7.5 \text{ N}$$

$$200 - 7.5 - F_{AB} = 5 \times 11.8$$

$$F_{AB} = 1.3 \times 10^2 \text{ N; opposite to motion.}$$

$$F_{BA} = 1.3 \times 10^2 \text{ N; in the direction of motion.}$$

- 5.35** (a) Maximum frictional force possible for opposing impending relative motion between the block and the trolley = $150 \times 0.18 = 27$ N, which is more than the frictional force of $15 \times 0.5 = 7.5$ N needed to accelerate the box with the trolley. When the trolley moves with uniform velocity, there is no force of friction acting on the block.

(b) For the accelerated (non-inertial) observer, frictional force is opposed by the pseudo-force of the same magnitude, keeping the box at rest relative to the observer. When the trolley moves with uniform velocity there is no pseudo-force for the moving (inertial) observer and no friction.

- 5.36** Acceleration of the box due to friction = $\mu g = 0.15 \times 10 = 1.5 \text{ m s}^{-2}$. But the acceleration of the truck is greater. The acceleration of the box relative to the truck is 0.5 m s^{-2}

towards the rear end. The time taken for the box to fall off the truck = $\sqrt{\frac{2 \times 5}{0.5}} = \sqrt{20} \text{ s}$.

During this time, the truck covers a distance = $\frac{1}{2} \times 2 \times 20 = 20 \text{ m}$.

5.37 For the coin to revolve with the disc, the force of friction should be enough to provide the necessary centripetal force, i.e. $\frac{mv^2}{r} \leq \mu mg$. Now $v = r\omega$, where $\omega = \frac{2\pi}{T}$ is the angular frequency of the disc. For a given μ and ω , the condition is $r \leq \mu g / \omega^2$. The condition is satisfied by the nearer coin (4 cm from the centre).

5.38 At the uppermost point, $N + mg = \frac{mv^2}{R}$, where N is the normal force (downwards) on the motorcyclist by the ceiling of the chamber. The minimum possible speed at the uppermost point corresponds to $N = 0$.

$$\text{i.e. } v_{\min} = \sqrt{Rg} = \sqrt{25 \times 10} = 16 \text{ m s}^{-1}$$

5.39 The horizontal force N by the wall on the man provides the needed centripetal force: $N = mR\omega^2$. The frictional force f (vertically upwards) opposes the weight mg . The man remains stuck to the wall after the floor is removed if $mg = f < \mu N$ i.e. $mg < \mu mR\omega^2$. The minimum angular speed of rotation of the cylinder is $\omega_{\min} = \sqrt{g / \mu R} = 5 \text{ s}^{-1}$

5.40 Consider the free-body diagram of the bead when the radius vector joining the centre of the wire makes an angle θ with the vertical downward direction. We have $mg = N \cos \theta$ and $mR \sin \theta \omega^2 = N \sin \theta$. These equations give $\cos \theta = g / R\omega^2$. Since $\cos \theta \leq 1$,

the bead remains at its lowermost point for $\omega \leq \sqrt{\frac{g}{R}}$.

$$\text{For } \omega = \sqrt{\frac{2g}{R}}, \quad \cos \theta = \frac{1}{2} \quad \text{i.e. } \theta = 60^\circ.$$

Chapter 6

6.1 (a) +ve (b) -ve (c) -ve (d) +ve (e) -ve

6.2 (a) 882 J ; (b) -247 J; (c) 635 J ; (d) 635 J;
Work done by the net force on a body equals change in its kinetic energy.

6.3 (a) $x > a$; 0 (c) $x < a$, $x > b$; $-V_1$
(b) $-\infty < x < \infty$; V_1 (d) $-b/2 < x < -a/2$, $a/2 < x < b/2$; $-V_1$

6.5 (a) rocket; (b) For a conservative force work done over a path is minus of change in potential energy. Over a complete orbit, there is no change in potential energy; (c) K.E. increases, but P.E. decreases, and the sum decreases due to dissipation against friction; (d) in the second case.

6.6 (a) decrease; (b) kinetic energy; (c) external force; (d) total linear momentum, and also total energy (if the system of two bodies is isolated).

6.7 (a) F ; (b) F ; (c) F ; (d) F (true usually but not always, why?)

6.8 (a) No
(b) Yes
(c) Linear momentum is conserved during an inelastic collision, kinetic energy is, of course, not conserved even after the collision is over.
(d) elastic.

6.9 (b) t

- 6.10** (c) $t^{3/2}$
- 6.11** 12 J
- 6.12** The electron is faster, $v_e / v_p = 13.5$
- 6.13** 0.082 J in each half; -0.163 J
- 6.14** Yes, momentum of the molecule + wall system is conserved. The wall has a recoil momentum such that the momentum of the wall + momentum of the outgoing molecule equals momentum of the incoming molecule, assuming the wall to be stationary initially. However, the recoil momentum produces negligible velocity because of the large mass of the wall. Since kinetic energy is also conserved, the collision is elastic.
- 6.15** 43.6 kW
- 6.16** (b)
- 6.17** It transfers its entire momentum to the ball on the table, and does not rise at all.
- 6.18** 5.3 m s^{-1}
- 6.19** 27 km h^{-1} (no change in speed)
- 6.20** 50 J
- 6.21** (a) $m = \rho A v t$ (b) $K = \rho A v^3 t / 2$ (c) $P = 4.5 \text{ kW}$
- 6.22** (a) 49,000 J (b) $6.45 \times 10^{-3} \text{ kg}$
- 6.23** (a) 200 m^2 (b) comparable to the roof of a large house of dimension $14 \text{ m} \times 14 \text{ m}$.
- 6.24** 21.2 cm, 28.5 J
- 6.25** No, the stone on the steep plane reaches the bottom earlier; yes, they reach with the same speed v , [since $mgh = (1/2) m v^2$]
 $v_B = v_C = 14.1 \text{ m s}^{-1}$, $t_B = 2\sqrt{2} \text{ s}$, $t_C = 2\sqrt{2} \text{ s}$
- 6.26** 0.125
- 6.27** 8.82 J for both cases.
- 6.28** The child gives an impulse to the trolley at the start and then runs with a constant relative velocity of 4 m s^{-1} with respect to the trolley's new velocity. Apply momentum conservation for an observer outside. 10.36 m s^{-1} , 25.9 m.
- 6.29** All except (V) are impossible.

Chapter 7

- 7.1** The geometrical centre of each. No, the CM may lie outside the body, as in case of a ring, a hollow sphere, a hollow cylinder, a hollow cube etc.
- 7.2** Located on the line joining H and C1 nuclei at a distance of $1.24 \text{ \AA}$ from the H end.
- 7.3** The speed of the CM of the (trolley + child) system remains unchanged (equal to v) because no external force acts on the system. The forces involved in running on the trolley are internal to this system.
- 7.6** $l_z = x p_y - y p_x$, $l_x = y p_z - z p_y$, $l_y = z p_x - x p_z$
- 7.8** 72 cm
- 7.9** 3675 N on each front wheel, 5145 N on each back wheel.
- 7.10** (a) $7/5 MR^2$ (b) $3/2 MR^2$

- 7.11** Sphere
- 7.12** Kinetic Energy = 3125 J; Angular Momentum = 62.5 J s
- 7.13** (a) 100 rev/min (use angular momentum conservation).
(b) The new kinetic energy is 2.5 times the initial kinetic energy of rotation. The child uses his internal energy to increase his rotational kinetic energy.
- 7.14** 25 s^{-2} ; 10 m s^{-2}
- 7.15** 36 kW
- 7.16** at $R/6$ from the center of original disc opposite to the center of cut portion.
- 7.17** 66.0 g
- 7.18** (a) Yes; (b) Yes, (c) the plane with smaller inclination ($\because a \propto \sin \theta$)
- 7.19** 4J
- 7.20** $6.75 \times 10^{12} \text{ rad s}^{-1}$
- 7.21** (a) 3.8 m (b) 3.0 s
- 7.22** Tension = 98 N, $N_B = 245 \text{ N}$, $N_C = 147 \text{ N}$.
- 7.23** (a) 59 rev/min, (b) No, the K.E. is increased and it comes from work done by man in the process.
- 7.24** 0.625 rad s^{-1}
- 7.27** (a) By angular momentum conservation, the common angular speed

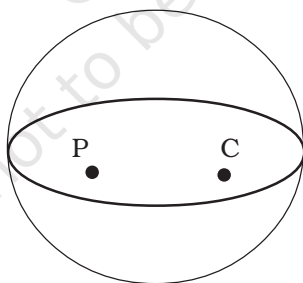
$$\omega = (I_1 \omega_1 + I_2 \omega_2) / (I_1 + I_2)$$

 (b) The loss is due to energy dissipation in frictional contact which brings the two discs to a common angular speed ω . However, since frictional torques are internal to the system, angular momentum is unaltered.
- 7.28** Velocity of A = $\omega_0 R$ in the same direction as the arrow; velocity of B = $\omega_0 R$ in the opposite direction to the arrow; velocity of C = $\omega_0 R/2$ in the same direction as the arrow. The disc will not roll on a frictionless plane.
- 7.29** (a) Frictional force at B opposes velocity of B. Therefore, frictional force is in the same direction as the arrow. The sense of frictional torque is such as to oppose angular motion. ω_0 and τ are both normal to the paper, the first into the paper, and the second coming out of the paper.
 (b) Frictional force decreases the velocity of the point of contact B. Perfect rolling ensues when this velocity is zero. Once this is so, the force of friction is zero.
- 7.30** Frictional force causes the CM to accelerate from its initial zero velocity. Frictional torque causes retardation in the initial angular speed ω_0 . The equations of motion are : $\mu_k mg = ma$ and $\mu_k mgR = -I\alpha$, which yield $v = \mu_k g t$, $\omega = \omega_0 - \mu_k mgRt / I$. Rolling begins when $v = R\omega$. For a ring, $I = mR^2$, and rolling begins at $t = \omega_0 R / 2\mu_k g$. For a disc, $I = \frac{1}{2}mR^2$ and rolling starts at break line $t = R\omega_0 / 3\mu_k g$. Thus, the disc begins to roll earlier than the ring, for the same R and ω_0 . The actual times can be obtained for $R = 10 \text{ cm}$, $\omega_0 = 10 \pi \text{ rad s}^{-1}$, $\mu_k = 0.2$

- 7.31** (a) 16.4 N
 (b) Zero
 (c) 37° approx.

Chapter 8

- 8.1** (a) No.
 (b) Yes, if the size of the space ship is large enough for him to detect the variation in g .
 (c) Tidal effect depends inversely on the cube of the distance unlike force, which depends inversely on the square of the distance.
- 8.2** (a) decreases; (b) decreases; (c) mass of the body; (d) more.
- 8.3** Smaller by a factor of 0.63.
- 8.5** 3.54×10^8 years.
- 8.6** (a) Kinetic energy, (b) less,
- 8.7** (a) No, (b) No, (c) No, (d) Yes
 [The escape velocity is independent of mass of the body and the direction of projection. It depends upon the gravitational potential at the point from where the body is launched. Since this potential depends (slightly) on the latitude and height of the point, the escape velocity (speed) depends (slightly) on these factors.]
- 8.8** All quantities vary over an orbit except angular momentum and total energy.
- 8.9** (b), (c) and (d)
- 8.10** and **8.11** For these two problems, complete the hemisphere to sphere. At both P, and C, potential is constant and hence intensity = 0. Therefore, for the hemisphere, (c) and (e) are correct.



- 8.12** 2.6×10^8 m
8.13 2.0×10^{30} kg
8.14 1.43×10^{12} m
8.15 28 N
8.16 125 N
8.17 8.0×10^6 m from the earth's centre
8.18 31.7 km/s
8.19 5.9×10^9 J

- 8.20** $2.6 \times 10^6 \text{ m/s}$
- 8.21** $0, 2.7 \times 10^{-8} \text{ J/kg}$; an object placed at the mid point is in an unstable equilibrium
- 8.22** $-9.4 \times 10^6 \text{ J/kg}$
- 8.23** $GM/R^2 = 2.3 \times 10^{12} \text{ m s}^{-2}$, $\omega^2 R = 1.1 \times 10^6 \text{ m s}^{-2}$; here ω is the angular speed of rotation. Thus in the rotating frame of the star, the inward force is much greater than the outward centrifugal force at its equator. The object will remain stuck (and not fly off due to centrifugal force). Note, if angular speed of rotation increases say by a factor of 2000, the object will fly off.
- 8.24** $3 \times 10^{11} \text{ J}$
- 8.25** 495 km

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APPENDICES

APPENDIX A 1 THE GREEK ALPHABET

Alpha	A	α	Iota	I	ι	Rho	P	ρ
Beta	B	β	Kappa	K	κ	Sigma	Σ	σ
Gamma	Γ	γ	Lambda	Λ	λ	Tau	T	τ
Delta	Δ	δ	Mu	M	μ	Upsilon	Y	υ
Epsilon	E	ε	Nu	N	ν	Phi	Φ	ϕ, φ
Zeta	Z	ζ	Xi	Ξ	ξ	Chi	X	χ
Eta	H	η	Omicron	O	o	Psi	Ψ	ψ
Theta	Θ	θ	Pi	Π	π	Omega	Ω	ω

APPENDIX A 2 COMMON SI PREFIXES AND SYMBOLS FOR MULTIPLES AND SUB-MULTIPLES

Multiple			Sub-Multiple		
Factor	Prefix	Symbol	Factor	Prefix	symbol
10^{18}	Exa	E	10^{-18}	atto	a
10^{15}	Peta	P	10^{-15}	femto	f
10^{12}	Tera	T	10^{-12}	pico	p
10^9	Giga	G	10^{-9}	nano	n
10^6	Mega	M	10^{-6}	micro	μ
10^3	kilo	k	10^{-3}	milli	m
10^2	Hecto	h	10^{-2}	centi	c
10^1	Deca	da	10^{-1}	deci	d

APPENDIX A 3

SOME IMPORTANT CONSTANTS

Name	Symbol	Value
Speed of light in vacuum	c	$2.9979 \times 10^8 \text{ m s}^{-1}$
Charge of electron	e	$1.602 \times 10^{-19} \text{ C}$
Gravitational constant	G	$6.673 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Planck constant	h	$6.626 \times 10^{-34} \text{ J s}$
Boltzmann constant	k	$1.381 \times 10^{-23} \text{ J K}^{-1}$
Avogadro number	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Universal gas constant	R	$8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
Mass of electron	m_e	$9.110 \times 10^{-31} \text{ kg}$
Mass of neutron	m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Electron-charge to mass ratio	e/m_e	$1.759 \times 10^{11} \text{ C/kg}$
Faraday constant	F	$9.648 \times 10^4 \text{ C/mol}$
Rydberg constant	R	$1.097 \times 10^7 \text{ m}^{-1}$
Bohr radius	a_0	$5.292 \times 10^{-11} \text{ m}$
Stefan-Boltzmann constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Wien's Constant	b	$2.898 \times 10^{-3} \text{ m K}$
Permittivity of free space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
	$1/4\pi \epsilon_0$	$8.987 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Permeability of free space	μ_0	$4\pi \times 10^{-7} \text{ T m A}^{-1}$
		$\cong 1.257 \times 10^{-6} \text{ Wb A}^{-1} \text{ m}^{-1}$

Other useful constants

Name	Symbol	Value
Mechanical equivalent of heat	J	4.186 J cal^{-1}
Standard atmospheric pressure	1 atm	$1.013 \times 10^5 \text{ Pa}$
Absolute zero	0 K	$-273.15 \text{ }^\circ\text{C}$
Electron volt	1 eV	$1.602 \times 10^{-19} \text{ J}$
Unified Atomic mass unit	1 u	$1.661 \times 10^{-27} \text{ kg}$
Electron rest energy	mc^2	0.511 MeV
Energy equivalent of 1 u	$1 \text{ u } c^2$	931.5 MeV
Volume of ideal gas(0 °C and 1atm)	V	22.4 L mol^{-1}
Acceleration due to gravity (sea level, at equator)	g	9.78049 m s^{-2}

APPENDIX A 4 CONVERSION FACTORS

Conversion factors are written as equations for simplicity.

Length

$$\begin{aligned} 1 \text{ km} &= 0.6215 \text{ mi} \\ 1 \text{ mi} &= 1.609 \text{ km} \\ 1 \text{ m} &= 1.0936 \text{ yd} = 3.281 \text{ ft} = 39.37 \text{ in} \\ 1 \text{ in} &= 2.54 \text{ cm} \\ 1 \text{ ft} &= 12 \text{ in} = 30.48 \text{ cm} \\ 1 \text{ yd} &= 3 \text{ ft} = 91.44 \text{ cm} \\ 1 \text{ lightyear} &= 1 \text{ ly} = 9.461 \times 10^{15} \text{ m} \\ 1 \text{ \AA} &= 0.1 \text{ nm} \end{aligned}$$

Area

$$\begin{aligned} 1 \text{ m}^2 &= 10^4 \text{ cm}^2 \\ 1 \text{ km}^2 &= 0.3861 \text{ mi}^2 = 247.1 \text{ acres} \\ 1 \text{ in}^2 &= 6.4516 \text{ cm}^2 \\ 1 \text{ ft}^2 &= 9.29 \times 10^{-2} \text{ m}^2 \\ 1 \text{ m}^2 &= 10.76 \text{ ft}^2 \\ 1 \text{ acre} &= 43,560 \text{ ft}^2 \\ 1 \text{ mi}^2 &= 460 \text{ acres} = 2.590 \text{ km}^2 \end{aligned}$$

Volume

$$\begin{aligned} 1 \text{ m}^3 &= 10^6 \text{ cm}^3 \\ 1 \text{ L} &= 1000 \text{ cm}^3 = 10^{-3} \text{ m}^3 \\ 1 \text{ gal} &= 3.786 \text{ L} \\ 1 \text{ gal} &= 4 \text{ qt} = 8 \text{ pt} = 128 \text{ oz} = 231 \text{ in}^3 \\ 1 \text{ in}^3 &= 16.39 \text{ cm}^3 \\ 1 \text{ ft}^3 &= 1728 \text{ in}^3 = 28.32 \text{ L} = 2.832 \times 10^4 \text{ cm}^3 \end{aligned}$$

Speed

$$\begin{aligned} 1 \text{ km h}^{-1} &= 0.2778 \text{ m s}^{-1} = 0.6215 \text{ mi h}^{-1} \\ 1 \text{ mi h}^{-1} &= 0.4470 \text{ m s}^{-1} = 1.609 \text{ km h}^{-1} \\ 1 \text{ mi h}^{-1} &= 1.467 \text{ ft s}^{-1} \end{aligned}$$

Magnetic Field

$$\begin{aligned} 1 \text{ G} &= 10^{-4} \text{ T} \\ 1 \text{ T} &= 1 \text{ Wb m}^{-2} = 10^4 \text{ G} \end{aligned}$$

Angle and Angular Speed

$$\begin{aligned} \pi \text{ rad} &= 180^\circ \\ 1 \text{ rad} &= 57.30^\circ \\ 1^\circ &= 1.745 \times 10^{-2} \text{ rad} \\ 1 \text{ rev min}^{-1} &= 0.1047 \text{ rad s}^{-1} \\ 1 \text{ rad s}^{-1} &= 9.549 \text{ rev min}^{-1} \end{aligned}$$

Mass

$$\begin{aligned} 1 \text{ kg} &= 1000 \text{ g} \\ 1 \text{ tonne} &= 1000 \text{ kg} = 1 \text{ Mg} \\ 1 \text{ u} &= 1.6606 \times 10^{-27} \text{ kg} \\ 1 \text{ kg} &= 6.022 \times 10^{26} \text{ u} \\ 1 \text{ slug} &= 14.59 \text{ kg} \\ 1 \text{ kg} &= 6.852 \times 10^{-2} \text{ slug} \\ 1 \text{ u} &= 931.50 \text{ MeV}/c^2 \end{aligned}$$

Density

$$1 \text{ g cm}^{-3} = 1000 \text{ kg m}^{-3} = 1 \text{ kg L}^{-1}$$

Force

$$\begin{aligned} 1 \text{ N} &= 0.2248 \text{ lbf} = 10^5 \text{ dyn} \\ 1 \text{ lbf} &= 4.4482 \text{ N} \\ 1 \text{ kgf} &= 2.2046 \text{ lbf} \end{aligned}$$

Time

$$\begin{aligned} 1 \text{ h} &= 60 \text{ min} = 3.6 \text{ ks} \\ 1 \text{ d} &= 24 \text{ h} = 1440 \text{ min} = 86.4 \text{ ks} \\ 1 \text{ y} &= 365.24 \text{ d} = 31.56 \text{ Ms} \end{aligned}$$

Pressure

$$\begin{aligned} 1 \text{ Pa} &= 1 \text{ N m}^{-2} \\ 1 \text{ bar} &= 100 \text{ kPa} \\ 1 \text{ atm} &= 101.325 \text{ kPa} = 1.01325 \text{ bar} \\ 1 \text{ atm} &= 14.7 \text{ lbf/in}^2 = 760 \text{ mm Hg} \\ &= 29.9 \text{ in Hg} = 33.8 \text{ ft H}_2\text{O} \\ 1 \text{ lbf in}^{-2} &= 6.895 \text{ kPa} \\ 1 \text{ torr} &= 1 \text{ mm Hg} = 133.32 \text{ Pa} \end{aligned}$$

Energy

$$1 \text{ kW h} = 3.6 \text{ MJ}$$

$$1 \text{ cal} = 4.186 \text{ J}$$

$$1 \text{ ft lbf} = 1.356 \text{ J} = 1.286 \times 10^{-3} \text{ Btu}$$

$$1 \text{ L atm} = 101.325 \text{ J}$$

$$1 \text{ L atm} = 24.217 \text{ cal}$$

$$1 \text{ Btu} = 778 \text{ ft lb} = 252 \text{ cal} = 1054.35 \text{ J}$$

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

$$1 \text{ u } c^2 = 931.50 \text{ MeV}$$

$$1 \text{ erg} = 10^{-7} \text{ J}$$

Power

$$1 \text{ horsepower (hp)} = 550 \text{ ft lbf/s} \\ = 745.7 \text{ W}$$

$$1 \text{ Btu min}^{-1} = 17.58 \text{ W}$$

$$1 \text{ W} = 1.341 \times 10^{-3} \text{ hp} \\ = 0.7376 \text{ ft lbf/s}$$

Thermal Conductivity

$$1 \text{ W m}^{-1} \text{ K}^{-1} = 6.938 \text{ Btu in/hft}^2 \text{ } ^\circ\text{F}$$

$$1 \text{ Btu in/hft}^2 \text{ } ^\circ\text{F} = 0.1441 \text{ W/m K}$$

APPENDIX A 5 MATHEMATICAL FORMULAE

Geometry

Circle of radius r : circumference = $2\pi r$;

$$\text{area} = \pi r^2$$

Sphere of radius r : area = $4\pi r^2$;

$$\text{volume} = \frac{4}{3}\pi r^3$$

Right circular cylinder of radius r and height h : area = $2\pi r^2 + 2\pi r h$;

$$\text{volume} = \pi r^2 h;$$

Triangle of base a and altitude h .

$$\text{area} = \frac{1}{2} a h$$

Quadratic Formula

If $ax^2 + bx + c = 0$,

$$\text{then } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

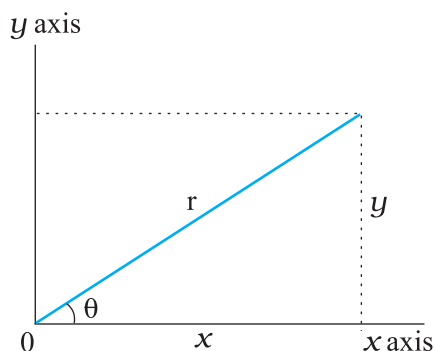
Trigonometric Functions of Angle θ 

Fig. A 5.1

$$\begin{aligned} \sin \theta &= \frac{y}{r} & \cos \theta &= \frac{x}{r} \\ \tan \theta &= \frac{y}{x} & \cot \theta &= \frac{x}{y} \\ \sec \theta &= \frac{r}{x} & \csc \theta &= \frac{r}{y} \end{aligned}$$

Pythagorean Theorem

In this right triangle, $a^2 + b^2 = c^2$

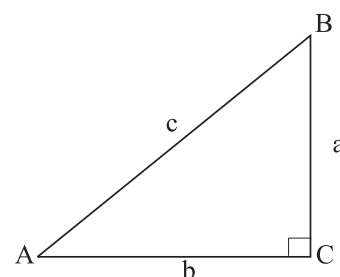


Fig. A 5.2

Triangles

Angles are A, B, C

Opposite sides are a, b, c

Angles $A + B + C = 180^\circ$

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Exterior angle $D = A + C$

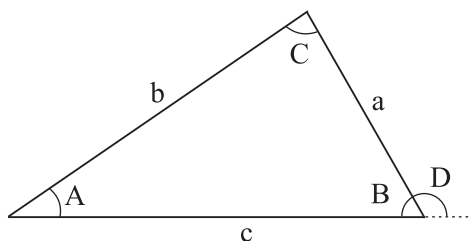


Fig. A 5.3

Mathematical Signs and Symbols

= equals

 $\cong$ equals approximately $\sim$ is the order of magnitude of $\neq$ is not equal to $\equiv$ is identical to, is defined as $>$ is greater than ($>>$ is much greater than) $<$ is less than ($<<$ is much less than) $\geq$ is greater than or equal to (or, is no less than) $\leq$ is less than or equal to (or, is no more than) $\pm$ plus or minus $\propto$ is proportional to Σ the sum of $\bar{x}$ or $\langle x \rangle$ or x_{av} the average value of x **Trigonometric Identities**

$$\sin(90^\circ - \theta) = \cos \theta$$

$$\cos(90^\circ - \theta) = \sin \theta$$

$$\sin \theta / \cos \theta = \tan \theta$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\csc^2 \theta - \cot^2 \theta = 1$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2\cos^2 \theta - 1 \\ = 1 - 2\sin^2 \theta$$

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

$$\sin \alpha \pm \sin \beta = 2 \sin \frac{1}{2}(\alpha \pm \beta) \cos \frac{1}{2}(\alpha \mp \beta)$$

$$\cos \alpha + \cos \beta$$

$$= 2 \cos \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta)$$

$$\cos \alpha - \cos \beta$$

$$= -2 \sin \frac{1}{2}(\alpha + \beta) \sin \frac{1}{2}(\alpha - \beta)$$

Binomial Theorem

$$(1 - x)^n = 1 - \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \dots (x^2 < 1)$$

$$(1 - x)^{-n} = 1 + \frac{nx}{1!} + \frac{n(n+1)x^2}{2!} + \dots (x^2 < 1)$$

Exponential Expansion

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

Logarithmic Expansion

$$\ln(1 + x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \dots (|x| < 1)$$

Trigonometric Expansion**(θ in radians)**

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$$

$$\cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots$$

$$\tan \theta = \theta + \frac{\theta^3}{3} + \frac{2\theta^5}{15} - \dots$$

Products of Vectors

Let $\hat{i}$, $\hat{j}$ and $\hat{k}$ be unit vectors in the x , y and z directions. Then

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

$$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0, \hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$$

Any vector $\mathbf{a}$ with components a_x , a_y , and a_z along the x , y , and z axes can be written,

$$\mathbf{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

Let **a**, **b** and **c** be arbitrary vectors with magnitudes a , b and c . Then

$$\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$$

$$(\mathbf{s}\mathbf{a}) \times \mathbf{b} = \mathbf{a} \times (\mathbf{s}\mathbf{b}) = s(\mathbf{a} \times \mathbf{b}) \quad (s \text{ is a scalar})$$

Let θ be the smaller of the two angles between **a** and **b**. Then

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a} = a_x b_x + a_y b_y + a_z b_z = ab \cos \theta$$

$$|\mathbf{a} \times \mathbf{b}| = ab \sin \theta$$

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= -\mathbf{b} \times \mathbf{a} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} \\ &= (a_y b_z - b_y a_z) \hat{\mathbf{i}} + (a_z b_x - b_z a_x) \hat{\mathbf{j}} + (a_x b_y - b_x a_y) \hat{\mathbf{k}} \\ \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) &= \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) \\ \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c} \end{aligned}$$

APPENDIX A 6

SI DERIVED UNITS

A 6.1 Some SI Derived Units expressed in SI Base Units

Physical quantity	SI Unit	
	Name	Symbol
Area	square metre	m ²
Volume	cubic metre	m ³
Speed, velocity	metre per second	m/s or m s ⁻¹
Angular velocity	radian per second	rad/s or rad s ⁻¹
Acceleration	metre per second square	m/s ² or m s ⁻²
Angular acceleration	radian per second square	rad/s ² or rad s ⁻²
Wave number	per metre	m ⁻¹
Density, mass density	kilogram per cubic metre	kg/m ³ or kg m ⁻³
Current density	ampere per square metre	A/m ² or A m ⁻²
Magnetic field strength, magnetic intensity, magnetic moment density	ampere per metre	A/m or A m ⁻¹
Concentration (of amount of substance)	mole per cubic metre	mol/m ³ or mol m ⁻³
Specific volume	cubic metre per kilogram	m ³ /kg or m ³ kg ⁻¹
Luminance, intensity of illumination	candela per square metre	cd/m ² or cd m ⁻²
Kinematic viscosity	square metre per second	m ² /s or m ² s ⁻¹
Momentum	kilogram metre per second	kg m s ⁻¹
Moment of inertia	kilogram square metre	kg m ²
Radius of gyration	metre	m
Linear/superficial/volume expansivities	per kelvin	K ⁻¹
Flow rate	cubic metre per second	m ³ s ⁻¹

A 6.2 SI Derived Units with special names

Physical quantity	SI Unit			
	Name	Symbol	Expression in terms of other units	Expression in terms of SI base units
Frequency	hertz	Hz	-	s^{-1}
Force	newton	N	-	$kg\ m\ s^{-2}$ or $kg\ m/s^2$
Pressure, stress	pascal	Pa	N/m^2 or $N\ m^{-2}$	$kg\ m^{-1}\ s^{-2}$ or $kg /s^2\ m$
Energy, work, quantity of heat	joule	J	$N\ m$	$kg\ m^2\ s^{-2}$ or $kg\ m^2/s^2$
Power, radiant flux	watt	W	J/s or $J\ s^{-1}$	$kg\ m^2\ s^{-3}$ or $kg\ m^2/s^3$
Quantity of electricity, electric charge	coulomb	C	-	$A\ s$
Electric potential, potential difference, electromotive force	volt	V	W/A or $W\ A^{-1}$	$kg\ m^2\ s^{-3}\ A^{-1}$ or $kg\ m^2/s^3\ A$
Capacitance	farad	F	C/V	$A^2\ s^4\ kg^{-1}\ m^{-2}$
Electric resistance	ohm	Ω	V/A	$kg\ m^2\ s^{-3}\ A^{-2}$
Conductance	siemens	S	A/V	$m^{-2}\ kg^{-1}\ s^3\ A^2$
Magnetic flux	weber	Wb	$V\ s$ or J/A	$kg\ m^2\ s^{-2}\ A^{-1}$
Magnetic field, magnetic flux density, magnetic induction	tesla	T	Wb/m^2	$kg\ s^{-2}\ A^{-1}$
Inductance	henry	H	Wb/A	$kg\ m^2\ s^{-2}\ A^{-2}$
Luminous flux, luminous power	lumen	lm	-	cd /sr
Illuminance	lux	lx	lm/m^2	$m^{-2}\ cd\ sr^{-1}$
Activity (of a radio nuclide/radioactive source)	becquerel	Bq	-	s^{-1}
Absorbed dose, absorbed dose index	gray	Gy	J/kg	m^2/s^2 or $m^2\ s^{-2}$

A 6.3 Some SI Derived Units expressed by means of SI Units with special names

Physical quantity	SI Unit		
	Name	Symbol	Expression in terms of SI base units
Magnetic moment	joule per tesla	$J\ T^{-1}$	$m^2\ A$
Dipole moment	coulomb metre	$C\ m$	$s\ A\ m$
Dynamic viscosity	poiseulles or pascal second or newton second per square metre	Pl or $Pa\ s$ or $N\ s\ m^{-2}$	$m^{-1}\ kg\ s^{-1}$
Torque, couple, moment of force	newton metre	$N\ m$	$m^2\ kg\ s^{-2}$
Surface tension	newton per metre	N/m	$kg\ s^{-2}$
Power density, irradiance, heat flux density	watt per square metre	W/m^2	$kg\ s^{-3}$

Heat capacity, entropy	joule per kelvin	J/K	$\text{m}^2 \text{kg s}^{-2} \text{K}^{-1}$
Specific heat capacity, specific entropy	joule per kilogram kelvin	J/kg K	$\text{m}^2 \text{s}^{-2} \text{K}^{-1}$
Specific energy, latent heat	joule per kilogram	J/kg	$\text{m}^2 \text{s}^{-2}$
Radiant intensity	watt per steradian	W sr^{-1}	$\text{kg m}^2 \text{s}^{-3} \text{sr}^{-1}$
Thermal conductivity	watt per metre kelvin	$\text{W m}^{-1} \text{K}^{-1}$	$\text{m kg s}^{-3} \text{K}^{-1}$
Energy density	joule per cubic metre	J/m^3	$\text{kg m}^{-1} \text{s}^{-2}$
Electric field strength	volt per metre	V/m	$\text{m kg s}^{-3} \text{A}^{-1}$
Electric charge density	coulomb per cubic metre	C/m^3	$\text{m}^{-3} \text{A s}$
Electric flux density	coulomb per square metre	C/m^2	$\text{m}^{-2} \text{A s}$
Permittivity	farad per metre	F/m	$\text{m}^{-3} \text{kg}^{-1} \text{s}^4 \text{A}^2$
Permeability	henry per metre	H/m	$\text{m kg s}^{-2} \text{A}^{-2}$
Molar energy	joule per mole	J/mol	$\text{m}^2 \text{kg s}^{-2} \text{mol}^{-1}$
Angular momentum, Planck's constant	joule second	J s	$\text{kg m}^2 \text{s}^{-1}$
Molar entropy, molar heat capacity	joule per mole kelvin	J/mol K	$\text{m}^2 \text{kg s}^{-2} \text{K}^{-1} \text{mol}^{-1}$
Exposure (x-rays and γ -rays)	coulomb per kilogram	C/kg	$\text{kg}^{-1} \text{s A}$
Absorbed dose rate	gray per second	Gy/s	$\text{m}^2 \text{s}^{-3}$
Compressibility	per pascal	Pa^{-1}	$\text{m kg}^{-1} \text{s}^2$
Elastic moduli	newton per square metre	N/m^2 or N m^{-2}	$\text{kg m}^{-1} \text{s}^{-2}$
Pressure gradient	pascal per metre	Pa/m or N m^{-3}	$\text{kg m}^{-2} \text{s}^{-2}$
Surface potential	joule per kilogram	J/kg or N m/kg	$\text{m}^2 \text{s}^{-2}$
Pressure energy	pascal cubic metre	Pa m^3 or N m	$\text{kg m}^2 \text{s}^{-2}$
Impulse	newton second	N s	kg m s^{-1}
Angular impulse	newton metre second	N m s	$\text{kg m}^2 \text{s}^{-1}$
Specific resistance	ohm metre	Ωm	$\text{kg m}^3 \text{s}^{-3} \text{A}^{-2}$
Surface energy	joule per square metre	J/m^2 or N/m	kg s^{-2}

APPENDIX A 7

GENERAL GUIDELINES FOR USING SYMBOLS FOR PHYSICAL QUANTITIES, CHEMICAL ELEMENTS AND NUCLIDES

- Symbols for physical quantities are normally single letters and printed in italic (or sloping) type. However, in case of the two letter symbols, appearing as a factor in a product, some spacing is necessary to separate this symbol from other symbols.
- Abbreviations, i.e., shortened forms of names or expressions, such as p.e. for potential energy, are not used in physical equations. These abbreviations in the text are written in ordinary normal/roman (upright) type.
- Vectors are printed in bold and normal/roman (upright) type. However, in class room situations, vectors may be indicated by an arrow on the top of the symbol.
- Multiplication or product of two physical quantities is written with some spacing between them. Division of one physical quantity by another may be indicated with a horizontal bar or with

solidus, a slash or a short oblique stroke mark (/) or by writing it as a product of the numerator and the inverse first power of the denominator, using brackets at appropriate places to clearly distinguish between the numerator and the denominator.

- Symbols for chemical elements are written in normal/roman (upright) type. The symbol is not followed by a full stop.
For example, Ca, C, H, He, U, etc.
- The attached numerals specifying a nuclide are placed as a left subscript (atomic number) and superscript (mass number).

For example, a U-235 nuclide is expressed as ${}_{92}^{235}\text{U}$ (with 235 expressing the mass number and 92 as the atomic number of uranium with chemical symbol U).

- The right superscript position is used, if required, for indicating a state of ionisation (in case of ions).

For example, Ca^{2+} , PO_4^{3-}

APPENDIX A 8

GENERAL GUIDELINES FOR USING SYMBOLS FOR SI UNITS, SOME OTHER UNITS, AND SI PREFIXES

- Symbols for units of physical quantities are printed/written in Normal/Roman (upright) type.
- Standard and recommended symbols for units are written in lower case roman (upright) type, starting with small letters. The shorter designations for units such as kg, m, s, cd, etc., are symbols and not the abbreviations. The unit names are never capitalised. However, the unit symbols are capitalised only if the symbol for a unit is derived from a proper name of scientist, beginning with a capital, normal/roman letter.

For example, m for the unit 'metre', d for the unit 'day', atm for the unit 'atmospheric pressure', Hz for the unit 'hertz', Wb for the unit 'weber', J for the unit 'joule', A for the unit 'ampere', V for the unit 'volt', etc. The single exception is L, which is the symbol for the unit 'litre'. This exception is made to avoid confusion of the lower case letter l with the Arabic numeral 1.

- Symbols for units do not contain any final full stop at the end of recommended letter and remain unaltered in the plural, using only singular form of the unit.
For example, for a length of 25 centimetres the unit symbol is written as 25 cm and not 25 cms or 25 cm. or 25 cms., etc.

- Use of solidus (/) is recommended only for indicating a division of one letter unit symbol by another unit symbol. Not more than one solidus is used.

For example :

m/s^2 or m s^{-2} (with a spacing between m and s^{-2}) but not m/s/s ;

$1 \text{ Pl} = 1 \text{ N s m}^{-2} = 1 \text{ N s/m}^2 = 1 \text{ kg/s m} = 1 \text{ kg m}^{-1} \text{ s}^{-1}$, but not 1 kg/m/s ;

J/K mol or $\text{J K}^{-1} \text{ mol}^{-1}$, but not J/K/mol ; etc.

- Prefix symbols are printed in normal/roman (upright) type without spacing between the prefix symbol and the unit symbol. Thus certain approved prefixes written very close to the unit symbol are used to indicate decimal fractions or multiples of a SI unit, when it is inconveniently small or large.

For example :

megawatt ($1 \text{ MW} = 10^6 \text{ W}$);

centimetre ($1 \text{ cm} = 10^{-2} \text{ m}$);

kilometre ($1 \text{ km} = 10^3 \text{ m}$);

millivolt ($1 \text{ mV} = 10^{-3} \text{ V}$);

nanosecond ($1 \text{ ns} = 10^{-9} \text{ s}$);

picoFarad ($1 \text{ pF} = 10^{-12} \text{ F}$);

microsecond ($1 \mu\text{s} = 10^{-6} \text{ s}$);

gigahertz ($1 \text{ GHz} = 10^9 \text{ Hz}$);

kilowatt-hour ($1 \text{ kW h} = 10^3 \text{ W h} = 3.6 \text{ MJ} = 3.6 \times 10^6 \text{ J}$);
 microampere ($1 \mu\text{A} = 10^{-6} \text{ A}$); micron ($1 \mu\text{m} = 10^{-6} \text{ m}$);
 angstrom ($1 \text{ \AA} = 0.1 \text{ nm} = 10^{-10} \text{ m}$); etc.

The unit 'micron' which equals 10^{-6} m , i.e. a micrometre, is simply the name given to convenient sub-multiple of the metre. In the same spirit, the unit 'fermi', equal to a femtometre or 10^{-15} m has been used as the convenient length unit in nuclear studies. Similarly, the unit 'barn', equal to 10^{-28} m^2 , is a convenient measure of cross-sectional areas in sub-atomic particle collisions. However, the unit 'micron' is preferred over the unit 'micrometre' to avoid confusion of the 'micrometre' with the length measuring instrument called 'micrometer'. These newly formed multiples or sub-multiples (cm, km, μm , μs , ns) of SI units, metre and second, constitute a new composite inseparable symbol for units.

- When a prefix is placed before the symbol of a unit, the combination of prefix and symbol is considered as a new symbol, for the unit, which can be raised to a positive or negative power without using brackets. These can be combined with other unit symbols to form compound unit. Rules for binding-in indices are not those of ordinary algebra.

For example :

cm^3 means always $(\text{cm})^3 = (0.01 \text{ m})^3 = (10^{-2} \text{ m})^3 = 10^{-6} \text{ m}^3$, but never 0.01 m^3 or 10^{-2} m^3 or 1 cm^3 (prefix c with a spacing with m^3 is meaningless as prefix c is to be attached to a unit symbol and it has no physical significance or independent existence without attachment with a unit symbol).

Similarly, mA^2 means always $(\text{mA})^2 = (0.001 \text{ A})^2 = (10^{-3} \text{ A})^2 = 10^{-6} \text{ A}^2$, but never 0.001 A^2 or 10^{-3} A^2 or m A^2 ;

$1 \text{ cm}^{-1} = (10^{-2} \text{ m})^{-1} = 10^2 \text{ m}^{-1}$, but not 1 c m^{-1} or 10^{-2} m^{-1} ;

$1 \mu\text{s}^{-1}$ means always $(10^{-6} \text{ s})^{-1} = 10^6 \text{ s}^{-1}$, but not $1 \times 10^{-6} \text{ s}^{-1}$;

1 km^2 means always $(\text{km})^2 = (10^3 \text{ m})^2 = 10^6 \text{ m}^2$, but not 10^3 m^2 ;

1 mm^2 means always $(\text{mm})^2 = (10^{-3} \text{ m})^2 = 10^{-6} \text{ m}^2$, but not 10^{-3} m^2 .

- A prefix is never used alone. It is always attached to a unit symbol and written or fixed before (pre-fix) the unit symbol.

For example :

$10^3/\text{m}^3$ means $1000/\text{m}^3$ or 1000 m^{-3} , but not k/m^3 or k m^{-3} .

$10^6/\text{m}^3$ means $10,00,000/\text{m}^3$ or $10,00,000 \text{ m}^{-3}$, but not M/m^3 or M m^{-3}

- Prefix symbol is written very close to the unit symbol without spacing between them, while unit symbols are written separately with spacing when units are multiplied together.

For example :

m s^{-1} (symbols m and s^{-1} , in lower case, small letter m and s, are separate and independent unit symbols for metre and second respectively, with spacing between them) means 'metre per second', but not 'milli per second'.

Similarly, ms^{-1} [symbol m and s are written very close to each other, with prefix symbol m (for prefix milli) and unit symbol s, in lower case, small letter (for unit 'second') without any spacing between them and making ms as a new composite unit] means 'per millisecond', but never 'metre per second'.

mS^{-1} [symbol m and S are written very close to each other, with prefix symbol m (for prefix milli) and unit symbol S, in capital roman letter S (for unit 'siemens') without any spacing between them, and making mS as a new composite unit] means 'per millisiemens', but never 'per millisecond'.

C m [symbol C and m are written separately, representing unit symbols C (for unit 'coulomb') and m (for unit 'metre'), with spacing between them] means 'coulomb metre', but never 'centimetre', etc.

- The use of double prefixes is avoided when single prefixes are available.

For example :

$10^{-9} \text{ m} = 1 \text{ nm}$ (nanometre), but not $1 \text{ m}\mu\text{m}$ (millimicrometre),
 $10^{-6} \text{ m} = 1 \mu\text{m}$ (micron), but not 1 mmm (millimillimetre),
 $10^{-12} \text{ F} = 1 \text{ pF}$ (picofarad), but not $1 \mu\mu\text{F}$ (micromicrofarad),
 $10^9 \text{ W} = 1 \text{ GW}$ (giga watt), but not 1 kMW (kilomegawatt), etc.

- The use of a combination of unit and the symbols for units is avoided when the physical quantity is expressed by combining two or more units.

For example :

joule per mole kelvin is written as J/mol K or $\text{J mol}^{-1} \text{ K}^{-1}$, but not joule/mole K or J/ mol kelvin or J/mole K, etc.

joule per tesla is written as J/T or J T^{-1} , but not joule /T or J per tesla or J/tesla, etc.

newton metre second is written as N m s , but not Newton m second or N m second or N metre s or newton metre s, etc.

joule per kilogram kelvin is written as J/kg K or $\text{J kg}^{-1} \text{ K}^{-1}$, but not J/kilog K or joule/kg K or J/kg kelvin or J/kilogram K, etc.

- To simplify calculations, the prefix symbol is attached to the unit symbol in the numerator and not to the denominator.

For example :

10^6 N/m^2 is written more conveniently as MN/m^2 , in preference to N/mm^2 .

A preference has been expressed for multiples or sub-multiples involving the factor 1000, $10^{\pm 3n}$ where n is the integer.

- Proper care is needed when same symbols are used for physical quantities and units of physical quantities.

For example :

The physical quantity weight (W) expressed as a product of mass (m) and acceleration due to gravity (g) may be written in terms of symbols W , m and g printed in italic (or sloping) type as $W = m g$, preferably with a spacing between m and g . It should not be confused with the unit symbols for the units watt (W), metre (m) and gram (g). However, in the equation $W = m g$, the symbol W expresses the weight with a unit symbol J, m as the mass with a unit symbol kg and g as the acceleration due to gravity with a unit symbol m/s^2 . Similarly, in equation $F = m a$, the symbol F expresses the force with a unit symbol N, m as the mass with a unit symbol kg, and a as the acceleration with a unit symbol m/s^2 . These symbols for physical quantities should not be confused with the unit symbols for the units 'farad' (F), 'metre' (m) and 'are' (a).

Proper distinction must be made while using the symbols h (prefix hecto, and unit hour), c (prefix centi, and unit carat), d (prefix deci and unit day), T (prefix tera, and unit tesla), a (prefix atto, and unit are), da (prefix deca, and unit deciare), etc.

- SI base unit 'kilogram' for mass is formed by attaching SI prefix (a multiple equal to 10^3) 'kilo' to a cgs (centimetre, gram, second) unit 'gram' and this may seem to result in an anomaly. Thus, while a thousandth part of unit of length (metre) is called a millimetre (mm), a thousandth part of the unit of mass (kg) is not called a millikilogram, but just a gram. This appears to give the impression that the unit of mass is a gram (g) which is not true. Such a situation has arisen because we are unable to replace the name 'kilogram' by any other suitable unit. Therefore, as an exception, name of the multiples and sub-multiples of the unit of mass are formed by attaching prefixes to the word 'gram' and not to the word 'kilogram'.

For example :

$10^3 \text{ kg} = 1 \text{ megagram (1 Mg)}$, but not 1 kilo kilogram (1 kkg);

$10^{-6} \text{ kg} = 1 \text{ milligram (1 mg)}$, but not 1 microkilogram (1 μkg);

$10^{-3} \text{ kg} = 1 \text{ gram (1g)}$, but not 1 millikilogram (1 mkg), etc.

It may be emphasised again that you should use the internationally approved and recommended symbols only. Continual practice of following general rules and guidelines in unit symbol writing would make you learn mastering the correct use of SI units, prefixes and related symbols for physical quantities in a proper perspective.

APPENDIX A 9
DIMENSIONAL FORMULAE OF PHYSICAL QUANTITIES

S.No	Physical quantity	Relationship with other physical quantities	Dimensions	Dimensional formula
1.	Area	Length $\times$ breadth	$[L^2]$	$[M^0 L^2 T^0]$
2.	Volume	Length $\times$ breadth $\times$ height	$[L^3]$	$[M^0 L^3 T^0]$
3.	Mass density	Mass/volume	$[M]/[L^3]$ or $[M L^{-3}]$	$[M L^{-3} T^0]$
4.	Frequency	1/time period	$1/[T]$	$[M^0 L^0 T^{-1}]$
5.	Velocity, speed	Displacement/time	$[L]/[T]$	$[M^0 L T^{-1}]$
6.	Acceleration	Velocity /time	$[L T^{-1}]/[T]$	$[M^0 L T^{-2}]$
7.	Force	Mass $\times$ acceleration	$[M][L T^{-2}]$	$[M L T^{-2}]$
8.	Impulse	Force $\times$ time	$[M L T^{-2}][T]$	$[M L T^{-1}]$
9.	Work, Energy	Force $\times$ distance	$[M L T^{-2}][L]$	$[M L^2 T^{-2}]$
10.	Power	Work/time	$[M L^2 T^{-2}]/[T]$	$[M L^2 T^{-3}]$
11.	Momentum	Mass $\times$ velocity	$[M][L T^{-1}]$	$[M L T^{-1}]$
12.	Pressure, stress	Force/area	$[M L T^{-2}]/[L^2]$	$[M L^{-1} T^{-2}]$
13.	Strain	$\frac{\text{Change in dimension}}{\text{Original dimension}}$	$[L]/[L]$ or $[L^3]/[L^3]$	$[M^0 L^0 T^0]$
14.	Modulus of elasticity	Stress/strain	$\frac{[M L^{-1} T^{-2}]}{[M^0 L^0 T^0]}$	$[M L^{-1} T^{-2}]$
15.	Surface tension	Force/length	$[M L T^{-2}]/[L]$	$[M L^0 T^{-2}]$
16.	Surface energy	Energy/area	$[M L^2 T^{-2}]/[L^2]$	$[M L^0 T^{-2}]$
17.	Velocity gradient	Velocity/distance	$[L T^{-1}]/[L]$	$[M^0 L^0 T^{-1}]$
18.	Pressure gradient	Pressure/distance	$[M L^{-1} T^{-2}]/[L]$	$[M L^{-2} T^{-2}]$
19.	Pressure energy	Pressure $\times$ volume	$[M L^{-1} T^{-2}][L^3]$	$[M L^2 T^{-2}]$
20.	Coefficient of viscosity	Force/area $\times$ velocity gradient	$\frac{[M L T^{-2}]}{[L^2][L T^{-1}]/[L]}$	$[M L^{-1} T^{-1}]$
21.	Angle, Angular displacement	Arc/radius	$[L]/[L]$	$[M^0 L^0 T^0]$
22.	Trigonometric ratio ($\sin\theta$, $\cos\theta$, $\tan\theta$, etc.)	Length/length	$[L]/[L]$	$[M^0 L^0 T^0]$
23.	Angular velocity	Angle/time	$[L^0]/[T]$	$[M^0 L^0 T^{-1}]$

24.	Angular acceleration	Angular velocity/time	$[T^{-1}]/[T]$	$[M^0 L^0 T^{-2}]$
25.	Radius of gyration	Distance	$[L]$	$[M^0 L T^0]$
26.	Moment of inertia	Mass $\times$ (radius of gyration) ²	$[M] [L^2]$	$[ML^2 T^0]$
27.	Angular momentum	Moment of inertia $\times$ angular velocity	$[ML^2] [T^{-1}]$	$[ML^2 T^{-1}]$
28.	Moment of force, moment of couple	Force $\times$ distance	$[MLT^{-2}] [L]$	$[ML^2 T^{-2}]$
29.	Torque	Angular momentum/time, Or Force $\times$ distance	$[ML^2 T^{-1}] / [T]$ or $[MLT^{-2}] [L]$	$[ML^2 T^{-2}]$
30.	Angular frequency	$2\pi \times$ Frequency	$[T^{-1}]$	$[M^0 L^0 T^{-1}]$
31.	Wavelength	Distance	$[L]$	$[M^0 L T^0]$
32.	Hubble constant	Recession speed/distance	$[LT^{-1}]/[L]$	$[M^0 L^0 T^{-1}]$
33.	Intensity of wave	(Energy/time)/area	$[ML^2 T^{-2}/T]/[L^2]$	$[ML^0 T^{-3}]$
34.	Radiation pressure	$\frac{\text{Intensity of wave}}{\text{Speed of light}}$	$[MT^{-3}]/[LT^{-1}]$	$[ML^{-1} T^{-2}]$
35.	Energy density	Energy/volume	$[ML^2 T^{-2}] / [L^3]$	$[ML^{-1} T^{-2}]$
36.	Critical velocity	$\frac{\text{Reynold's number} \times \text{coefficient of viscosity}}{\text{Mass density} \times \text{radius}}$	$\frac{[M^0 L^0 T^0][ML^{-1} T^{-1}]}{[ML^{-3}][L]}$	$[M^0 L T^{-1}]$
37.	Escape velocity	$(2 \times \text{acceleration due to gravity} \times \text{earth's radius})^{1/2}$	$[LT^{-2}]^{1/2} \times [L]^{1/2}$	$[M^0 L T^{-1}]$
38.	Heat energy, internal energy	Work (= Force $\times$ distance)	$[MLT^{-2}] [L]$	$[ML^2 T^{-2}]$
39.	Kinetic energy	$(1/2) \text{ mass} \times (\text{velocity})^2$	$[M] [LT^{-1}]^2$	$[ML^2 T^{-2}]$
40.	Potential energy	Mass $\times$ acceleration due to gravity $\times$ height	$[M] [LT^{-2}] [L]$	$[ML^2 T^{-2}]$
41.	Rotational kinetic energy	$\frac{1}{2} \times \text{moment of inertia} \times (\text{angular velocity})^2$	$[M^0 L^0 T^0] [ML^2] \times [T^{-1}]^2$	$[ML^2 T^{-2}]$
42.	Efficiency	$\frac{\text{Output work or energy}}{\text{Input work or energy}}$	$\frac{[ML^2 T^{-2}]}{[ML^2 T^{-2}]}$	$[M^0 L^0 T^0]$
43.	Angular impulse	Torque $\times$ time	$[ML^2 T^{-2}] [T]$	$[ML^2 T^{-1}]$
44.	Gravitational constant	$\frac{\text{Force} \times (\text{distance})^2}{\text{mass} \times \text{mass}}$	$\frac{[MLT^{-2}][L^2]}{[M][M]}$	$[M^{-1} L^3 T^{-2}]$
45.	Planck constant	Energy/frequency	$[ML^2 T^{-2}] / [T^{-1}]$	$[ML^2 T^{-1}]$

46.	Heat capacity, entropy	Heat energy / temperature	$[ML^2 T^{-2}]/[K]$	$[ML^2 T^{-2} K^{-1}]$
47.	Specific heat capacity	$\frac{\text{Heat Energy}}{\text{Mass} \times \text{temperature}}$	$[ML^2 T^{-2}]/[M] [K]$	$[M^0 L^2 T^{-2} K^{-1}]$
48.	Latent heat	Heat energy/mass	$[ML^2 T^{-2}]/[M]$	$[M^0 L^2 T^{-2}]$
49.	Thermal expansion coefficient or Thermal expansivity	$\frac{\text{Change in dimension}}{\text{Original dimension} \times \text{temperature}}$	$[L] / [L][K]$	$[M^0 L^0 K^{-1}]$
50.	Thermal conductivity	$\frac{\text{Heat energy} \times \text{thickness}}{\text{Area} \times \text{temperature} \times \text{time}}$	$\frac{[ML^2 T^{-2}][L]}{[L^2] [K] [T]}$	$[MLT^{-3} K^{-1}]$
51.	Bulk modulus or (compressibility) ⁻¹	$\frac{\text{Volume} \times (\text{change in pressure})}{(\text{change in volume})}$	$\frac{[L^3] [ML^{-1} T^{-2}]}{[L^3]}$	$[ML^{-1} T^{-2}]$
52.	Centripetal acceleration	(Velocity) ² / radius	$[LT^{-1}]^2 / [L]$	$[M^0 LT^{-2}]$
53.	Stefan constant	$\frac{(\text{Energy} / \text{area} \times \text{time})}{(\text{Temperature})^4}$	$\frac{[ML^2 T^{-2}]}{[L^2] [T] [K]^4}$	$[ML^0 T^{-3} K^{-4}]$
54.	Wien constant	Wavelength $\times$ temperature	$[L] [K]$	$[M^0 LT^0 K]$
55.	Boltzmann constant	Energy/temperature	$[ML^2 T^{-2}]/[K]$	$[ML^2 T^{-2} K^{-1}]$
56.	Universal gas constant	$\frac{\text{Pressure} \times \text{volume}}{\text{mole} \times \text{temperature}}$	$\frac{[ML^{-1} T^{-2}][L^3]}{[\text{mol}] [K]}$	$[ML^2 T^{-2} K^{-1} \text{mol}^{-1}]$
57.	Charge	Current $\times$ time	$[A] [T]$	$[M^0 L^0 TA]$
58.	Current density	Current / area	$[A] / [L^2]$	$[M^0 L^{-2} T^0 A]$
59.	Voltage, electric potential, electromotive force	Work/charge	$[ML^2 T^{-2}]/[AT]$	$[ML^2 T^{-3} A^{-1}]$
60.	Resistance	$\frac{\text{Potential difference}}{\text{Current}}$	$\frac{[ML^2 T^{-3} A^{-1}]}{[A]}$	$[ML^2 T^{-3} A^{-2}]$
61.	Capacitance	Charge/potential difference	$\frac{[AT]}{[ML^2 T^{-3} A^{-1}]}$	$[M^{-1} L^{-2} T^4 A^2]$
62.	Electrical resistivity or (electrical conductivity) ⁻¹	$\frac{\text{Resistance} \times \text{area}}{\text{length}}$	$\frac{[ML^2 T^{-3} A^{-2}]}{[L^2]/[L]}$	$[ML^3 T^{-3} A^{-2}]$
63.	Electric field	Electrical force/charge	$[MLT^{-2}]/[AT]$	$[MLT^{-3} A^{-1}]$
64.	Electric flux	Electric field $\times$ area	$[MLT^{-3} A^{-1}][L^2]$	$[ML^3 T^{-3} A^{-1}]$

65.	Electric dipole moment	Torque/electric field	$\frac{[ML^2 T^{-2}]}{[MLT^{-3} A^{-1}]}$	$[M^0 LTA]$
66.	Electric field strength or electric intensity	$\frac{\text{Potential difference}}{\text{distance}}$	$\frac{[ML^2 T^{-3} A^{-1}]}{[L]}$	$[MLT^{-3} A^{-1}]$
67.	Magnetic field, magnetic flux density, magnetic induction	$\frac{\text{Force}}{\text{Current} \times \text{length}}$	$[MLT^{-2}]/[A] [L]$	$[ML^0 T^{-2} A^{-1}]$
68.	Magnetic flux	Magnetic field $\times$ area	$[MT^{-2} A^{-2}] [L^2]$	$[ML^2 T^{-2} A^{-1}]$
69.	Inductance	$\frac{\text{Magnetic flux}}{\text{Current}}$	$\frac{[ML^2 T^{-2} A^{-1}]}{[A]}$	$[ML^2 T^{-2} A^{-2}]$
70.	Magnetic dipole moment	Torque/magnetic field or current $\times$ area	$[ML^2 T^{-2}] / [MT^{-2} A^{-1}]$ or $[A] [L^2]$	$[M^0 L^2 T^0 A]$
71.	Magnetic field strength, magnetic intensity or magnetic moment density	$\frac{\text{Magnetic moment}}{\text{Volume}}$	$\frac{[L^2 A]}{[L^3]}$	$[M^0 L^{-1} T^0 A]$
72.	Permittivity constant (of free space)	$\frac{\text{Charge} \times \text{charge}}{4 \pi \times \text{electric force} \times (\text{distance})^2}$	$\frac{[AT][AT]}{[MLT^{-2}][L]^2}$	$[M^{-1} L^{-3} T^4 A^2]$
73.	Permeability constant (of free space)	$\frac{2 \pi \times \text{force} \times \text{distance}}{\text{current} \times \text{current} \times \text{length}}$	$\frac{[M^0 L^0 T^0][MLT^{-2}][L]}{[A][A][L]}$	$[MLT^{-2} A^{-2}]$
74.	Refractive index	$\frac{\text{Speed of light in vacuum}}{\text{Speed of light in medium}}$	$[LT^{-1}]/[LT^{-1}]$	$[M^0 L^0 T^0]$
75.	Faraday constant	Avogadro constant $\times$ elementary charge	$[AT]/[\text{mol}]$	$[M^0 L^0 TA \text{ mol}^{-1}]$
76.	Wave number	$2\pi/\text{wavelength}$	$[M^0 L^0 T^0] / [L]$	$[M^0 L^{-1} T^0]$
77.	Radiant flux, Radiant power	Energy emitted/time	$[ML^2 T^{-2}]/[T]$	$[ML^2 T^{-3}]$
78.	Luminosity of radiant flux or radiant intensity	$\frac{\text{Radiant power or radiant flux of source}}{\text{Solid angle}}$	$[ML^2 T^{-3}] / [M^0 L^0 T^0]$	$[ML^2 T^{-3}]$
79.	Luminous power or luminous flux of source	$\frac{\text{Luminous energy emitted}}{\text{time}}$	$[ML^2 T^{-2}]/[T]$	$[ML^2 T^{-3}]$

80.	Luminous intensity or illuminating power of source	$\frac{\text{Luminous flux}}{\text{Solid angle}}$	$\frac{[\text{ML}^2 \text{T}^{-3}]}{[\text{M}^0 \text{L}^0 \text{T}^0]}$	$[\text{ML}^2 \text{T}^{-3}]$
81.	Intensity of illumination or luminance	$\frac{\text{Luminous intensity}}{(\text{distance})^2}$	$[\text{ML}^2 \text{T}^{-3}]/[\text{L}^2]$	$[\text{ML}^0 \text{T}^{-3}]$
82.	Relative luminosity	$\frac{\text{Luminous flux of a source of given wavelength}}{\text{luminous flux of peak sensitivity wavelength (555 nm) source of same power}}$	$\frac{[\text{ML}^2 \text{T}^{-3}]}{[\text{ML}^2 \text{T}^{-3}]}$	$[\text{M}^0 \text{L}^0 \text{T}^0]$
83.	Luminous efficiency	$\frac{\text{Total luminous flux}}{\text{Total radiant flux}}$	$[\text{ML}^2 \text{T}^{-3}] / [\text{ML}^2 \text{T}^{-3}]$	$[\text{M}^0 \text{L}^0 \text{T}^0]$
84.	Illuminance or illumination	$\frac{\text{Luminous flux incident}}{\text{area}}$	$[\text{ML}^2 \text{T}^{-3}]/[\text{L}^2]$	$[\text{ML}^0 \text{T}^{-3}]$
85.	Mass defect	(sum of masses of nucleons) - (mass of the nucleus)	$[\text{M}]$	$[\text{ML}^0 \text{T}^0]$
86.	Binding energy of nucleus	Mass defect $\times$ (speed of light in vacuum) ²	$[\text{M}] [\text{L T}^{-1}]^2$	$[\text{ML}^2 \text{T}^{-2}]$
87.	Decay constant	0.693/half life	$[\text{T}^{-1}]$	$[\text{M}^0 \text{L}^0 \text{T}^{-1}]$
88.	Resonant frequency	$(\text{Inductance} \times \text{capacitance})^{-\frac{1}{2}}$	$[\text{ML}^2 \text{T}^{-2} \text{A}^{-2}]^{-\frac{1}{2}} \times$ $[\text{M}^{-1} \text{L}^{-2} \text{T}^4 \text{A}^2]^{-\frac{1}{2}}$	$[\text{M}^0 \text{L}^0 \text{A}^0 \text{T}^{-1}]$
89.	Quality factor or Q-factor of coil	$\frac{\text{Resonant frequency} \times \text{inductance}}{\text{Resistance}}$	$\frac{[\text{T}^{-1}][\text{ML}^2 \text{T}^{-2} \text{A}^{-2}]}{[\text{ML}^2 \text{T}^{-3} \text{A}^{-2}]}$	$[\text{M}^0 \text{L}^0 \text{T}^0]$
90.	Power of lens	(Focal length) ⁻¹	$[\text{L}^{-1}]$	$[\text{M}^0 \text{L}^{-1} \text{T}^0]$
91.	Magnification	$\frac{\text{Image distance}}{\text{Object distance}}$	$[\text{L}] / [\text{L}]$	$[\text{M}^0 \text{L}^0 \text{T}^0]$
92.	Fluid flow rate	$\frac{(\pi/8) (\text{pressure}) \times (\text{radius})^4}{(\text{viscosity coefficient}) \times (\text{length})}$	$\frac{[\text{ML}^{-1} \text{T}^{-2}] [\text{L}^4]}{[\text{ML}^{-1} \text{T}^{-1}] [\text{L}]}$	$[\text{M}^0 \text{L}^3 \text{T}^{-1}]$
93.	Capacitive reactance	(Angular frequency $\times$ capacitance) ⁻¹	$[\text{T}^{-1}]^{-1} [\text{M}^{-1} \text{L}^{-2} \text{T}^4 \text{A}^2]^{-1}$	$[\text{ML}^2 \text{T}^{-3} \text{A}^{-2}]$
94.	Inductive reactance	(Angular frequency $\times$ inductance)	$[\text{T}^{-1}][\text{ML}^2 \text{T}^{-2} \text{A}^{-2}]$	$[\text{ML}^2 \text{T}^{-3} \text{A}^{-2}]$